Cognitive Development in the Collaborative Perinatal Project: Twenty-Five Analyses Demonstrating an Underutilized Dataset

Jordan Lasker

Texas Tech University

jlasker@ttu.edu

August 2026

Abstract

The Collaborative Perinatal Project (CPP, 1959–1974) enrolled approximately 60,000 pregnancies across 12 U.S. medical centers and followed children through age 7. Despite its large sample, rich sibling structure (MZ/DZ twins, full and half-siblings), and detailed prenatal records, the CPP has seen little modern use because its data were distributed as poorly documented fixed-width ASCII files. Working from a recently modernized extract, we report seventeen original analyses and eight replications spanning behavior genetics, developmental psychology, and social epidemiology. Multi-level ACE decomposition yields WISC FSIQ heritability of $a^2 = 0.71$, while multivariate Independent Pathway modeling reveals structurally different genetic and environmental loading patterns across twelve cognitive measures. Mother fixed-effects analyses show that cross-sectional breastfeeding and prenatal-smoking associations do not replicate within mothers, while birth weight remains positive. Observed-score birth-order coefficients become positive, but cannot be separated cleanly from maternal age and calendar time. Measurement-invariance tests and replications of landmark CPP studies demonstrate both the research value of the cohort and the importance of documented, reproducible recoding.

1 Introduction

Between 1959 and 1974, the Collaborative Perinatal Project (CPP) enrolled approximately 60,000 pregnancies across 12 U.S. medical centers and followed children through age 7 with cognitive, medical, and socioeconomic assessments (Niswander & Gordon, 1972; Broman et al., 1975, 1987). Because many mothers had multiple children in the study, and because twin zygosity was determined through blood-group typing, the CPP offers an unusual combination of within-family comparisons and behavior genetic designs—all linked to detailed obstetric records and standardized testing at ages 4 (Stanford-Binet) and 7 (WISC).

Few modern researchers have used the CPP. The data were distributed as fixed-width ASCII files keyed to scanned microfiche codebooks from the 1960s, and no unified analytic extract existed. A companion data paper (Lasker, 2026) describes our rescue of this dataset: extraction of all 1,233 documented variables, construction of 27,721 pairwise kinship links—14,208 sibling and twin pairs with multi-tiered zygosity classification plus 13,513 extended relative pairs (cousins, nieces/nephews, and more distant kin)—and parsing of all 309 assessment card types from the 6.1-million-record master file.

Here we report seventeen original analyses and eight replications of published CPP findings. The original analyses cover behavior genetics, developmental psychology, and social epidemiology; the replications check our data extraction against known benchmarks and confirm continuity with earlier work. None of these analyses are meant to be the final word on their respective topics—they are intended to show what the CPP can do. We lean heavily on within-family designs that use the sibling and twin structure to deal with confounding, and we take advantage of historical features of the 1960s cohort—like the reversed SES–breastfeeding gradient—that create natural experiments no longer available in modern data.

2 Data and Methods

2.1 Sample

We restrict to live births with known sex ($N = 53{,}640$). The sample is approximately evenly split between White ($n = 24{,}693$; 46.0%) and Black ($n = 24{,}991$; 46.6%) children, with smaller numbers of Puerto Rican, Oriental, and other-race children.[1] Mean maternal age is 24.2 years; 46.7% of mothers smoked during pregnancy (recorded at prenatal registration).

The registration codebook assigns 88 to "no prior pregnancy" and 99 to "unknown" for

[1]"Puerto Rican" and "Oriental" are the CPP's original racial classification labels. The former reflects the predominantly Puerto Rican Hispanic population at the participating urban hospitals; the latter corresponds to what would now be termed "Asian." We retain the original terms throughout.

parity, prior perinatal loss, and prior live births. We map 88 to the structural value zero and retain 99 as missing. In the live-birth analysis file, 40,033 WISC records have observed live-birth rank; the WISC sibling-FE sample contains 13,111 children in 5,876 mothers. Mother fixed-effect regression uncertainty is clustered by mother, and pair-correlation sensitivity analyses use mother or extended-family blocks.

Case identifiers are read as character strings and standardized to the source-linked nine-character keys before merging. Because the terminal character is a linkage component rather than an outcome indicator, child-level analyses separately require the relevant observed birth outcome and known sex.

Cognitive outcomes include the Stanford-Binet IQ at age 4 (n = 38,651; M = 97.1, SD = 16.6) and the Wechsler Intelligence Scale for Children (WISC) at age 7 (n = 40,486; Full Scale IQ M = 95.6, SD = 14.8; Verbal IQ M = 91.9, SD = 13.6, n = 31,423; Performance IQ M = 96.4, SD = 15.0, n = 31,421). VIQ and PIQ are extracted directly from the raw assessment card using column positions verified against the JHU SAS programs (see Lasker, 2026); sample sizes are lower than for FSIQ because these composites come from card-level data rather than the summary variable file. We also extract a latent g factor as the first principal component (PCA PC1) from twelve age-7 cognitive measures (seven WISC scaled subtests, three WRAT raw scores, reversed Bender-Gestalt errors, and Goodenough Draw-a-Person raw score; n = 31,031; variance explained: 43.7%).

2.2 Family Structure and Kinship

The dataset contains 14,208 classified pairwise sibling links: 10,948 full-sibling pairs and 1,457 half-sibling pairs (classified by whether the study recorded the same or different biological father), 640 twin pairs, and 1,163 ambiguous pairs (siblings that could not be definitively classified as full or half). Twin zygosity is classified using a multi-tiered algorithm incorporating blood group determinations from Myrianthopoulos (1975), same-sex concordance, and placental examination, yielding 208 monozygotic (MZ) and 383 dizygotic (DZ) classified pairs (including 42 probable DZ).

In addition to within-family links, we constructed 13,513 extended kinship pairs from the NARA W17C–M work files, which record cross-family relationships identified by the original CPP field staff. These include 9,060 first-cousin pairs, 1,702 second-cousin pairs, 1,266 first-cousins-once-removed, 720 half-first-cousins, 435 half-nieces/nephews, and 330 nieces/nephews. The total kinship dataset thus comprises 27,721 classified pairs spanning genetic relatedness from $R = 1.0$ (MZ twins) through $R = 0.03125$ (second cousins).

For sibling-pair analyses of IQ, we have 9,242 pairs where both members completed the

WISC: 132 MZ twin pairs, 236 DZ twin pairs (including 10 probable DZ), 7,328 full-sibling pairs, 926 half-sibling pairs, and 620 ambiguous pairs. For extended kin, 6,916 first-cousin pairs and 1,291 second-cousin pairs have WISC data for both members. This multi-level relatedness structure (genetic relatedness $R = 1.0, 0.5, 0.5, 0.25, 0.125, 0.0625, 0.03125$) is unusual in behavioral science and enables behavior genetic decomposition beyond what classical twin-only designs allow.

2.3 Statistical Approach

We use three main analytical strategies. First, **ordinary least squares (OLS)** regression provides cross-sectional associations. Second, **mother fixed effects** (estimated via `fixest::feols` with `mother_id` as the fixed effect) exploit within-family variation, controlling for all time-invariant maternal characteristics including genetics, stable socioeconomic conditions, and neighborhood effects. Standard errors are clustered at the mother level. Third, **behavior genetic decomposition** uses the multi-level relatedness structure to partition variance into additive genetic (a^2), shared environmental (c^2), and non-shared environmental (e^2) components.

All IQ outcomes are standardized to z-scores (mean 0, SD 1) within the analysis sample. Coefficients are therefore interpretable in standard deviation units.

2.4 Cognitive Measures and g Factor Scores

The data release includes precomputed latent g factor scores computed via multiple extraction methods (PCA, PAF, CFA) and variable sets of increasing breadth, plus an IRT-derived g score from the age-4 Stanford-Binet with basal/ceiling reconstruction, and PCA/IRT factor scores from six developmental and behavioral item batteries. Full details of these derived variables—including the SB adaptive reconstruction procedure, inter-correlation matrices, and criterion validity statistics—are provided in the companion data paper (Lasker, 2026). For the analyses in this paper, we extract g as the first principal component (PCA PC1) from the full 12-measure age-7 battery—the same battery used in the pathway models: seven WISC scaled subtests (Information, Comprehension, Vocabulary, Digit Span, Picture Arrangement, Block Design, Coding), three WRAT raw scores (Reading, Spelling, Arithmetic), reversed Bender-Gestalt errors, and Goodenough Draw-a-Person raw score. PCA scores are used rather than ML factor scores because they are unit-weighted linear combinations without the shrinkage toward zero inherent in regression-based factor scores, providing a more transparent composite ($n = 31{,}031$ complete cases; PC1 variance explained: 43.7%). To examine sensitivity to battery composition, we also extract PCA PC1 from the seven WISC subtests alone ($n = 31{,}312$; 45.1% variance explained) and from the five non-WISC measures ($n = 31{,}252$; 65.5% variance explained).

3 Behavior Genetics

3.1 IQ Correlations by Relatedness

Table 1 presents pairwise IQ correlations by relatedness level. The pattern is highly consistent across outcomes: MZ correlations are highest ($r = 0.57$–0.86), followed by DZ twins and full siblings ($r = 0.44$–0.68), with half-siblings lowest ($r = 0.26$–0.35). The monotonic increase in IQ similarity with genetic relatedness is consistent with heritable variation, subject to the environmental and selection differences among kinship groups discussed below.

Table 1: Pairwise IQ Correlations by Relatedness Level

Outcome	Group	n	r	SE
WISC FSIQ	MZ ($R = 1.0$)	132	0.810	0.038
	DZ twin ($R = 0.5$)	236	0.611	0.050
	Full sibling ($R = 0.5$)	7,328	0.556	0.013
	Half-sibling ($R = 0.25$)	926	0.349	0.031
WISC VIQ	MZ ($R = 1.0$)	105	0.736	0.044
	DZ twin ($R = 0.5$)	176	0.649	0.048
	Full sibling ($R = 0.5$)	4,637	0.524	0.014
	Half-sibling ($R = 0.25$)	868	0.289	0.036
WISC PIQ	MZ ($R = 1.0$)	105	0.572	0.089
	DZ twin ($R = 0.5$)	176	0.484	0.077
	Full sibling ($R = 0.5$)	4,637	0.438	0.015
	Half-sibling ($R = 0.25$)	868	0.260	0.034
SB IQ (4yr)	MZ ($R = 1.0$)	111	0.771	0.035
	DZ twin ($R = 0.5$)	196	0.615	0.048
	Full sibling ($R = 0.5$)	5,731	0.540	0.011
	Half-sibling ($R = 0.25$)	671	0.348	0.037
PCA g^a	MZ ($R = 1.0$)	102	0.860	0.031
	DZ twin ($R = 0.5$)	175	0.684	0.053
	Full sibling ($R = 0.5$)	4,558	0.528	0.014
	Half-sibling ($R = 0.25$)	846	0.299	0.034

[a]PCA PC1 from the 12-measure age-7 battery described in Section 2.4 (n = 31,031; 43.7% variance explained). All displayed SEs come from 1,000 connected-pedigree bootstrap replications; maternal families joined by any documented extended-kin link are resampled as one block.

Notably, DZ twin correlations are modestly higher than full-sibling correlations for every outcome (e.g., 0.611 vs. 0.556 for FSIQ; 0.649 vs. 0.524 for VIQ; 0.484 vs. 0.438 for PIQ), consistent with twins sharing more prenatal environment and experiencing more similar postnatal environments and timing.

Extended Kinship Correlations

Table 2 extends the relatedness gradient to cousin and more distant relative pairs, using WISC FSIQ residualized on race and institution dummies. Residualization is necessary for cross-family pairs because, unlike within-family siblings who share race and institution by definition (so that race/institution effects are genuine shared environment captured by c^2), extended kin are drawn from different families and often different institutions, making raw correlations inflated by population stratification. Race and institution jointly explain 25.5% of WISC FSIQ variance; raw first-cousin correlations ($r = 0.30$) are 3.6× the residualized values ($r = 0.085$). The sibling/twin correlations in Table 1 are raw values, which is appropriate because race and institution represent genuine shared environment for within-family pairs.

Table 2: Extended Kinship WISC FSIQ Correlations (Residualized on Race + Institution)

Group	R	n	r_{adj}	SE	r_{raw}
First cousin + half-niece/nephew	0.125	7,269	0.087	0.012	0.305
1st cousin 1× rem. + half-1st cousin	0.0625	1,514	0.068	0.026	0.265
Second cousin[a]	0.03125	1,291	0.138	0.027	0.372

r_{adj}: correlation of WISC FSIQ residualized on race and institution dummies. r_{raw}: unadjusted correlation for comparison. Groups at the same R level are pooled.

[a]Second-cousin correlations are anomalously high (exceeding the $R = 0.0625$ group) and likely reflect misclassification or selective ascertainment in the original field records; they should be treated with caution.

The residualized cousin gradient is monotonically declining from $R = 0.125$ ($r = 0.087$) to $R = 0.0625$ ($r = 0.068$), consistent with the declining within-family gradient in Table 1. Second cousins break monotonicity with a residualized $r = 0.138$ that exceeds the $R = 0.0625$ group, suggesting these pairs are unreliable—likely due to misclassification in the original field records or selective identification of second cousins from the same institutional catchment area. The first-cousin correlation ($r = 0.087$) closely matches the genetic expectation: with $h^2 \approx 0.56$ from the IP model (below) and $R = 0.125$, the expected cousin IQ correlation from additive genetics alone is $0.125 \times 0.56 = 0.070$; the modestly higher observed value is consistent with a small contribution of shared extended-family environment and/or assortment.

Within-race correlations provide a cross-check: White first cousins ($r = 0.173$, $n = 3{,}071$) and Black first cousins ($r = 0.112$, $n = 3{,}664$) are higher than the fully residualized estimate because race dummies alone do not remove institution-level confounding. Further residualizing on institution within race gives $r = 0.112$ for Whites and $r = 0.069$ for Blacks, bracketing the pooled estimate. Cross-race cousin pairs are rare ($< 1\%$ of all cousin pairs), confirming that the extended kinship network is racially stratified.

We can test the stratification claim directly. Because the CPP enrolled entire urban hospital populations, two unrelated children from the same race and institution share between-group IQ components (the race mean, the institution mean, and their interaction) without sharing any within-group component—no genes, no family, no household. If population stratification is the dominant source of raw cross-family IQ correlation, these unrelated pairs should be correlated at roughly the R^2 of the residualization model ($= 0.255$). We draw 100,000 random pairs of children from different mothers and compute raw IQ correlations under progressively stricter matching. Fully random pairs yield $r \approx 0$ ($n \approx 100{,}000$), confirming that the population-level IQ correlation without matching is nil. Restricting to same-race pairs raises this to $r = 0.188$ ($n = 46{,}661$); same-institution pairs give $r = 0.249$ ($n = 11{,}930$); and same-race *plus* same-institution pairs give $r = 0.282$ ($n = 9{,}544$). That last figure is 93% of the raw first-cousin $r = 0.302$, meaning the vast majority of the raw cousin correlation can be attributed to stratification rather than kinship. Residualizing the same-race-and-institution pairs on race and institution dummies drops their correlation to $r = 0.018$—near zero, as expected when no familial connection remains. The staircase—random < race < institution < race + institution—tracks the progressive addition of between-group confounders and confirms that residualization on both dimensions removes the non-genetic stratification component from cross-family pairs.

3.2 ACE Decomposition

We estimate ACE components using two methods: (1) multi-level Falconer regression, which regresses observed IQ correlations on genetic relatedness ($r = a^2 R + c^2$) using weighted least squares with all four relatedness levels; and (2) classical twin-only Falconer formulas ($a^2 = 2(r_{MZ} - r_{DZ})$). Table 3 presents the results.

Table 3: ACE Estimates by Method

	Multi-Level Falconer			Twin-Only Falconer		
Outcome	a^2	c^2	e^2	a^2	c^2	e^2
WISC FSIQ	0.71	0.20	0.09	0.40	0.41	0.19
WISC VIQ	0.76	0.14	0.10	0.17	0.56	0.26
WISC PIQ	0.55	0.15	0.29	0.18	0.40	0.43
SB IQ	0.64	0.22	0.14	0.31	0.46	0.23
PCA g	0.83	0.11	0.06	0.35	0.51	0.14

Heritability increases from age 4 to age 7, consistent with the Wilson effect and/or declining measurement error. Twin-only Falconer heritability rises from $a^2 = 0.31$ (SB, age 4) to $a^2 = 0.40$ (WISC, age 7); multi-level Falconer estimates show a parallel increase from $a^2 = 0.64$ to $a^2 = 0.71$, and the FS–HS contrast yields $a^2 = 0.77$ (SB) vs. $a^2 = 0.83$ (WISC).

All kinship correlations are higher at age 7 than age 4 for MZ twins (0.81 vs. 0.77) and full siblings (0.56 vs. 0.54), while half-sibling correlations are essentially unchanged (0.35 vs. 0.35), consistent with increasing genetic influence on IQ as children develop.

Adding the half-sibling anchor at $R = 0.25$ pulls heritability up and shared environment down relative to the twin-only estimates. With a mother-block bootstrap, WISC FSIQ has multi-level $a^2 = 0.71$ [0.54, 0.86]. For the 12-measure PCA g factor, $a^2 = 0.83$ [0.67, 1.00] with $c^2 = 0.11$. These are assumption-dependent Falconer decompositions: DZ twins, full siblings, and half-siblings differ in more than nominal genetic relatedness, and the four-equation model below explicitly allows a twin-specific shared environment.

A methodological caveat applies to the half-sibling anchor. Half-sibling pairs come from disrupted families and are substantially lower in SES than other kinship types (mean SEI = 3.2 vs. 5.0 for full siblings, 4.6 for DZ twins). If the Scarr–Rowe interaction (Scarr-Salapatek, 1971) is real—heritability increases and shared environment decreases with SES—then the HS correlation is inflated relative to what it would be if half-siblings had the same SES distribution as full siblings, because at low SES, c^2 is large and dominates the HS correlation ($r_{\mathrm{HS}} = a^2/4 + c^2$). Using the SES-moderated estimates from Section 8.2, we calculate that the HS correlation at the FS mean SES would be approximately 0.31 rather than the observed 0.35. This inflated r_{HS} flattens the Falconer regression slope, producing a conservative bias: a^2 is underestimated by approximately 0.10 and c^2 overestimated by a similar amount. For WISC FSIQ, removing HS and estimating from MZ-vs-DZ+FS alone gives $a^2 = 0.50$, $c^2 = 0.31$; with HS, $a^2 = 0.71$, $c^2 = 0.20$; and with a counterfactual SES-matched $r_{\mathrm{HS}} = 0.31$, $a^2 = 0.81$, $c^2 = 0.15$. The bias is thus conservative—it makes heritability appear lower than it likely is—and does not affect within-family analyses, which do not rely on relatedness coefficients. This SES confound also applies to the four-equation system (Section 8.2), where an inflated r_{HS} would push the estimated spousal correlation (μ) downward and c^2_{fam} upward, again in the conservative direction. However, the multivariate IP model (Section 3.4) and between-group decomposition (Section 4.1) are essentially unaffected: refitting the 12-measure IP without HS yields A-common and C-common loading vectors with Tucker congruence > 0.999 relative to the full model, and the proportion of common-factor variance that is genetic changes from 85.1% to 84.9%. The post-hoc Shapley decomposition depends on loading *profiles* rather than absolute variance levels, so the HS SES confound does not alter the between-group attribution.

The simple multi-level Falconer regression ($r = a^2R + c^2$) does not model twin-specific shared environment (c^2_{twin}) separately, so the excess DZ correlation above FS (0.611 vs. 0.556 for FSIQ) acts as a leverage point that can inflate a^2 and compress c^2 relative to the exactly identified four-equation solution in Section 8.2 (which estimates $a^2 = 0.61$,

$c^2_{\text{fam}} = 0.14$, $c^2_{\text{twin}} = 0.06$). The multi-level Falconer's $a^2 = 0.71$ exceeds the analytical solution's $a^2 = 0.61$ in part because twin-specific environment is misattributed. The four-equation ACET decomposition (Table 23) provides a more accurate partition when four kinship levels are available.

3.3 Heritability of Birth Weight and Head Circumference

The same ACE framework applied to birth weight and head circumference (at birth and at age 7) gives the results in Table 4.

Table 4: ACE Estimates for Birth Weight and Head Circumference

	Multi-Level Falconer			Twin-Only Falconer		
Outcome	a^2	c^2	e^2	a^2	c^2	e^2
Birth weight	0.43	0.32	0.26	−0.10	0.78	0.32
HC at birth	0.43	0.23	0.34	−0.09	0.71	0.38
HC at age 7	0.90	−0.04	0.14	0.41	0.39	0.20

Multi-level heritability for birth weight is moderate ($a^2 = 0.43$ [0.24, 0.61]), with a large shared-environment component ($c^2 = 0.32$); birth head circumference is similar ($a^2 = 0.43$ [0.24, 0.61], $c^2 = 0.23$). The large c^2 makes sense: the intrauterine environment matters a great deal for size at birth. The twin-only Falconer estimates yield *negative* heritabilities for both birth measures. The reason is well known: monochorionic MZ twins share a placenta and compete for blood supply, so their birth-weight correlation ($r = 0.68$) is actually *lower* than that of dichorionic DZ twins ($r = 0.73$). The resulting MZ $<$ DZ pattern makes the classical Falconer formula give nonsensical answers—and this is exactly the case where non-twin siblings and half-siblings are needed to identify the genetic component.

By age 7, head circumference is highly heritable ($a^2 = 0.90$ [0.69, 1.11]) with negligible c^2, as one would expect for a skeletal trait approaching adult size. Even the twin-only estimate is positive by this age ($a^2 = 0.41$), indicating that the prenatal placental confound has mostly washed out by age 7.

3.4 Common vs. Independent Pathway Models

The univariate ACE analyses establish that WISC IQ is substantially heritable, but they do not address the *structure* of genetic and environmental influences across cognitive subtests. We fit multivariate behavior genetic models to the same twelve cognitive measures used for

g extraction (Section 2.4) to test whether genetic and environmental variance is mediated through a common latent factor (Common Pathway, CP) or operates through structurally distinct pathways (Independent Pathway, IP).

Model Specification

The four-group design (MZ, DZ, FS, HS) decomposes variance into three components: additive genetic (A, scaled by relatedness R), family shared environment (C, shared by all pair types), and unique environment (E). We omit the twin-specific shared environment (T) because the twin sample is small (48 White MZ, 107 White DZ pairs) and T is only weakly identified; all results below use ACE models. The between-pair covariance structure is: MZ $= \Sigma_A + \Sigma_C$; DZ $= 0.5\Sigma_A + \Sigma_C$; FS $= 0.5\Sigma_A + \Sigma_C$; HS $= 0.25\Sigma_A + \Sigma_C$. We fit five models via FIML in OpenMx:

1. **Cholesky ACE**: Lower-triangular 12×12 matrices for each variance component. Fully saturated ACE decomposition.

2. **Independent Pathway (IP)**: One common factor per variance component (A, C, E) with 12 free loadings each, plus subtest-specific diagonal residuals.

3. **Bifactor**: A general g factor loading on all subtests plus an orthogonal Verbal/Crystallized group factor, with its own ACE decomposition. The Verbal factor loads on Information, Comprehension, Vocabulary, Digit Span, WRAT Reading, WRAT Spelling, and WRAT Arithmetic; Performance subtests (Picture Arrangement, Block Design, Coding, Bender-Gestalt, and Goodenough) load on g only. A Performance group factor was initially included but collapsed under strict measurement invariance (negative variance), and was removed from the final specification.

4. **Higher-Order**: Subtests load on Verbal/Crystallized and Performance/Non-verbal group factors, which in turn load on a second-order g factor (Rijsdijk et al., 2002). ACE is estimated at the g level, group-residual level, and specific level.

5. **Common Pathway (CP)**: One latent g factor with 12 loadings and an ACE decomposition of the factor variance (fixed to 1), plus subtest-specific ACE residuals. The CP model constrains genetic and environmental influences to have proportional loading patterns.

These five models form a hierarchy of proportionality constraints on genetic and environmental covariance (Franić et al., 2013; Panizzon et al., 2014). The CP imposes exact proportionality: the genetic and environmental common-factor covariance matrices are both $\boldsymbol{\lambda}\boldsymbol{\lambda}'$ scaled by a_g^2 and c_g^2, respectively—the same loading vector $\boldsymbol{\lambda}$ routes both sources, so the Dolan between-group decomposition is unidentified by construction. The Higher-Order and Bifactor models impose a weaker constraint: genetic and environmental covariance both flow through the same phenotypic factor structure $\boldsymbol{\Lambda}$, differing only in the variance allocated at each factor level. The IP model alone allows structurally different loading vectors for each variance component ($\mathbf{a}_c \neq k \cdot \mathbf{c}_c$), making it the only model that permits the genetic and environmental common factors to load on subtests in entirely different patterns.

Results

All five models converged cleanly (OpenMx status = 0). The design includes 8,651 pairs (MZ: 131, DZ: 238, FS: 7,343, HS: 939). Table 5 presents the model comparison.

Table 5: Pathway Model Fit Comparison (4-Group Design, 12 Measures)

Model	k	-2LL	AIC	BIC
Cholesky ACE	246	842,748	507,394	844,978
Independent Pathway	84	845,039	509,361	845,800
Bifactor	78	846,946	511,256	847,653
Higher-Order	68	857,331	521,621	857,948
Common Pathway	62	859,172	523,450	859,734

With twelve diverse measures, the saturated Cholesky ACE model is preferred by both AIC and BIC, indicating that the single-common-factor-per-component assumption of the structured models is too restrictive for a battery spanning verbal knowledge, reading achievement, and visuomotor processing. Among the structured models, the IP has the lowest model-based AIC and BIC and the CP the highest. The conventional all-pairs likelihood-ratio comparison is large ($\Delta\chi^2$ = 14,133, df = 22). To remove dyad reuse, we join maternal families connected by any documented extended-kin link and retain one randomly selected pair total from each connected component. This leaves 5,143 nonoverlapping documented-pedigree components (MZ = 94, DZ = 166, full siblings = 4,335, half siblings = 548). On this component-level sample the IP remains strongly preferred to the CP ($\Delta\chi^2$ = 8,170.3, df = 22, $p < 10^{-300}$). The fitted IP loading matrix is also nearly unchanged: its rank-1 share is 90.6% and Tucker's congruence between the genetic and shared-environment loading vectors is 0.851. Thus repeated individuals and large sibships inflate the all-pairs statistic but do not explain either the model ordering or the non-proportional loading pattern.

The model ordering—IP > Bifactor > Higher-Order > CP—reflects the advantage of allowing genetic and environmental loading vectors to differ freely. The battery includes both strong g indicators (WRAT Arithmetic: g-loading = 0.80; WRAT Spelling: 0.80; WRAT Reading: 0.79; Vocabulary: 0.71) and weaker ones (Coding: 0.37; Bender-Gestalt: 0.70). The Higher-Order model, which routes all covariance through a second-order g via Verbal and Performance group factors, imposes additional constraints that the Bifactor avoids.

The CP model's factor-level ACE is dominated by genetic variance ($a_g^2 = 0.81$, $c_g^2 = 0.16$, $e_g^2 = 0.03$).

Congruence with Phenotypic g-Loadings

The CP model's standardized factor loadings are congruent with the PCA g-loadings ($r = 0.865$; Tucker's congruence coefficient $\phi = 0.982$, where $\phi > 0.95$ indicates essential equality). The IP model's standardized common-factor loadings—$\sqrt{h^2_{\text{com},i}}$, where $h^2_{\text{com},i}$ is the communality from Table 6—are even more congruent with PCA g-loadings (Tucker $\phi = 0.996$; $r = 0.945$), indicating near-perfect descriptive agreement on which tests are most g-loaded. The IP–CP communality profiles also diverge meaningfully: the Tucker congruence between the two models' communality vectors is $\phi = 0.962$ ($r = 0.901$), and between their standardized loadings $\phi = 0.987$ ($r = 0.890$)—both above the fair-similarity threshold but below essential equality, indicating that the IP allocates common-factor variance across subtests differently than the CP. All resampling-based uncertainty below uses connected-pedigree blocks rather than pair rows.

A confirmatory factor analysis (CFA) single-factor model on the 12 measures provides a phenotypic benchmark that complements PCA. The CP model's standardized loadings are near-identical to CFA g-loadings ($\phi = 0.999$), while the IP's standardized loadings align slightly better with PCA g ($\phi = 0.996$) than CFA g ($\phi = 0.992$). This correspondence is natural: the CP imposes the same rank-1 proportionality constraint as a single-factor CFA, whereas the IP, like PCA, allows departures from exact proportionality. The same pattern holds for communalities: CP communality tracks CFA g^2 almost perfectly ($\phi = 0.997$, $r = 0.993$), while IP communality is slightly closer to PCA g^2 ($\phi = 0.985$) than to CFA g^2 ($\phi = 0.976$). That said, the differences are slight—all pairwise Tucker coefficients between IP, CP, PCA, and CFA loading profiles exceed 0.97—and the models agree far more than they disagree on which tests are most g-loaded. Both twin-based models clearly capture general cognitive ability.

Structure of the IP Loading Matrix

The IP model's departure from the CP raises a natural question: how far does the IP loading structure deviate from the rank-1 constraint imposed by the CP? In a CP model, the common-factor loading matrix $[\mathbf{a}_c,\ \mathbf{c}_c,\ \mathbf{e}_c]$ is rank 1 (all columns proportional to $\boldsymbol{\lambda}$). Singular value decomposition of the fitted IP loading matrix shows that the first singular value accounts for 89.6% of total loading variance: most of the fitted common-factor structure is CP-consistent, with 10.4% non-proportional. Tucker's congruence between the genetic and shared-environmental loading vectors is $\phi(\mathbf{a}_c, \mathbf{c}_c) = 0.821$ (angle = 34.8°; Pearson $r = 0.468$). Tucker congruence between the full vectorized CP and IP loading matrices (36 elements: 12 subtests × 3 components) is $\phi = 0.911$, and the variance-weighted Tucker across A, C, and E is 0.952. In the seven-WISC-subtest benchmark, a 100-replication connected-pedigree bootstrap gives a median rank-1 share of 91.0% (95% percentile interval: 90.0–92.1%) and median genetic–shared-environment congruence of 0.908 (0.862–0.932). Across 100 additional one-pair-per-component selections, every fit converges and the IP–CP statistic has median $\Delta\chi^2 = 478.5$ (central 95% selection range: 447.1–506.1; minimum 436.0; df = 12 throughout). The loading result and model rejection are therefore stable to both pedigree-level resampling and the choice of retained dyad.

Battery heterogeneity drives the CP–IP divergence. Restricting the analysis to only the seven WISC subtests—a more homogeneous battery—yields a smaller CP rejection ($\chi^2/\mathrm{df} = 67.6$ vs. 645 for the full battery), higher vectorized Tucker ($\phi = 0.952$), and an SVD rank-1 share of 91.1%. This mirrors the near-perfect CP fit of examiner ratings ($\chi^2/\mathrm{df} = 5.3$, discussed below), suggesting a gradient of CP-likeness: global rater impressions (near-perfect CP) → homogeneous subtest batteries (moderate departure) → heterogeneous batteries spanning verbal, achievement, and visuomotor domains (substantial departure).

Table 6 reports the subtest-level decomposition of the IP common factor. Under the CP, all rows would be identical; the IP allows each subtest its own ratio. Most tests show moderate-to-high common-factor heritability ($h^2_{\mathrm{com}} = 0.27$–$0.94$). Comprehension ($h^2_{\mathrm{com}} = 0.26$, $c^2_{\mathrm{com}} = 0.49$) and Vocabulary ($h^2_{\mathrm{com}} = 0.27$, $c^2_{\mathrm{com}} = 0.68$) have the lowest common-factor heritabilities and highest shared-environmental loadings, indicating that their covariance with other tests operates largely through the environmental common factor. The verbal mean ($h^2_{\mathrm{com}} = 0.58$, $c^2_{\mathrm{com}} = 0.28$) is lower than the performance mean ($h^2_{\mathrm{com}} = 0.75$, $c^2_{\mathrm{com}} = 0.19$), primarily because Comprehension and Vocabulary drag down the verbal heritability average.

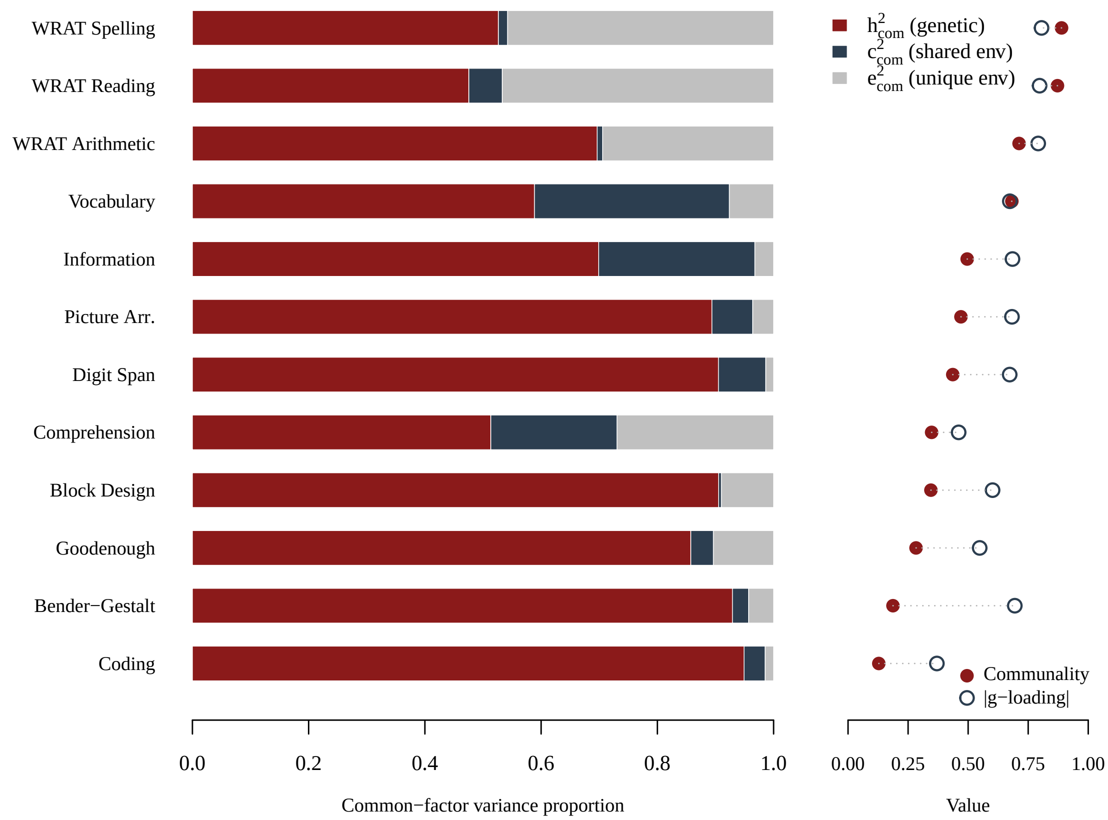
Table 6: Subtest-level ACE decomposition of the IP common factor.

Measure	Comm.	h^2_{com}	c^2_{com}	e^2_{com}	a_s	c_s
Information	.496	0.482	0.468	0.050	1.024	0.001
Comprehension	.348	0.255	0.485	0.260	0.630	0.001
Vocabulary	.682	0.267	0.678	0.055	0.240	0.001
Digit Span	.436	0.801	0.191	0.007	0.942	0.567
Picture Arr.	.470	0.534	0.411	0.055	1.245	0.001
Block Design	.345	0.544	0.346	0.109	0.916	0.694
Coding	.128	0.928	0.060	0.012	1.267	0.766
WRAT Reading	.872	0.657	0.062	0.281	3.215	0.001
Bender-Gestalt	.360	0.944	0.024	0.032	0.616	0.001
WRAT Spelling	.889	0.697	0.032	0.271	0.001	0.001
WRAT Arithmetic	.712	0.881	0.015	0.103	1.454	0.001
Goodenough	.283	0.942	0.008	0.050	2.785	1.732
Verbal mean	.563	0.577	0.276	0.147		
Performance mean	.283	0.747	0.189	0.064		

Note: $h^2_{\text{com},i} = a^2_{c,i}/(a^2_{c,i} + c^2_{c,i} + e^2_{c,i})$ gives the fraction of common-factor variance that is genetic for each subtest. *Comm.* is the communality: the fraction of total variance attributable to the common factor (the remainder is specific + error). a_s and c_s are subtest-specific (diagonal) standard deviations; values at 0.001 are at the lower boundary and effectively zero. Verbal = Info, Comp, Vocab, Digit, WRAT Read, WRAT Spell, WRAT Arith; Performance = PicArr, Block, Coding, Bender, Goodenough.

Figure 1 visualizes the subtest-level ACE decomposition of the common factor alongside communality and PCA g-loadings.

Figure 1: IP Common-Factor ACE Decomposition by Subtest.



Left: stacked proportions of genetic (h^2_{com}), shared-environmental (c^2_{com}), and unique-environmental (e^2_{com}) variance within the common factor, ordered by communality. Right: communality (filled circles) and absolute PCA g-loading (open circles) for each subtest.

These results illuminate the IP's advantage over the CP. Comprehension and Vocabulary show substantially different common-factor ACE profiles ($h^2_{\text{com}} = 0.26$–0.27, $c^2_{\text{com}} = 0.49$–0.68) from the core performance subtests and WRAT scales ($h^2_{\text{com}} = 0.66$–0.94), indicating that verbal crystallized measures' between-test covariance operates more through the environmental common factor. The IP model also reveals substantial c^2 heterogeneity: Vocabulary shows high environmental common-factor variance ($c^2_{\text{com}} = 0.68$), while Goodenough has near-zero c^2 (< 0.01). The CP model, which requires all subtests to share a single ACE decomposition of the common factor, cannot accommodate these measure-specific departures. With $N > 8{,}000$ pairs, even moderate departures from exact proportionality are overwhelmingly significant, confirming that the IP–CP distinction is a pervasive feature of cognitive ability batteries.

The communality column in Table 6 is essential for interpretation. Coding's striking common-factor heritability ($h^2_{\text{com}} = 0.93$) reflects the fact that only 12.8% of Coding's variance flows through the common factor at all—the remaining 87.2% is specific. For that thin slice, the genetic component dominates, but the total Coding heritability (Table 7: IP $h^2 = 0.30$, Cholesky $h^2 = 0.27$) is modest, consistent with Coding's low g-loading (0.37). More generally, performance subtests (mean communality = .28) have much lower communality than verbal subtests (.56), meaning their h^2_{com} values describe a smaller fraction of total variance. The variance-weighted average common-factor heritability across all 12 subtests is $\bar{h}^2_{\text{com}} = 0.62$, compared to the CP model's $a^2_g = 0.81$—both indicate substantial genetic influence on g, though the IP permits more heterogeneity.

CP vs. IP Variance Decomposition

Despite sharing 99.4% of common-factor subspace (RV coefficient; Tucker congruence of implied covariance matrices $\phi = 0.995$), the CP and IP models attribute different shares of total variance to genetic and environmental sources. We compute the total (common + specific) variance decomposition for each model by summing across all 12 subtests. The IP model yields mean per-test $h^2 = 0.42$, $c^2 = 0.14$, $e^2 = 0.45$, close to the saturated Cholesky benchmark ($\bar{h}^2 = 0.39$). The CP model, constrained to a single common-factor ACE ratio ($a^2_g = 0.80$, $c^2_g = 0.16$, $e^2_g = 0.03$), produces higher total heritability because the proportionality constraint forces high-h^2 subtests' ratio onto all subtests. This pattern arises because the CP's proportionality constraint forces all subtests to share the same $a^2_g/c^2_g/e^2_g$ ratio at the common-factor level; when some subtests (e.g., Comprehension, Vocabulary) have genuinely higher shared-environmental loadings, the CP absorbs this into a compromised common-factor decomposition that overstates the genetic share for the majority of subtests. The IP, unconstrained, allows each subtest its own common-factor ACE ratio, producing a total decomposition closer to the saturated Cholesky benchmark ($\bar{h}^2 = 0.40$). Importantly, despite these quantitative differences, the high convergence between the models (implied covariance Tucker $\phi = 0.996$; standardized-loading Tucker $\phi = 0.987$; RV = 0.995) means that the downstream substantive conclusions—including the between-group decomposition, gene–environment interaction tests, and the identification of g as the dominant source of covariance—are robust to the choice between the CP and IP. The primary difference is the magnitude, not the direction, of variance component estimates: the CP gives a somewhat more genetic account, the IP a somewhat more environmental one, but both support the same qualitative story.

Pruned IP

To assess the IP model's effective complexity, we examine which specific (diagonal) residual parameters are at or near their lower boundary. Of the 36 specific residuals—12 each for a_s, c_s, and e_s—9 are at the lower boundary (0.001): one a_s (WRAT Spelling), eight c_s (Info, Comp, Vocab, PicArr, WRAT Read, Bender, WRAT Spell, WRAT Arith), and zero e_s. The effective IP model retains eleven non-boundary a_s loadings, four non-boundary c_s loadings (Digit, Block, Coding, Goodenough), and all twelve e_s parameters.

Cholesky Benchmark

As a benchmark for the IP's implied per-test variance components, we fit the saturated Cholesky ACE model ($k = 246$, $-2\text{LL} = 842{,}748$), which imposes no rank constraint and provides unbiased subtest-level estimates. Table 7 compares the two decompositions. The IP slightly overstates heritability relative to the Cholesky: mean IP $\bar{h}^2 = 0.43$ vs. Cholesky $\bar{h}^2 = 0.40$ (mean difference = +0.03). The distortion is modest and varies by subtest: Goodenough shows the largest overstatement ($\Delta h^2 = +0.10$), while Vocabulary shows understatement ($\Delta h^2 = -0.07$). With twelve measures, the rank-1 genetic common factor has more degrees of freedom to accommodate the heterogeneous battery, producing less distortion than with fewer tests.

Table 7: Per-test ACE estimates from the saturated Cholesky and the IP model.

	Cholesky			IP (total)			
Measure	h^2	c^2	e^2	h^2	c^2	e^2	Δh^2
Information	.39	.20	.41	.36	.23	.41	−.03
Comprehension	.11	.19	.71	.14	.17	.69	+.03
Vocabulary	.26	.43	.31	.19	.46	.35	−.07
Digit Span	.36	.18	.46	.45	.12	.43	+.09
Picture Arr.	.32	.26	.42	.41	.19	.39	+.09
Block Design	.24	.24	.52	.30	.18	.51	+.07
Coding	.27	.09	.64	.30	.07	.62	+.03
WRAT Reading	.71	.08	.21	.65	.05	.29	−.06
Bender-Gestalt	.36	.17	.47	.55	.05	.40	+.19
WRAT Spelling	.63	.06	.31	.62	.03	.35	−.01
WRAT Arithmetic	.64	.05	.31	.72	.01	.27	+.08
Goodenough	.36	.12	.52	.46	.08	.46	+.10
Mean	.40	.16	.43	.43	.14	.43	+.03

Note: $\Delta h^2 = h^2_{\text{IP}} - h^2_{\text{Chol}}$; positive values indicate the IP overstates heritability relative to the unconstrained decomposition.

Implications for g Theory

These results clarify the relationship between the IP structure and the general factor of intelligence. The SVD analysis (90% rank-1 consistency) and the dominance of the common factor support g as the primary source of individual differences: most between-test covariance in this battery flows through a single shared pathway. The IP's rejection of the CP—with 10% of loading structure non-proportional—establishes that genetic and environmental influences on g are structurally similar but not identical.[2] The genetic common factor loads broadly

[2] The statistical rejection of the CP does not undermine the construct validity of g. The CP model's standardized loadings are congruent with phenotypic PCA g-loadings (Tucker $\phi = 0.982$), and the IP genetic common-factor loadings also track the phenotypic g structure closely: the genetic loading vector has uniformly positive entries and its profile correlates with PCA g-loadings. This pattern—where genetic influences on cognition organize around a general factor matching observed g, even when the full variance structure is more complex than the CP allows—provides construct evidence that g indexes a biologically coherent dimension of individual differences, not merely a statistical artifact of test overlap. The CP and IP differ in whether *all* genetic variance flows through a single latent factor (CP) or whether domain-specific genetic variance also exists alongside the general factor (IP); the latter is more consistent with the data, but the genetic common factor in the IP is recognizably g. A more comprehensive exploration of model space—including bifactor, higher-order, and hierarchical CP models with theoretically grounded factor structures—is deferred to future

and positively across all twelve tests (a_c range: 0.33 to 0.72), consistent with a biologically general g. The environmental common factor has a mixed profile (c_c range: -0.11 to 0.37), with the largest positive loadings on Vocabulary and Information and small negative loadings on Bender-Gestalt and Goodenough.

The Cholesky benchmark shows that the IP's rank-1 constraint produces only modest distortion of per-test heritabilities: mean $\Delta h^2 = +0.03$, with twelve tests providing sufficient degrees of freedom to accommodate the heterogeneous battery. For the between-group decomposition, the common-variance Shapley gives 64.8% genetic attribution, above the Cholesky mean within-group $h^2 = 0.40$ and comparable to the IP mean per-test $h^2 = 0.42$. The total decomposition including specific factors (Section 4.1) gives 69.0% genetic.

By contrast, examiner ratings of cognitive functioning (14 PSY25 items aggregated into 3 domain parcels—comprehension, reading, and personality/behavior) show a near-perfect CP structure: $\Delta\chi^2 = 21.2$ (df $= 4$, $p < .001$), an improvement of only 5.3 per degree of freedom vs. 533 for the 12-measure battery. (The personality/behavior parcel is retained because the IP model requires ≥ 3 indicators to identify separate A-common and C-common loading vectors; with only two parcels, the common-factor loadings would be exactly identified and the IP–CP comparison would have zero degrees of freedom.) Tucker's $\phi(\mathbf{a}_c, \mathbf{c}_c) = 0.996$ and the SVD rank-1 share is 89%. The CP common factor is highly heritable ($a_g^2 = 0.83$, $c_g^2 = 0.09$, $e_g^2 = 0.08$). When the same rater evaluates global cognitive ability rather than distinct cognitive skills, genetic and environmental influences operate through a single proportional pathway—precisely the structure the CP model assumes.

Twin-Only Sensitivity

In the standard twin-only design (MZ + DZ only), the IP model is again strongly preferred over the CP ($\Delta\chi^2 = 406$, df $= 22$, $p < .001$). The twin-only CP factor ACE estimates show higher shared environment than the 4-group estimates, consistent with twins sharing more environment than ordinary siblings.

Molenaar et al. (1993) Fourth-Order Moment Test for G×E

The IP model's common-factor structure provides a natural setting for testing gene–environment interaction. Under the standard linear ACE model, the latent A, C, and E factors are independent and normally distributed, implying that their standardized fourth-order moments satisfy $E[X^4] = 3$ (mesokurtic) and $E[X^2Y^2] = 1$ (independent). Under a fully multiplicative interaction, these benchmarks shift to 9 and 3, respectively. Departures from the null can be

work.

expressed as "interactivity percentages": 0% corresponds to a pure linear factor model, and 100% to full multiplicative interaction (Molenaar et al., 1993).

Unlike univariate ACE models where regression factor scores for A, C, and E are all linear functions of a single observed variable, the IP model identifies each common factor through 12 indicators, providing genuine separation of the latent components. We extract regression factor scores from the fitted IP model for all complete pairs and compute the six fourth-order moment statistics.

Because regression factor scores are not exactly orthogonal (even under the null), raw departures from the theoretical benchmarks overstate interactivity. We calibrate against simulation baselines by generating 50,000 pairs per kinship group from the fitted normal IP model and computing the expected moments under the null. Adjusted interactivity is $(M_{\text{obs}} - M_{\text{null}})/(M_{\max} - M_{\text{null}}) \times 100$.

Table 8 reports the simulation-calibrated results for the full-sibling group ($N = 4{,}314$ pairs), which provides the largest and most representative sample.

Table 8: Simulation-calibrated fourth-order moment test for G×E interaction (FS pairs, $N = 4{,}558$).

Moment	Null	Observed	Adj. %	Interpretation
$E[A^4]$	3.004	3.551	9.1%	Genetic factor kurtosis
$E[C^4]$	3.039	3.720	11.4%	Shared-env factor kurtosis
$E[E^4]$	3.007	5.405	40.0%	Unique-env factor kurtosis
$E[A^2C^2]$	1.126	1.286	8.6%	A × C cross-moment
$E[E^2A^2]$	1.084	1.769	35.8%	G × E interaction
$E[E^2C^2]$	1.023	1.248	11.4%	E × C cross-moment

The canonical G×E interaction ($E[E^2A^2]$) shows 36.1% calibrated interactivity in the FS group under the standard model ($\mu = 0$), and 62.6% in the HS group. The unique-environmental factor also shows high kurtosis (40.0%), indicating that E is non-normally distributed, consistent with rare large environmental perturbations. The genetic factor kurtosis is modest (9.1%), suggesting the genetic common factor is approximately normally distributed.

However, assortative mating correction substantially attenuates the G×E signal. A supplementary analysis re-fitting the IP model under four levels of spousal correlation ($\mu = 0, 0.20, 0.35, 0.55$) shows the calibrated $\mathrm{E}^2\mathrm{A}^2$ interactivity declining monotonically: 36.1% ($\mu = 0$) → 20.9% ($\mu = 0.20$) → 11.2% ($\mu = 0.35$) → 3.9% ($\mu = 0.55$) in the FS group, with a similar pattern in HS pairs (62.6% → 13.4%). This attenuation occurs because the standard model ($\mu = 0$) misattributes genetic variance to the shared-environment factor, inflating the C-common factor scores and creating spurious variance in the E factor that correlates with A. AM correction redirects this variance to the genetic channel, cleaning up the E factor and removing the artifact. At the data-estimated spousal correlation ($\mu = 0.55$), the G×E interactivity is near zero (3.9%), consistent with the classical additive model. At moderate AM levels spanning the meta-analytic range ($\mu = 0.20$–0.35), the interactivity is 11–21%—modest but non-negligible.

The conclusion depends on the degree of assortative mating assumed. Under the standard model, there is evidence for gene–environment interaction; under full AM correction, the interaction largely disappears. Pair reuse is material: 8,622 WISC pair rows come from 5,724 mothers, which collapse to 5,125 connected extended-family components; 436 components contain more than one analysis mother. Resampling those entire components gives correlations and bootstrap SEs of 0.810 (0.038) for MZ twins, 0.611 (0.050) for DZ twins, 0.556 (0.013) for full siblings, and 0.349 (0.031) for half siblings. Retaining only one pair per kinship group per connected pedigree yields 0.808, 0.616, 0.561, and 0.364, respectively. The core correlation gradient and IP–CP comparison are therefore robust to documented pedigree dependence; the fourth-order interaction percentages remain model- and assortative-mating-dependent sensitivity statistics rather than identified causal effects.

4 Group Differences

4.1 Between-Group Decomposition of Racial Mean Differences

The IP model's structurally distinct genetic ($\mathbf{a}_c$) and environmental ($\mathbf{c}_c$) common-factor loading profiles enable a decomposition of group mean differences into genetic and environmental components (Dolan et al., 1989, 1992, 1994a; Dolan & Molenaar, 1994b; Rowe et al., 1994; Rowe & Cleveland, 1996; Rowe, 2005; Dolan et al., 2020). The theoretical foundation, developed by Dolan et al. (1994a), rests on the Pearson–Lawley selection rules: if two groups differ because of differences in the means of their latent genetic and environmental factors (i.e., "latent selection"), the loading matrix $\mathbf{A}$ remains invariant while the factor means and covariances change in a predictable way. This yields a testable "common causation" assumption—that the same factors responsible for within-group individual differences also produce between-group mean differences—which can be evaluated by testing whether the structured means model fits alongside the covariance model (Dolan et al., 1989, 1992). Rowe & Cleveland (1996) applied this method to Black–White achievement gaps in the NLSY using full and half-siblings, finding that the structured means model fit well ($\chi^2(37) = 46.6$,

$p = .13$) with no loss of fit relative to the covariance-only model ($\Delta\chi^2(7) = 7.2$, $p = .42$), supporting the common-causation assumption. Under the Dolan decomposition, the White mean vector is expressed as

$$\boldsymbol{\mu}_W = \boldsymbol{\mu}_B + \mathbf{a}_c\kappa_A + \mathbf{c}_c\kappa_C,$$

where κ_A and κ_C are scalar shifts in the genetic and shared-environmental common-factor means, respectively. This identification strategy requires that $\mathbf{a}_c$ and $\mathbf{c}_c$ have sufficiently different profiles across subtests; in the CP model, all loadings are proportional by construction ($\mathbf{a}_c \propto \mathbf{c}_c \propto \boldsymbol{\lambda}$) and the decomposition is impossible. More generally, under any model dominated by a strong general factor, both $\mathbf{a}_c$ and $\mathbf{c}_c$ will point in approximately the same direction (all-positive entries reflecting g-loadings), producing near-collinearity that limits the precision of the decomposition.

We fit an 8-group ACE IP model (White $\times$ Black $\times$ {MZ, DZ, FS, HS}; $N = 8{,}451$ pairs) with covariance parameters constrained equal across races and all subtests standardized to z-scores. The model includes the same twelve cognitive measures used in the pathway analysis. With the 12-measure battery, the genetic and environmental loading vectors achieve an angle of 51° (condition number = 3.5), providing moderate identifiability for the decomposition. However, the within-SEM κ_A and κ_C estimates are poorly identified because the loading vectors, while not exactly proportional, are sufficiently correlated that many linear combinations of $\mathbf{a}_c\kappa_A + \mathbf{c}_c\kappa_C$ produce nearly identical predicted gap profiles. Sensitivity analysis confirms this: removing half-sibling pairs flips the sign of $\hat{\kappa}_A$ while leaving the predicted gaps virtually unchanged. We therefore use a post-hoc projection approach that yields stable estimates and handles the collinearity transparently.

Post-Hoc Decomposition

Let $\boldsymbol{\delta} = \hat{\boldsymbol{\mu}}_W - \hat{\boldsymbol{\mu}}_B$ denote the $p \times 1$ vector of estimated racial mean differences (in z-score units) and let $\hat{\mathbf{a}}_c$ and $\hat{\mathbf{c}}_c$ denote the estimated IP loading vectors from the 8-group model. We project $\boldsymbol{\delta}$ onto the column space of $\mathbf{X} = [\hat{\mathbf{a}}_c \;\; \hat{\mathbf{c}}_c]$ via ordinary least squares:

$$\begin{bmatrix} \hat{\kappa}_A \\ \hat{\kappa}_C \end{bmatrix} = (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\boldsymbol{\delta}.$$

The predicted gap for each subtest is then $\hat{\boldsymbol{\delta}} = \hat{\kappa}_A\hat{\mathbf{a}}_c + \hat{\kappa}_C\hat{\mathbf{c}}_c$, with residual $\boldsymbol{\varepsilon} = \boldsymbol{\delta} - \hat{\boldsymbol{\delta}}$. The coefficient of determination $R^2 = 1 - \|\boldsymbol{\varepsilon}\|^2/\|\boldsymbol{\delta}\|^2$ measures how well the two loading profiles jointly account for the pattern of mean differences.

Because $\hat{\mathbf{a}}_c$ and $\hat{\mathbf{c}}_c$ are correlated, the total R^2 cannot be uniquely attributed to either channel. We use Shapley value decomposition (Lipovetsky & Conklin, 2001), which assigns each predictor its average marginal contribution across all orderings. With two predictors, the Shapley value for the genetic channel is

$$\phi_A = \tfrac{1}{2}\big(R_A^2 + R^2 - R_C^2\big),$$

where R_A^2 is the R^2 from regressing $\boldsymbol{\delta}$ on $\hat{\mathbf{a}}_c$ alone and R_C^2 from $\hat{\mathbf{c}}_c$ alone. The between-group genetic share is ϕ_A/R^2 and the environmental share is $\phi_C/R^2 = 1 - \phi_A/R^2$.

Results

Table 9 reports the decomposition for the 12-measure battery. The overall $R^2 = 0.856$, with the genetic loading profile contributing more than the environmental one. The genetic scaling factor ($\hat{\kappa}_A = 0.710$, SE $= 0.154$) is comparable to the environmental factor ($\hat{\kappa}_C = 0.825$, SE $= 0.437$). Shapley decomposition attributes 64.8% of the explained gap profile to the genetic channel and 35.2% to the environmental channel. Of the total R^2, 30.5% is uniquely genetic (beyond what environment explains alone), 5.1% is uniquely environmental, and the remainder is shared.

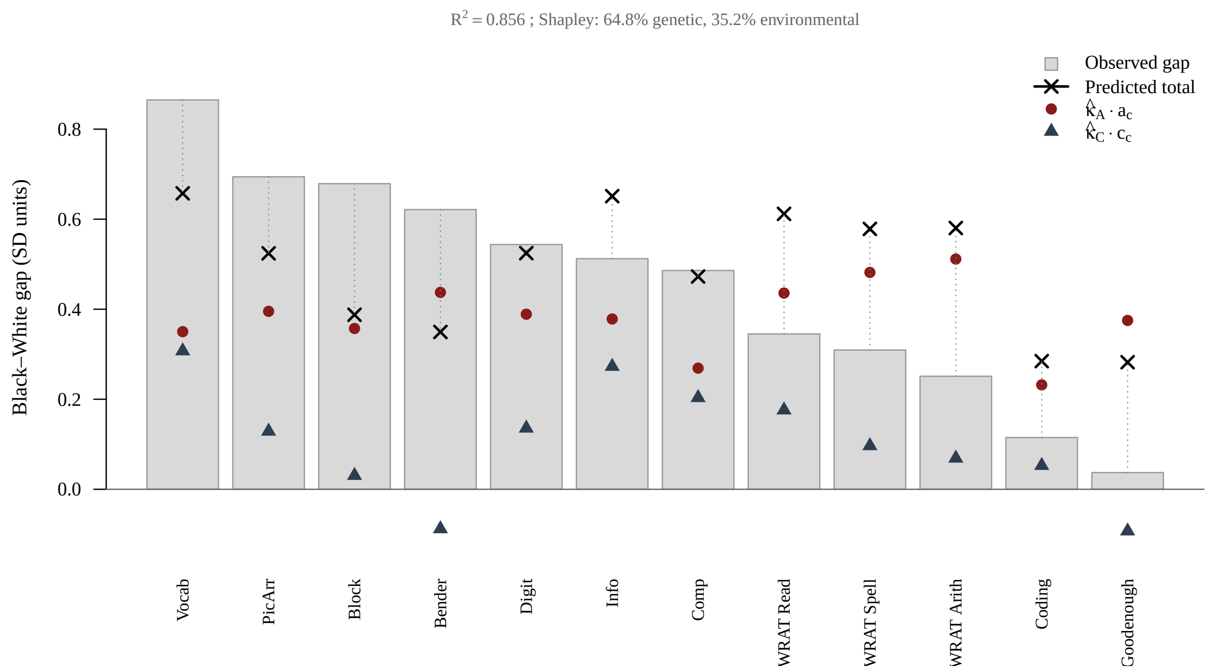
Table 9: Dolan decomposition of Black–White mean differences (12 measures, standardized).

Measure	δ	$\hat{\kappa}_A \hat{a}_{ci}$	$\hat{\kappa}_C \hat{c}_{ci}$	Predicted	Residual
Information	0.513	0.378	0.273	0.651	−0.138
Comprehension	0.486	0.269	0.203	0.473	0.013
Vocabulary	0.865	0.350	0.307	0.657	0.208
Digit Span	0.545	0.389	0.136	0.525	0.021
Picture Arr.	0.793	0.395	0.129	0.524	0.269
Block Design	0.756	0.357	0.031	0.388	0.368
Coding	0.165	0.232	0.053	0.285	−0.119
WRAT Reading	0.435	0.436	0.176	0.612	−0.177
Bender-Gestalt	0.625	0.437	−0.088	0.349	0.275
WRAT Spelling	0.394	0.482	0.097	0.578	−0.184
WRAT Arithmetic	0.342	0.511	0.069	0.580	−0.239
Goodenough	0.084	0.375	−0.093	0.282	−0.199

$\hat{\kappa}_A = 0.710$ (SE $= 0.154$); $\hat{\kappa}_C = 0.825$ (SE $= 0.437$); $R^2 = 0.856$.
Shapley attribution: 64.8% genetic, 35.2% environmental.

Figure 2 visualizes the decomposition.

Figure 2: Dolan Between-Group Decomposition.



Observed Black–White gaps (bars) with predicted genetic ($\hat{\kappa}_A \hat{a}_{c,i}$, filled circles) and environmental ($\hat{\kappa}_C \hat{c}_{c,i}$, triangles) contributions per subtest. Crosses show predicted totals.

Both loading profiles are coherent and mostly positive. The genetic common factor ($\mathbf{a}_c$) has uniformly positive loadings, and the environmental common factor ($\mathbf{c}_c$) also has mostly positive loadings, with small negative loadings on Bender-Gestalt and Goodenough. The genetic channel dominates the Shapley attribution (65%/35%) because the genetic loading vector has both higher magnitude and greater unique explanatory power (unique $R^2_A = 0.31$, unique $R^2_C = 0.05$), meaning the genetic profile captures aspects of the gap pattern that the environmental profile cannot, but not vice versa.

Total Decomposition with Specific Factors

The common-variance decomposition uses only the common-factor loadings ($\mathbf{a}_c$ and $\mathbf{c}_c$), which capture between-test covariance but not test-specific genetic and environmental variance. The IP model also includes subtest-specific residuals ($a_{s,i}$ and $c_{s,i}$) that contribute to total variance but not to inter-test covariance. The *total decomposition* extends the projection to include these specific loadings:

$$\delta_i = a_{c,i}\kappa_A + c_{c,i}\kappa_C + a_{s,i}\kappa_{A_s} + c_{s,i}\kappa_{C_s} + \varepsilon_i,$$

where κ_{A_s} and κ_{C_s} are uniform shifts applied to all specific genetic and environmental factors, respectively. This total decomposition (12 subtests, 4 predictors, 8 residual df) asks whether tests with more specific genetic or environmental variance show systematically larger or smaller gaps beyond what the common factors predict.

The total decomposition explains modestly more variance ($R^2 = 0.865$ vs. 0.856), with the specific channels accounting for the improvement. Shapley decomposition attributes 39.8% of R^2 to the genetic common factor, 24.3% to the environmental common factor, 29.2% to specific genetics, and 6.7% to specific environment. Aggregating by source, the total genetic share is 69.0% and the total environmental share is 31.0%—consistent with the common-variance decomposition. The specific genetic channel (29.2%) reflects tests whose specific genetic residuals covary with the gap profile.

Within-Group Heritability Comparison

The between-group genetic share should be compared with within-group heritability estimates. The saturated Cholesky gives mean per-test $h^2 = 0.39$ and $c^2 = 0.15$. The CP model estimates g-level heritability at $a_g^2 = 0.81$ with modest shared environment ($c_g^2 = 0.16$). The multi-level Falconer yields $a^2 = 0.71$ for FSIQ and $a^2 = 0.83$ for PCA g (Table 3), and the analytical 4-equation solution gives $a^2 = 0.61$ (Table 23). The total between-group genetic share (69.0% uncorrected) falls within the range of within-group estimates: above the Cholesky per-test mean ($h^2 = 0.39$) and comparable to the IP mean per-test $h^2 = 0.42$, but below the multi-level Falconer range ($a^2 = 0.71$–0.83). This is consistent with the general expectation that between-group and within-group heritabilities need not coincide, since they measure different quantities.

The between-group decomposition measures which loading *profile* the gap pattern aligns with, while within-group heritability partitions the variance of individual differences. The gap profile is coherent and g-loaded, and both the genetic and environmental common-factor loading vectors are coherent and mostly positive. The moderate angle between the loading vectors (51°) provides adequate identifiability for the 12-test battery. The genetic channel's Shapley advantage (65%/35%) reflects its greater unique explanatory power: the genetic loading profile captures substantially more of the gap pattern that the environmental profile cannot explain than vice versa.

These results should be interpreted in light of model choice. If the CP held, the Dolan decomposition would be unidentified since $\mathbf{a}_c \propto \mathbf{c}_c$; the strong preference for the IP (Section 3.4) and the moderate proportionality of the loading vectors (Tucker $\phi = 0.63$, angle $= 51^\circ$) mean the decomposition has adequate identifiability. The uncorrected post-hoc Shapley split (65/35 by common variance, 69/31 total) represents the baseline estimate before MI or assortative mating correction. The caveats below show that assortative mating correction shifts the balance further in the genetic direction.

Measurement Invariance

The higher-order MI analysis (Section 4.1) found that full metric invariance holds for all 12 indicators (ΔCFI = .009), meaning the covariance parameters (factor loadings) are equivalent across racial groups. Only the Goodenough intercept required freeing at the scalar level. Because metric invariance holds without any freed loadings, the full-invariance IP model's loading vectors are psychometrically appropriate for the Dolan decomposition—no MI correction is needed. The 65/35 Shapley split reported above is the final estimate before assortative mating correction.

Caveat: Assortative Mating

The standard IP model assumes genetic correlations of $R_{\text{DZ}} = R_{\text{FS}} = 0.5$ and $R_{\text{HS}} = 0.25$. Phenotypic assortative mating inflates these values: with spousal IQ correlation μ and heritability a^2, the first-generation approximation gives $R^* = (1 + \mu a^2)/2$ for DZ/FS pairs and $(1 + \mu a^2)/4$ for HS pairs. The analytical four-equation solution (Section 8.2) estimates $\mu = 0.55$, implying $R^*_{\text{FS}} = 0.67$ and $R^*_{\text{HS}} = 0.33$—substantially above nominal.

When the IP model does not correct for assortative mating, it systematically misattributes genetic variance to the shared-environment factor. AM inflates sibling resemblance on heritable traits; the model, constrained to $R = 0.5$, interprets the excess similarity as shared environment. The effect is largest for subtests with high genetic loadings on crystallized/verbal ability—precisely the traits most subject to spousal assortment—causing the environmental common factor ($\mathbf{c}_c$) to absorb what is partly genetic variance. Under the standard model, $\mathbf{c}_c$ has Tucker congruence $\phi = 0.67$ with the phenotypic g factor. With AM correction at $\mu = 0.55$, $\phi(g, \mathbf{c}_c)$ drops to 0.14, while $\phi(g, \mathbf{a}_c)$ remains near unity (0.997 to 0.993), indicating that the genetic factor becomes the sole carrier of the g-like loading pattern.

To assess sensitivity, we refit the full-invariance IP model at three levels of spousal correlation ($\mu = 0.20, 0.35, 0.55$) with $a^2 = 0.71$ and recompute the post-hoc Shapley decomposition at each level. Because metric invariance holds without any freed loadings, no MI correction is needed; the full-invariance model is the appropriate baseline. The lower bound ($\mu = 0.19$) corresponds to spousal IQ correlation predicted by assortment on education alone ($r_{\text{edu}} = 0.61 \times 0.56^2$); the midpoint ($\mu = 0.33$–$0.44$) spans the meta-analytic range (Horwitz et al., 2023; Bouchard & McGue, 1981); and the upper bound ($\mu = 0.55$) is the data-estimated value from the four-equation solution. All models converge cleanly (OpenMx status GREEN), pass parameter validation (no Heywood cases, positive-definite implied covariance matrices), and remain well-identified (condition number < 4.4, angle between loading vectors $> 58°$). The common-variance Shapley genetic share increases monotonically: 64.8% ($\mu = 0$) $\to$ 80.1% ($\mu = 0.20$) $\to$ 89.6% ($\mu = 0.35$) $\to$ 97.0% ($\mu = 0.55$). The total genetic share (including specific factors) follows a similar trajectory: 69.0% $\to$ 80.2% $\to$ 86.6% $\to$ 91.0%. The quantitative balance shifts markedly: at $\mu = 0.55$, the genetic channel absorbs the vast majority of the explained gap profile.

The true partition likely lies between the baseline (65/35 common, 69/31 total at $\mu = 0$) and the fully AM-corrected estimate (97/3 common, 91/9 total at $\mu = 0.55$), depending on the degree to which the first-generation AM approximation captures equilibrium assortment.

The 4-way Shapley decomposition reveals where the genetic attribution originates. At $\mu = 0$, the common-factor (g) channels account for 64.1% of explained variance and are split 62.1%/37.9% (genetic/environmental), while the specific-factor (non-g) channels account for 35.9% and are split 81.3%/18.7%—overwhelmingly genetic. Both g-level and non-g channels lean genetic at baseline. AM correction selectively reshuffles the g-level split (62.1% $\to$ 91.5% genetic as μ rises from 0 to 0.55), because spousal assortment operates on g rather than test-specific abilities. At $\mu = 0.55$, both g and non-g channels are strongly genetic.

Twin-Only Robustness Check

As a robustness check, we refit the between-group decomposition using only MZ and DZ twin pairs (White: 48 MZ, 107 DZ; Black: 77 MZ, 118 DZ; total = 350 pairs). Both the free-means model and the structured-means model converge (status = 0), but with so few pairs the estimates are sensitive to model specification. An ACTE model (including twin-specific shared environment) gives a common-variance Shapley split of 46%/54% (genetic/environmental), while an ACE model (dropping the poorly identified T) gives 73%/27% and a total 4-way Shapley of 72%/28%. The instability across specifications—a 27-percentage-point swing from adding or dropping a single variance component—reflects the small sample: Tucker congruence between $\mathbf{a}_c$ and $\mathbf{c}_c$ is 0.80–0.91 (vs. 0.63 in the full model), indicating that the loading vectors are comparably or less separable but far noisier. We regard the twin-only estimates as too underpowered to be informative and prefer the multi-level design with 8,000+ sibling pairs as the appropriate benchmark.

Measurement Invariance and Spearman's Hypothesis

We test Spearman's hypothesis—that the Black–White gap is entirely a g-factor gap—by fitting a multi-group higher-order CFA to the 12-test battery ($N = 29{,}551$ complete cases;

White = 10,536, Black = 19,015). The model specifies four first-order group factors—Crystallized (Gc: Information, Comprehension, Vocabulary), Visual-Spatial (Gv: Picture Arrangement, Block Design, Goodenough), Reading/Writing (Grw: WRAT Reading, Spelling, Arithmetic), and Processing Speed (Gs: Digit Span, Bender-Gestalt, Coding)—loading on a second-order g, estimated with MLR treating all indicators as continuous. The Gs factor residual variance is fixed to zero, reflecting the fact that these three indicators share no group-factor variance beyond g (confirmed by exploratory factor analysis showing Coding loads $< .30$ on every factor and by negative Gs variance in an unconstrained model). This four-factor structure fits substantially better than a three-factor V/P/Ach split (configural CFI = .938 vs. .927) or the original two-factor V/P model (.843). The higher-order specification constrains all factor covariance to flow through g, providing a strict test of whether between-group differences are entirely g-mediated. Factor variances are fixed to 1.0 in the White reference group.

We apply hierarchical partial measurement invariance (Meredith, 1993), testing first-order invariance (metric, scalar, strict) sequentially with second-order loadings freed, then testing second-order invariance (equal $g \rightarrow$ group-factor loadings and equal group-factor disturbances) from the first-order strict base (Chen et al., 2005). When full invariance at a level fails (ΔCFI $>$ .010; Cheung & Rensvold, 2002; Chen, 2007), we conduct a forward search, iteratively freeing the most strained parameter until the threshold is met or no further improvement is possible. Freed parameters carry forward to subsequent levels. Table 10 presents the results.

Table 10: Measurement Invariance of the Higher-Order CFA (Gc + Gv + Grw + Gs → g, Gs ~~ 0) Across Race (MLR)

Model	k	χ^2	df	CFI	RMSEA
1. Configural	86	8,394	94	.937	.072
2. Metric	74	9,782	106	.927	.065
3. Partial Scalar[a]	68	10,962	112	.918	.086
4. Strict	56	11,833	124	.910	.086
5. + 2nd-order Metric[b]	53	12,081	127	.909	.076
6. + 2nd-order Strict[c]	50	12,081	130	.907	.076

Note: Gs factor residual variance fixed to zero (Digit Span, Bender-Gestalt, Coding share no group-factor variance beyond g). First-order MI (rows 1–4) is tested with second-order loadings ($g \rightarrow$ Gc, Gv, Grw, Gs) freed across groups; second-order MI (rows 5–6) then constrains these loadings and disturbances equal. Full metric invariance holds (ΔCFI = .009). [a]Goodenough intercept freed; full scalar ΔCFI = .022, partial scalar ΔCFI = .009. Full strict ΔCFI = .008 (holds without freeing). [b]Second-order loadings constrained equal. [c]Group-factor disturbances constrained equal.

Full metric invariance holds (ΔCFI = .009), meaning all indicator loadings are equivalent across racial groups without any freed parameters. At the scalar level, full invariance fails (ΔCFI = .022), but freeing only the Goodenough intercept achieves partial scalar invariance (ΔCFI = .009). Full strict invariance then holds without further freeing (ΔCFI = .008). From this first-order strict base, second-order metric and strict invariance both hold (ΔCFI = .002 each). The model thus achieves full second-order strict invariance with only one freed first-order parameter (the Goodenough intercept)—an exceptionally clean result for a 12-indicator battery spanning heterogeneous content domains. Latent g variance homogeneity also holds, confirming that mean differences are on a common scale.

Under the full second-order strict model, the latent means (Black − White, Hedges' g) are $\delta_g = -0.75$, $\delta_{\mathrm{Gc}} = -0.60$, $\delta_{\mathrm{Gv}} = -0.97$, $\delta_{\mathrm{Grw}} = +0.44$, and δ_{Gs} determined by g (since Gs ~~ 0). All Hedges' g values are computed using the pooled latent SD ($\sigma_{\mathrm{pool}} = 1.14$, reflecting Black latent g variance of 1.47 vs. White = 1.00). Table 11 presents the fit of all constraint configurations, each compared against the all-free baseline. Fixing any single group factor to zero produces negligible loss (ΔCFI = .000). Among two-parameter models, g + Grw fits best (ΔCFI = 0.000), while g + Gc fits worst (ΔCFI = 0.010). The g-only model (ΔCFI = 0.011) marginally exceeds the conventional threshold, indicating that Grw carries a small but non-negligible independent mean difference. Under the most parsimonious acceptable model (g + Grw, ΔCFI = 0.000), the latent gaps are $\delta_g = -1.33$ and $\delta_{\mathrm{Grw}} = +1.51$ (Hedges'

g). The positive Grw residual means that at matched g, Black children score higher on the Reading/Writing group factor—the WRAT gap is smaller than g alone would predict. The g gap under this constraint is larger than the all-free $\delta_g = -0.75$ because g must absorb the Gc and Gv residual differences. Under the g-only constraint (Spearman's hypothesis, ΔCFI = 0.011), $\delta_g = -1.15$. Models without g fit more poorly ($\Delta\text{CFI} \geq .007$), confirming that g is essential, but unlike Spearman's strict hypothesis, Grw contributes independently.

Table 11: Higher-Order Model: Latent Mean Constraint Configurations (Second-Order Strict Base, CFI = .918)

	g	Gc	Gv	Grw	CFI	ΔCFI
All free	free	free	free	free	.918	—
Fix any one factor	(3 free, 1 = 0)				.918	.000
g + Grw	free	0	0	free	.918	.000
g + Gv	free	0	free	0	.909	.009
g + Gc	free	free	0	0	.908	.010
g only (Spearman)	free	0	0	0	.907	.011
Gc + Gv (no g)	0	free	free	0	.911	.007
Gv + Grw (no g)	0	0	free	free	.893	.026
Gv only	0	0	free	0	.893	.026
No differences	0	0	0	0	.877	.041

Note: "Free" means the Black group mean is estimated; "0" means it is constrained to equal the White group (zero). Gs is omitted because its residual variance is zero, so its mean is fully determined by g. All models include full first-order strict and second-order strict invariance (with Goodenough intercept freed at the scalar level). ΔCFI is relative to the all-free baseline.

Under the g + Grw model, the g gap (Hedges' $g = -1.33$) exceeds the observed PCA g factor-score gap ($d = 0.72$) because the latent metric removes attenuation due to imperfect reliability and because g absorbs the Gc and Gv residual differences. The positive Grw residual (Hedges' $g = +1.51$) is interpretable: the WRAT achievement tests show a smaller Black–White gap than would be expected from g alone, consistent with literacy being partially sensitive to instructional exposure that partially compensates for the g deficit. The method of correlated vectors—correlating standardized g-loadings with subtest gap magnitudes—yields $r = 0.30$ for the 12-measure battery. Restricting to the seven WISC subtests alone gives a higher MCV ($r = 0.78$); the 12-measure MCV is lower because the WRAT raw scores and Goodenough have high g-loadings but moderate-to-small observed gaps, breaking the linear relationship between g-loading and gap magnitude that holds within the WISC. The low 12-measure MCV is consistent with the finding that Grw carries an independent mean difference: the gap profile is not perfectly proportional to g-loadings because the Grw-loaded tests (WRAT) deviate systematically.

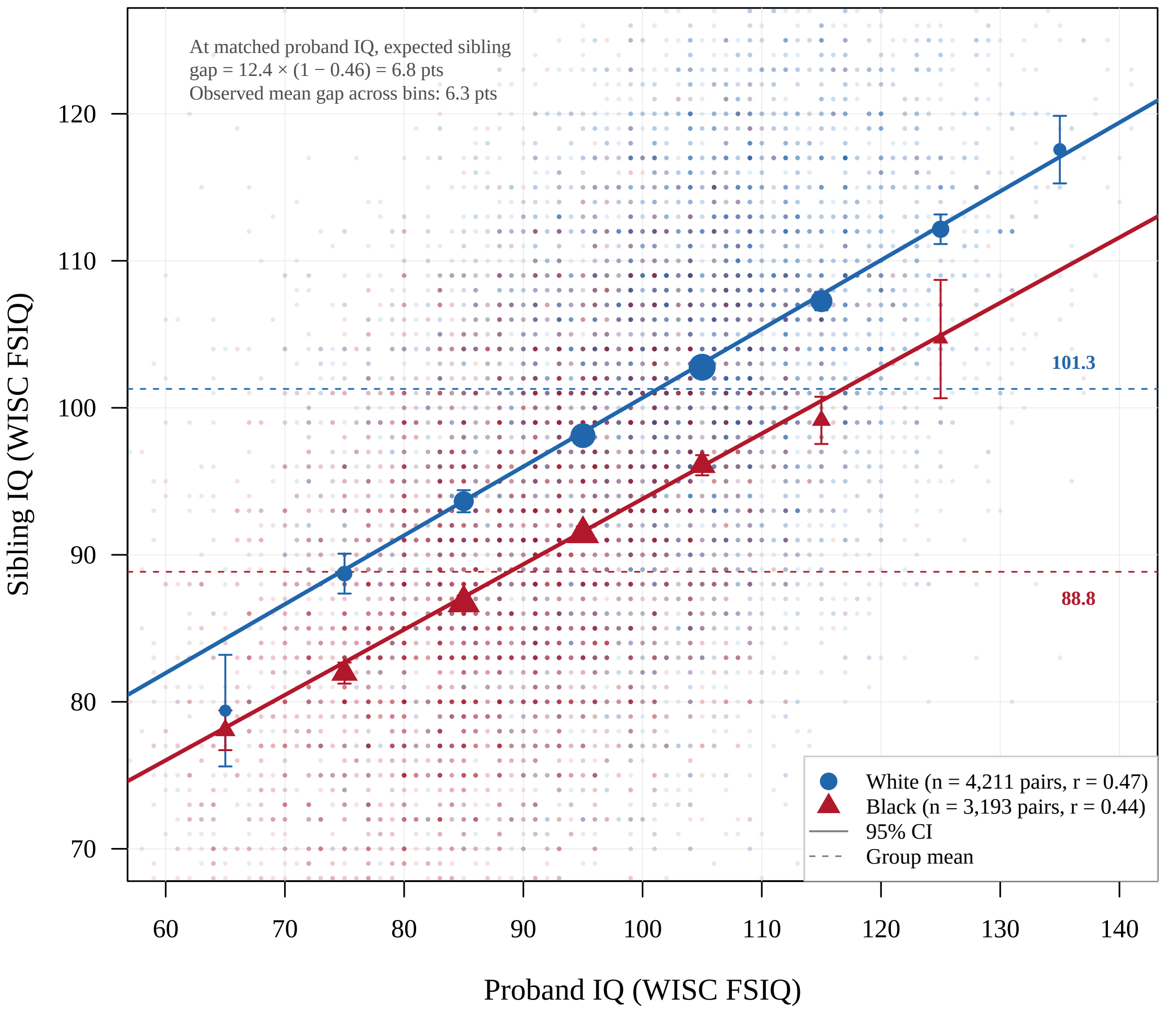
The only freed measurement parameter is the Goodenough Draw-a-Person intercept. Its dMACS effect size (Nye & Drasgow, 2011; Nye et al., 2019) is gap-deflating, meaning the observed Goodenough gap is smaller than what the latent factor structure implies. The latent-mean estimates reported above are derived from the second-order strict model with this intercept freed. Notably, metric invariance holds fully without any freed loadings—all 12 indicators have equivalent factor loadings across racial groups, indicating that the tests measure the same constructs with the same psychometric properties in both groups.

Differential Sibling Regression to the Mean

Jensen's differential regression prediction (Jensen, 1973, 1998) provides a simple test of whether racial group differences behave like individual differences within groups. Under standard regression to the mean, a proband who scores X has an expected sibling IQ of $\mu + r(X - \mu)$, where μ is the group mean and r is the sibling correlation. If group differences are "of the same nature" as individual differences (i.e., have partially genetic origins), siblings should regress toward their *own* group mean, not toward a common population mean. Concretely, at any matched proband IQ, the expected sibling gap equals $(\mu_W - \mu_B)(1 - r)$—the population gap attenuated by regression to the mean.

Figure 3 tests this prediction using 4,211 White and 3,193 Black same-race full-sibling and DZ twin pairs ($R = 0.5$). Each pair contributes two observations (each member serves as proband once), and proband IQ is binned in 10-point intervals. The sibling correlations are similar across races ($r_W = 0.47$, $r_B = 0.44$), yielding a predicted differential gap of $12.4 \times (1 - 0.46) = 6.8$ points. The observed mean gap across bins is 6.3 points, closely matching the prediction. At matched proband IQ = 85, for example, the White sibling averages 93.6 while the Black sibling averages 86.7 (gap = 6.9); at proband IQ = 105, the gap is 102.8 vs. 96.1 (gap = 6.7). The two regression lines are parallel and offset, with each group's siblings regressing toward their own group mean—exactly as predicted by Jensen's model. The consistency across the full proband IQ range (65–125) indicates that the group difference behaves as a constant additive shift operating through the same channels as within-group individual differences. As a robustness check, randomly assigning each sibling pair so that only one member serves as proband (eliminating the symmetric double-entry) yields identical slopes ($r_W = 0.47$, $r_B = 0.44$) and implied population means ($\hat{\mu}_W = 101.4$, $\hat{\mu}_B = 89.2$), confirming that the pattern is not an artifact of the paired design.

Figure 3: Differential sibling regression to the mean.



Binned mean sibling IQ (points, sized by n) as a function of proband IQ for White (●) and Black (▲) full-sibling and DZ twin pairs. Lines show theoretical regression $\mu + r(X - \mu)$. Both groups regress toward their own mean (dashed), with a constant offset of ∼6–7 points across the proband IQ range, matching the predicted $(\mu_W - \mu_B)(1 - r) = 6.8$ points.

We formalize the regression-to-the-mean analysis by predicting one sibling's WISC FSIQ from the other's, separately by race, among concordant-race full sibling pairs. The sibling regression slope estimates the reliability-weighted heritability, and the intercept divided by $(1 - \text{slope})$ gives the implied population mean toward which siblings regress. Among 4,101 White pairs, the slope is $b = 0.445$ (SE $= 0.013$) with an implied mean of 101.9; among 3,072 Black pairs, $b = 0.422$ (SE $= 0.015$) with an implied mean of 90.1. The slopes do not differ significantly ($p_{\text{interaction}} = 0.27$), but the intercepts do: White $= 56.6$, Black $= 52.1$, difference

= 4.5 (p = 0.021). Equal slopes with unequal intercepts is the signature of regression to different group means at a common rate. The implied means differ by 11.8 points, closely tracking the raw group mean difference.

Table 12 shows sibling IQ at matched proband IQ levels (±5 points). When Black and White children are matched at the same IQ, the gap between their siblings' IQs widens from 6.0 points at proband IQ = 80 to 9.1 points at proband IQ = 120. White siblings regress modestly toward their higher group mean; Black siblings regress further toward their lower group mean. The widening gap with proband IQ is the quantitative signature of regression toward different population means.

Table 12: Sibling IQ at Matched Proband IQ Levels (±5 points)

Proband IQ	Sib (White)	Sib (Black)	Gap	n_W	n_B
80	92.1	86.0	6.0	402	866
90	97.0	89.8	7.1	843	1,018
100	100.8	94.4	6.3	1,265	739
110	105.4	96.6	8.7	1,101	242
120	110.0	100.9	9.1	527	47

The equal sibling regression slopes indicate that the within-family process governing IQ transmission operates identically across racial groups. This parallels the IP model's loading structure: the common-factor and specific-factor loadings are fully invariant across groups (no freed loadings required at the metric level), meaning the same latent factors generate observed scores through the same pathways. Equal within-family regression rates combined with structural loading equivalence is the expected pattern if a single developmental architecture produces both groups' IQ distributions, with the groups differing primarily in latent factor means rather than in the processes by which latent ability translates into test performance.

The measurement-invariance and differential-sibling-regression results constrain the measurement description of the group gap, but neither identifies its causes. Approximate invariance says that the fitted items relate to the modeled latent constructs similarly across groups; it makes gross group-specific measurement distortions less plausible. Equal sibling-regression slopes say that the observed familial regression rates are similar. These findings do not exclude an unmeasured exposure that shifts latent means without materially changing loadings, an approximately uniform additive influence, or different causal mixtures that yield the same covariance structure. The results therefore support comparability of the modeled scores and a common descriptive regression rate, while leaving genetic, environmental, historical, and selection explanations for the latent mean difference unresolved.

4.2 SES–IQ by Race

Education–IQ slopes differ substantially by race. Among White children, each additional year of maternal education is associated with 2.06 FSIQ points; among Black children, the slope is 1.27 points ($p_{\text{interaction}} < 0.001$). The same pattern holds for all IQ outcomes. Occupational status (SEI) shows a similar but smaller racial difference.

The Black–White IQ gap *widens* at higher education levels (Table 13). At <9 years of education, the gap is 8.2 points ($d = 0.62$); at 16+ years, it is 14.4 points ($d = 1.13$). The differential sibling regression (Section 4.1) established that children in both groups regress toward their group-specific mean at a common rate ($\rho \approx 0.44$), with the intercept difference ($p = 0.021$) confirming regression toward means separated by $\Delta\mu = 11.8$ points. The SES–IQ widening is linked to this through the regression rate: under the regression-to-the-mean model, a child's expected IQ is $\mu_g + \rho(\text{parent IQ} - \mu_g)$, so the slope of child IQ on maternal education equals ρ times the parental IQ–education slope: $\beta_g = \rho \cdot \gamma_g$. Since ρ is equal across groups, the child-IQ slopes can differ only if the parental IQ–education slopes differ. The observed $\beta_W = 2.38$ and $\beta_B = 1.44$ imply $\gamma_W = 5.4$ and $\gamma_B = 3.3$—each year of education indexes a larger parental IQ difference among White than Black families, consistent with education being a noisier proxy for cognitive ability in the Black population. The gap widens at rate $\beta_W - \beta_B = \rho(\gamma_W - \gamma_B) \approx 0.94$ points per year of education, and the predicted gaps match the observed values across all education levels ($r = 0.97$). Evaluated at the mean education level ($\sim$12 years), the model predicts a gap close to the sibling-regression implied $\Delta\mu = 11.8$, confirming that both analyses recover the same population mean difference. Running the sibling regression within education terciles further confirms the pattern: the implied mean gap is 9.3 points among low-education families, 10.4 among middle, and 16.8 among high, and controlling for education amplifies the race intercept shift from -4.2 to -6.3 because at matched education, Black families are more positively selected relative to their group mean.

Table 13: Mean WISC FSIQ by Education Level and Race

	White		Black		
Education	FSIQ	n	FSIQ	n	Gap (d)
<9 years	93.8	2,308	85.6	3,899	8.2 (0.62)
9–11 years	98.3	5,764	88.8	9,475	9.6 (0.78)
12 years	103.7	6,702	92.9	5,602	10.8 (0.88)
13–15 years	110.8	2,042	97.0	769	13.8 (1.09)
16+ years	115.7	1,454	101.3	130	14.4 (1.13)

4.3 Examiner Ratings and Cognitive Ability

After administering the WISC and other standardized tests at the age-7 examination, examining psychologists completed a separate behavioral observation form (the PSY25 battery, card 41300) recording their clinical impressions of each child. The PSY25 master index labels ("comprehension," "reading," "personality") are misleading generic codes from the computerized indexing system; the actual item content, verified against the original PS-38 Test Summary form, consists of 5 clinical impression ratings (Intelligence, WRAT Spelling, WRAT Arithmetic, Abstract Language Thinking, and Behavioral—each coded Normal/Suspect/Abnormal or Superior/Average/Borderline/Defective), 8 binary physical/behavioral observations (deviant behaviors, physical defects), and 1 administrative item (repeat exam). The factor is dominated by the 5 clinical impression items. Each item is an ordinal rating; these are not test items with correct answers but subjective clinical judgments.

Critically, the WISC is an individually administered test: the psychologist sits across from the child, reads each question aloud, listens to the child's verbal response, scores it in real time, and decides when to stop each subtest based on consecutive failures. The examining psychologist therefore has maximal exposure to the child—having heard every answer, watched the child draw (Bender-Gestalt), observed their attention and frustration tolerance, and interacted with them for an extended session. The PSY25 impressions are not independent of test performance; they are informed by it. This makes the measurement invariance analysis a direct test of whether examiners translate the *same* observed performance into *different* ratings depending on the child's race.

4.3.1 Convergent Validity with Objective IQ

We fit a joint confirmatory factor model with two correlated latent factors: WISC g (indicated by four subtests: Information, Comprehension, Vocabulary, Digit Span) and examiner-rated g (indicated by three domain parcels formed from the original item groupings: clinical impressions, achievement ratings, and behavioral observations). In the pooled sample ($N = 27{,}644$), the latent correlation between the two factors is $r = 0.888$ (SE = 0.004), indicating that examiner ratings share approximately 79% of their variance with the objective g factor. The observed correlation between examiner IRT factor scores and WISC FSIQ is $r = 0.80$; correcting for attenuation using the marginal IRT reliability (0.66) yields a disattenuated correlation of approximately 1.0, confirming that the two measures capture essentially the same construct.

By race, the examiner-IRT correlation with WISC FSIQ is $r = 0.79$ for White children ($N = 10{,}720$) and $r = 0.77$ for Black children ($N = 19{,}307$). The latent correlation between examiner-rated g and WISC g is $r = 0.895$ for White and $r = 0.910$ for Black children; constraining this correlation equal across groups produces negligible deterioration in fit (ΔCFI $= 0.0003$). Both groups' examiner ratings tap the same underlying cognitive construct with the same fidelity.

These correlations are remarkably high by the standards of the broader literature on subjective judgments of cognitive ability. A meta-analysis of teacher judgments of student intelligence found a mean correlation of only $r = .50$ with measured IQ (Machts et al., 2016); teacher judgments of academic achievement correlate somewhat higher at $r = .58$–$.66$ (Hoge & Coladarci, 1989; Südkamp et al., 2012). Parent estimates of their children's cognitive ability correlate $r = .46$ with IQ tests (Mack et al., 2025), and self-estimates of one's own intelligence correlate only $r = .30$ (Freund & Holling, 2024). The CPP examiners' latent correlation of $r = .89$ exceeds all of these substantially, which is expected given that these are trained psychologists who had just spent an extended session individually administering cognitive tests to the child—providing far richer behavioral evidence than classroom teachers or parents typically observe. The finding that even under these conditions of maximal exposure, the examiner ratings exhibit full measurement invariance across race is particularly informative.

4.3.2 Cross-Validation Against Non-Observed Tests

The high convergent validity with WISC g is potentially tautological: the WISC was individually administered by the same psychologist who then filled out the PSY25 form, and the "Intelligence" clinical impression item is explicitly defined relative to WISC IQ. To disentangle examiner competence from same-method contamination, we correlate examiner ratings with three tiers of tests differentiated by the degree of examiner observation:

1. **Directly observed** (WISC subtests): The examining psychologist read each question aloud, heard the child's verbal response, scored it in real time, and decided when to stop each subtest.

2. **Scored from written/drawn work** (WRAT Reading, Spelling, Arithmetic; Bender Gestalt; Goodenough Draw-a-Person): The examiner was present during administration but the scores derive from the child's written or drawn work products—the objective scoring is independent of the examiner's impression.

3. **Completely independent** (Stanford-Binet IQ, age 4): Administered three years earlier, in a different session, by a different examining psychologist. The age-7 examiner had no exposure whatsoever to the child's age-4 test performance.

At the observed level (restricting to live births with known sex, as throughout this paper), the examiner IRT score correlates $|r| = 0.80$ with WISC FSIQ ($N = 31{,}569$), $|r| = 0.73$ with written-output g ($N = 30{,}523$), and $|r| = 0.61$ with Stanford-Binet IQ ($N = 27{,}156$). The gradient—highest with directly observed tests, lower with objectively scored written work, lowest with a completely independent examiner—follows the expected pattern of decreasing method overlap, but even the most conservative comparison (age-4 SB, $r = 0.61$) exceeds the literature's best estimates for teacher judgments.

To estimate latent correlations free of measurement error, we fit a four-factor CFA model with latent WISC g (7 subtests: Information, Comprehension, Vocabulary, Digit Span, Picture Arrangement, Block Design, Coding), latent written-output g (5 tests), latent examiner g (3 domain parcels), and latent SB g (single-indicator, reliability fixed at 0.90). Absolute fit indices are moderate (CFI = 0.904, RMSEA = 0.084, $N = 23{,}766$); this is expected because the model is intentionally misspecified—g is broader than any single battery, and the four-factor structure is designed to estimate between-battery correlations rather than to fit perfectly within each battery. In the pooled sample, the latent correlations are:

Table 14: Latent Correlations Among Four Cognitive Factors

	WISC g	Written g	Examiner g	SB g
WISC g	1.000			
Written g	.755	1.000		
Examiner g	.897	.756	1.000	
SB g (age 4)	.803	.571	.686	1.000

The examiner g–SB g correlation of $r = 0.69$ is particularly informative: the age-7 examiner's global impression shares 47% of its latent variance with a completely independent cognitive assessment administered three years earlier by a different examiner. This cannot be attributed to shared method variance, halo effects, or same-session contamination—it reflects genuine cognitive ability assessment.

A higher-order model in which a single general factor G loads on all four first-order factors produces equivalent fit (CFI = 0.859, RMSEA = 0.101), as expected when the correlated-factors model is just-identified at the second-order level. The higher-order loadings are: WISC g (0.996), examiner g (0.917), written g (0.774), and SB g (0.785). The examiner factor's loading of 0.92 on G is only modestly below the WISC factor's loading (1.00) and substantially exceeds the written-output factor (0.77) and age-4 SB factor (0.79). In other words, the examiner's clinical impression captures higher-order g almost as well as the individually administered WISC, and better than the battery of paper-and-pencil tests scored from written work.

4.3.3 Measurement Invariance Across Race

Multi-group CFA with WLSMV estimation and theta parameterization on the 5 clinical impression items (comprehension ratings, each scored on 3–4 ordinal categories spanning broad IQ ranges) tests whether the examiner rating factor functions identically across Black (n = 19,360) and White (n = 10,701) children. The theta parameterization fixes residual variances to 1 for identification, so equal residual variances are a parameterization constraint rather than an empirical test; the MI hierarchy below tests configural, metric, and scalar invariance, with strict invariance holding by construction. Table 15 presents the results.

Table 15: Measurement Invariance of Examiner Clinical Impression Ratings Across Race (WLSMV, Theta)

Model	χ^2	df	CFI	RMSEA	ΔCFI
1. Configural	27.9	10	0.979	0.011	—
2. Metric	29.7	14	0.982	0.009	−0.003
3. Scalar	28.1	13	0.983	0.009	−0.001
4. Latent var. homogeneity	27.8	14	0.984	0.008	−0.002

Full measurement invariance—including scalar invariance (equal thresholds) and latent variance homogeneity—meets the approximate $|\Delta\text{CFI}| \leq 0.003$ criterion. This supports a similar fitted factor structure, loadings, thresholds, and latent variance for Black and White children; it is evidence against large rating-standard differences that would alter those parameters, but it cannot exclude every systematic additive or latent influence. We also test MI across sexes: configural CFI = 0.959, with metric ΔCFI = 0.001, scalar ΔCFI = 0.000, and strict ΔCFI = 0.000. The latent mean difference between boys and girls is effectively zero ($\Delta\mu < 0.01$ SD, favoring neither sex), and constraining the latent means equal produces no deterioration in fit (ΔCFI = 0.000). Boys show slightly greater latent variance (ratio = 1.07; LRT $p = 0.005$, ΔCFI = 0.000), a difference that is statistically detectable but practically negligible. Note that the 5 clinical impression items use coarse ordinal categories (e.g., "Average" spans IQ 80–119), which limits the precision with which latent mean differences can be estimated from these items alone; the gap estimates in Section 4.3.4 from the joint WISC–examiner model provide a more precise estimate of the cognitive component.

4.3.4 Race Gaps: Examiner Ratings vs. Standardized Tests

Under scalar invariance, latent mean differences can be estimated directly. A 7-subtest WISC-only strict MI model (all seven WISC scaled subtests loading on a single g factor, with loadings, intercepts, and residual variances constrained equal across race) achieves CFI = 0.860 and RMSEA = 0.091, and yields a latent mean gap of $\delta_g = -0.94$ SD. Adding latent variance homogeneity produces ΔCFI = 0.015. In the joint WISC–examiner model, the Black–White gap on latent WISC g is -0.862 SD; the gap on latent examiner-rated g is -0.778 SD, or 90% of the WISC gap. At the composite score level, the examiner IRT score gap ($d = 0.65$) is 75% of the WISC FSIQ gap ($d = 0.87$) and 86% of the Stanford-Binet gap ($d = 0.76$). The examiner gap's attenuation relative to the test gap is consistent with what lower measurement reliability predicts: the examiner IRT score (marginal reliability 0.66) is less reliable than WISC FSIQ, so observed-score gaps are attenuated toward zero. At the latent level—where reliability differences are removed—the examiner gap closes to 90% of the WISC gap.

5 Within-Family Analyses

5.1 Birth Order

5.1.1 Cross-Sectional vs. Within-Family Effects

A well-established finding in IQ research is the negative cross-sectional correlation between birth order and IQ: later-born children score lower. In the CPP extract, the OLS gradient is -0.075 SD per prior live birth for WISC FSIQ ($p < 0.001$). However, this cross-sectional gradient confounds birth order with family characteristics, since larger families tend to have lower SES. Under mother fixed effects, which compare siblings within the same family and cluster uncertainty by mother, the sign reverses for WISC FSIQ: $b = +0.032$ SD per rank (SE = 0.008, $p < 0.001$; N = 13,111), or approximately +0.48 IQ points. Four observed-score outcomes are positive under mother FE, whereas the PCA g score is slightly negative ($b = -0.023$, SE = 0.011); the outcomes therefore do not support a universal positive reversal (Table 16).

Table 16: Birth Order Effects on IQ: OLS vs. Mother Fixed Effects

	OLS		Mother FE		
Outcome	b	SE	b	SE	N (FE)
WISC FSIQ	−0.075	0.002	+0.032	0.008	13,111
WISC VIQ	−0.052	0.003	+0.041	0.010	9,179
WISC PIQ	−0.065	0.003	+0.045	0.011	9,179
SB IQ	−0.051	0.003	+0.103	0.010	11,289
PCA g	−0.072	0.003	−0.023	0.011	9,091

The categorical specification reveals that the within-family SB IQ effect is approximately linear and cumulative: relative to firstborns, secondborns score +0.17 SD, thirdborns +0.25 SD, and children with eight or more prior live births score +0.63 SD. Adding maternal age leaves the SB slope positive ($b = +0.089$ SD/rank), but the WISC estimate becomes less precise ($b = +0.031$, SE $= 0.017$, $p = 0.057$; approximate 95% CI $[-0.001, 0.064]$) because maternal age and birth order are strongly collinear within families. This interval describes the sibling-rank coefficient conditional on maternal age; it is neither an estimate nor a bound for secular cohort change.

5.1.2 Measurement Invariance

To test whether the WISC measures the same construct across birth order groups, we fit a single-factor g model with all seven WISC subtests (Information, Comprehension, Vocabulary, Digit Span, Picture Arrangement, Block Design, Coding) across four groups: only children (zero prior live births and one child registered in the CPP; n = 7,015), firstborns with an observed sibling (n = 1,804), secondborns (n = 6,592), and laterborns (n = 15,504). Table 17 presents the results.

Table 17: Measurement Invariance of WISC Across Birth Order Groups (4 Groups)

Model	χ^2	df	CFI	RMSEA	ΔCFI
1. Configural	2,230	56	0.960	0.071	—
2. Metric	2,431	74	0.956	0.064	0.003
3. Scalar	2,595	92	0.954	0.059	0.003
4. Strict	2,980	113	0.947	0.057	0.007
5. Equal variances	3,143	116	0.944	0.058	0.003
6. Equal means	3,565	119	0.936	0.061	0.008

Strict factorial invariance holds at every step by the ΔCFI < 0.01 criterion (Cheung & Rensvold, 2002; Chen, 2007), confirming that the WISC measures the same latent construct with the same loadings, intercepts, and residual variances across all four birth order groups. Latent variance homogeneity also meets this criterion (ΔCFI $= 0.003$), though laterborns show lower variance (0.77 vs. 1.00 for only children and 1.04 for firstborns). Equal latent means meet the approximate ΔCFI criterion (0.008) but are rejected by the LRT ($p < 0.001$): relative to only children, firstborns score -0.19 SD, secondborns -0.04 SD, and laterborns -0.27 SD on latent g. The "only child" classification remains imperfect because mothers could have had children after CPP follow-up. The WISC is therefore approximately measurement invariant across the observed birth-order groups, but this result does not identify the causes of their latent mean differences.

5.1.3 Secular Cohort Trend (Flynn Effect)

The Flynn-effect estimand is change in test performance across birth cohorts; it is distinct from the effect of occupying a later sibling rank. Because later-born siblings necessarily belong to later cohorts, an unadjusted mother-FE cohort coefficient is a reduced-form mixture of cohort, rank, maternal age, spacing, and other time-varying processes. Using birth cohort directly, the unadjusted WISC trend is +0.247 SD/decade. It shrinks to +0.015 (SE $= 0.024$) with institution fixed effects and becomes -0.074 (SE $= 0.024$) after additionally adjusting for race and SEI, showing that the aggregate trend is strongly compositional. Within mothers, the unadjusted cohort coefficient is +0.192 SD/decade (SE $= 0.047$); conditioning on sex and live-birth rank yields +0.074 (SE $= 0.107$; 95% CI $[-0.136, 0.283]$), while using study pregnancy order instead gives -0.057 (SE $= 0.112$). These sibling models provide no clear evidence of a WISC Flynn effect after accounting for rank, but they do not estimate a causal birth-order effect.

The first-cousin design compares children of different mothers within connected extended families, which supplies variation in child cohort that is not mechanically identical to maternal aging within one mother. Pair differences adjust for sex and live-birth rank, and uncertainty is clustered by extended family. For WISC, the estimate is +0.125 SD/decade (SE $= 0.086$) when every cousin pair receives equal weight and -0.072 (SE $= 0.091$) when each extended family receives equal total weight. In the preferred equal-family specification, adding maternal age gives -0.092 (SE $= 0.091$; 95% CI $[-0.271, 0.086]$); adding maternal education and SEI gives -0.146 (SE $= 0.094$; 95% CI $[-0.329, 0.038]$). Conditional on equal-family weighting, a linear maternal-age specification, and no residual mother-specific selection after the stated controls, +0.086 SD/decade is the 95% upper limit from the age-adjusted design. Older maternal age is positively selected on observed SEI and income within these genealogies; if residual age-related selection also weakly raises child IQ, the upper-limit interpretation is conservative. Without that directional assumption, it remains a model-based confidence limit rather than an assumption-free causal bound.

For Stanford–Binet IQ at age 4, the simpler cousin estimates are positive (+0.330, SE = 0.094 pair-weighted; +0.203, SE = 0.095 family-weighted), but they attenuate after maternal-age adjustment (+0.177, SE = 0.096) and after education and SEI adjustment (+0.102, SE = 0.099). Thus the NCPP provides no robust evidence of a WISC Flynn effect; the age-4 result is positive in simpler specifications but not in the fully adjusted equal-family model. The short 1959–1966 cohort span and residual mother-specific selection remain important limitations of both the sibling and cousin designs.

5.2 Resource Dilution

The resource dilution hypothesis (Blake, 1981; Downey, 1995) posits that each additional child reduces per-capita parental investment, lowering cognitive outcomes—particularly in resource-constrained families. Countervailing forces include parenting experience (later-born children benefit from more practiced parents) and returns to scale (marginal costs per child decline with family size). The CPP's sibling structure permits a comprehensive set of tests.

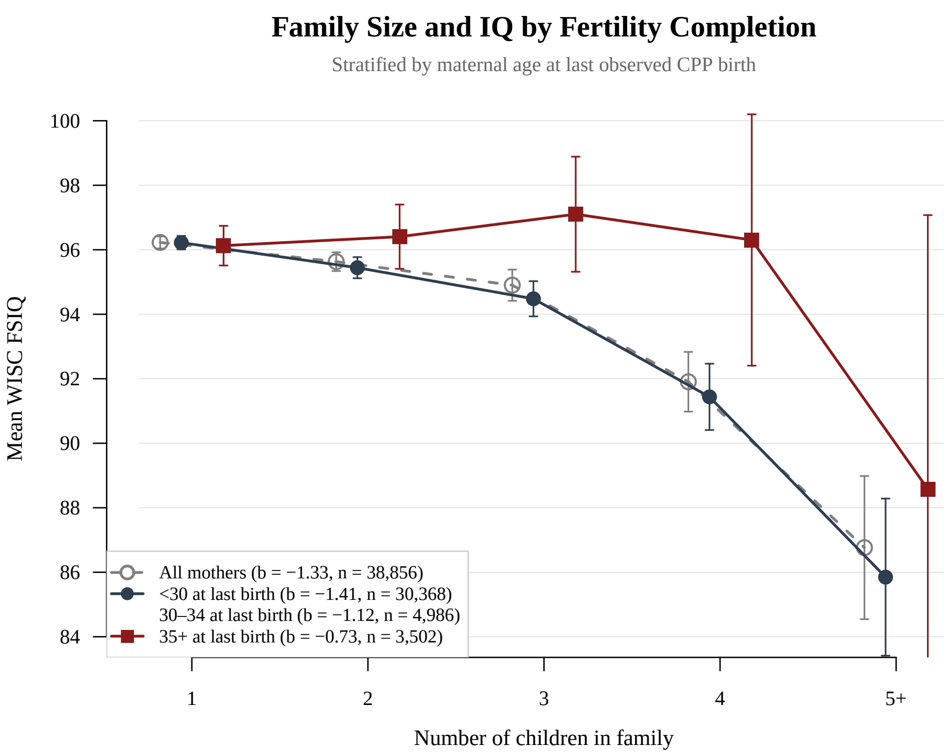
Between-family family size gradient. We define observed family size from all live-birth records linked to each mother, before restricting to children with IQ outcomes. Cross-sectionally, each additional observed child is associated with −1.34 WISC IQ points (SE = 0.09, $p < 0.001$); controlling for income and SEI attenuates the coefficient to −1.16. The gradients are nearly identical among White (−1.33) and Black (−1.34) children. These are between-family associations and remain vulnerable to socioeconomic, parental-ability, fertility-timing, and right-censoring confounding. Among firstborns, mean IQ is 97.9, 97.4, and 96.8 in observed families of one, two, and three children, respectively. The lower means in families of four (90.8) and five (86.6) are based on only 97 and 14 firstborns, so the tail does not provide a stable bound on dilution.

We use maternal age at the last observed CPP birth as an imperfect fertility-completion proxy, but not as a bound: older mothers are selected, observed fertility remains incomplete for some women, and the oldest/high-parity cells are sparse. The WISC family-size slope is −1.41 (SE = 0.10; n = 30,368) among mothers under 30 and −0.73 (SE = 0.31; n = 3,502) among mothers aged 35+, with an interaction of +0.53 ($p = 0.096$). Within the 35+ group, means are nearly flat for one through four observed children (96.1, 96.4, 97.1, 96.3), but the five-child cell averages 88.6 and contains only 19 observations; this sparse tail drives much of the linear slope. Adding SES controls eliminates the pooled interaction ($b = -0.17$,

$p = 0.57$). Race-specific interactions point in opposite directions (White +1.80, Black −1.03), and the Stanford–Binet interaction is null ($b = -0.06$, $p = 0.88$). Thus fertility timing and composition are plausible contributors, but this stratification neither identifies nor tightly bounds a causal family-size effect.

The implication for fertility research is correspondingly modest. Cross-sectional gradients based on children born so far mix fertility tempo, incomplete observation, and completed family size. The CPP can diagnose this mixture but cannot recover completed fertility for every mother, so the age-stratified estimates should be treated as sensitivity analyses rather than estimates of dysgenic selection or causal dilution.

Figure 4: Family size–IQ gradient by fertility completion proxy.



Mothers are stratified by age at their last observed CPP birth: <30, 30–34, and 35+. The WISC slope is smaller among mothers aged 35+ (−0.73 vs. −1.41), but the interaction is imprecise ($p = 0.096$), disappears with SES adjustment, and is sensitive to a 19-child cell at five observed births. Maternal age at last birth is a selected fertility-completion proxy, not a causal design or bound.

Within-family birth order. Under mother fixed effects with mother-clustered standard errors, later-born children score +0.48 WISC IQ points per rank (SE = 0.12) and +1.71 Stanford–Binet points (SE = 0.16; Table 16). The observed-score coefficients are positive, but birth order, calendar time, and maternal age are nearly collinear within families. They therefore reject a large negative rank association in this sample, not identify a beneficial causal effect of being later born.

Birth order $\times$ family size. If dilution accumulates, the birth-order gradient should become more negative in larger families. The mother-FE interaction is +0.13 (SE = 0.13, $p = 0.31$), providing no evidence of the predicted moderation.

Birth order $\times$ income. The mother-FE birth-order $\times$ income interaction is also null ($b = 0.0045$, SE = 0.0041, $p = 0.26$). Rank coefficients by income quartile are +0.47, +0.55, +0.56, and +0.32 IQ points; their uncertainty is substantial and they do not show the stronger negative gradient predicted for low-income families.

Income per current child. Dividing income by birth order+1 is only a proxy for resources per child at the focal birth, not completed family resources. It predicts WISC IQ no better than raw income in OLS ($b = 0.129$ vs. 0.139; $R^2 = 0.219$ vs. 0.231). Under mother FE, raw income is null ($b = -0.0002$, SE = 0.0091), and the proxy is small and negative ($b = -0.0168$, SE = 0.0090, $p = 0.062$).

Birth spacing. Longer spacing predicts higher IQ between families ($b = +0.77$ per year, SE = 0.15), but the mother-FE estimate is +0.19 (SE = 0.35, $p = 0.59$). It is likewise imprecise among White (+0.23, SE = 0.55) and Black (+0.25, SE = 0.44) families.

Gained siblings. In the reproducible birth-to-age-7 specification, change from Stanford–Binet to WISC IQ is unrelated to the number of younger siblings gained after the focal birth once baseline IQ and mother FE are included ($b = +0.17$, SE = 0.14, $p = 0.21$). This estimate rejects only effects more negative than approximately −0.10 points per gained sibling at the 95% level; it is not evidence that the effect is exactly zero.

We do not estimate family size at age 4 because doing so requires reconstruction from the full pregnancy history with explicit censoring rules. That age-specific exposure is a useful extension but is distinct from the observed-family-size analysis reported here.

Family size $\times$ SES. Between families, neither the family-size $\times$ low-SES interaction ($b = 0.07$, SE = 0.18, $p = 0.70$) nor the family-size $\times$ income interaction ($b = 0.0020$, SE = 0.0043, $p = 0.64$) supports stronger penalties in resource-poor families.

Measurement invariance across family size. A seven-subtest WISC model across observed family sizes 1, 2, 3, and 4+ (N = 31,297) meets conventional approximate-invariance thresholds: configural CFI = 0.960, with ΔCFI = 0.001, 0.001, and 0.004 for metric, scalar, and strict constraints. Under the strict model, latent means are −0.15, −0.28, and −0.62 SD relative to singletons. This weighs against a simple measurement-artifact account of the between-family score gradient; it does not make that gradient causal.

Sibling death as a natural experiment. The most direct test of resource dilution separates the birth event (pregnancy, delivery, maternal recovery) from ongoing resource sharing with a living child. The CPP obstetric history records the number and type of prior perinatal deaths for each pregnancy, creating within-family variation in whether prior siblings survived or died. If dilution operates, a prior living sibling—who competes for parental time, money, and attention for years—should be more harmful to the focal child's IQ than a prior dead sibling, who imposed only the cost of pregnancy and birth.

The obstetric history distinguishes code 88 (no prior pregnancy, a structural zero) from code 99 (unknown). We decompose biological rank into prior surviving and prior dead siblings. Cross-sectionally, both coefficients are negative (surviving −1.46, SE = 0.04; dead −0.51, SE = 0.09), consistent with substantial between-family selection. Under mother fixed effects with mother-clustered standard errors, the surviving coefficient is +0.45 (SE = 0.12, $p < 0.001$), whereas the dead-sibling coefficient is +0.24 (SE = 0.26, $p = 0.35$). The dead-sibling estimate is too imprecise to establish equivalence between the channels; it shows no evidence of a negative effect but does not tightly rule one out. For Stanford–Binet IQ, the corresponding estimates are +1.58 (SE = 0.15) and +0.51 (SE = 0.34), with an imprecise channel contrast ($p = 0.14$).

Separating deaths by timing likewise produces no ordered pattern: surviving +0.45 (SE = 0.12), fetal −0.13 (SE = 0.39), stillbirth −0.20 (SE = 0.69), and neonatal +0.71 (SE = 0.75). The death-type estimates are individually imprecise. Race-stratified estimates also remain uncertain (White: surviving +0.27, dead +0.31; Black: surviving +0.61, dead +0.21), and SES interactions are null. These checks do not reveal the negative living-sibling effect predicted by simple dilution, but power is limited for uncommon death categories.

Table 18 gives the descriptive cross-tabulation. At a fixed biological rank, moving right replaces a surviving prior sibling with a prior death. The sparse and non-monotone cells caution against treating this table alone as causal; the mother-FE decomposition is the identifying comparison, subject to the assumption that changes in perinatal-death risk are not themselves related to unmeasured time-varying determinants of later-child IQ.

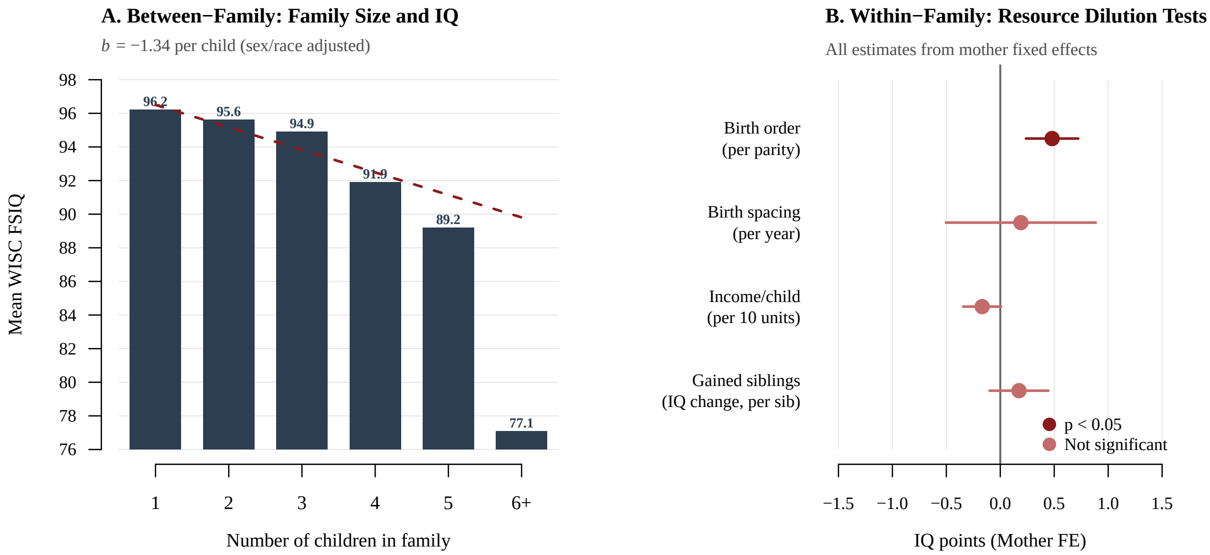
Table 18: Mean WISC FSIQ by Biological Birth Order and Number of Prior Sibling Deaths

	Prior siblings who died			
Bio BO	0	1	2	3+
1	97.7 (10,441)			
2	97.5 (7,101)	97.1 (746)		
3	96.1 (5,565)	96.8 (234)	97.7 (163)	
4	94.4 (2,896)	96.0 (1,259)	96.5 (252)	97.1 (46)
5	94.0 (1,986)	95.4 (1,022)	94.6 (101)	96.8 (91)

Note: Cell sizes in parentheses. At fixed biological rank, more prior deaths imply fewer surviving prior siblings. Mother-FE estimates are surviving +0.45 (SE = 0.12) and dead +0.24 (SE = 0.26). The latter is too imprecise to claim equivalence or tightly bound the contrast.

Summary. The within-family tests do not support the negative associations predicted by a simple resource-dilution model. Birth spacing, gained siblings, and moderation by income or family size are null; observed-score birth-order coefficients are positive; and prior-death coefficients are not negative. These estimates substantially weaken a causal reading of the -1.34-point between-family family-size gradient and favor a large compositional component. They do not prove that the entire gradient is compositional: birth order is entangled with calendar time and maternal age, death contrasts are imprecise, completed fertility is unavailable, and the age-specific family-size design must be rebuilt from the full pregnancy history.

Figure 5: Resource dilution tests.



Panel A shows the between-family association of observed family size with WISC FSIQ ($b = -1.34$ per child). Panel B shows mother-FE estimates with mother-clustered uncertainty: birth order +0.48 points/rank, birth spacing +0.19 points/year, income per current child −0.017 points per income unit, and IQ change per sibling gained +0.17 points. Only the birth-order coefficient differs from zero; it is not interpreted causally because rank, calendar time, and maternal age are nearly collinear.

5.3 Breastfeeding and IQ

In the CPP, breastfeeding refers to nursery feeding during the first 0–8 days postpartum, not long-term breastfeeding. The overall breastfeeding rate was 92.8%. Critically, breastfeeding in the 1960s was *negatively* associated with maternal education: 94.5% of mothers with <9 years of education breastfed, compared to only 75.0% of college-educated mothers. This reversed SES–breastfeeding gradient provides a natural experiment for testing whether the breastfeeding–IQ association is causal or confounded.

The bivariate cross-sectional association is large and negative: breastfed children score −0.65 SD lower in WISC FSIQ. Adjusting for demographics reduces this to −0.23 SD; full adjustment (adding birth weight, gestational age, smoking, and birth order) yields −0.20 SD. Under mother fixed effects the coefficient is +0.016 (SE = 0.041); restricting to the 464 breastfeeding-discordant families gives +0.009 (SE = 0.044; 1,053 effective observations). The other mother-FE estimates are also small and non-significant (VIQ +0.049, PIQ +0.045, SB −0.023, PCA g +0.026). The large reversed cross-sectional association therefore does not replicate within mothers.

We do not interpret the recorded 0–8 nursery-day measure as breastfeeding duration: it is mechanically entangled with length of neonatal hospital stay and therefore does not support an auditable causal duration–response claim.

5.3.1 Robustness to Neonatal Health Controls

Neonatal health problems (low Apgar, prematurity, malformations) could both prevent breastfeeding and lower IQ. Adding health controls leaves the mother-FE estimate near zero ($b = +0.022$, SE = 0.045). Restricting to healthy singletons—excluding low Apgar (<7), preterm births, very low birth weight (<1,500g), any malformation by age 1, and twins—also gives a null estimate ($b = -0.015$, SE = 0.065; 206 discordant families).

Twins are nearly universally breastfed in the CPP (98.0% vs. 92.1% for healthy singletons)—their longer hospital stays ensured nursery feeding. Only 4 of 311 twin pairs with IQ data are discordant for breastfeeding, far too few for within-pair fixed effects.

5.4 Prenatal Smoking

The CPP recorded smoking status at prenatal registration ("Cigarettes Per Day Now"), providing a direct measure of active smoking during pregnancy. Nearly half the sample's mothers (46.7%) smoked, averaging 12.3 cigarettes/day among smokers (5.7 overall). Smoking prevalence varied by race (White: 53.5%, Black: 42.4%) and was inversely related to education (49.8% for <12 years, 35.9% for >12 years).

Smoking is associated with a 167g reduction in birth weight in adjusted OLS. Under mother fixed effects, this attenuates to −38g (SE = 16g), suggesting that much of the cross-sectional association reflects stable family differences. The within-family dose–response is imprecise: −1.2 g/cigarette/day (SE = 1.1).

For WISC IQ, the adjusted OLS association with smoking is modest ($b = -0.059$ SD) and disappears under mother fixed effects ($b = +0.006$, SE = 0.033). The within-family dose-response is similarly near zero. Alternative outcomes produce small, specification-sensitive estimates, so the robust conclusion is the WISC null rather than a claim that every outcome is identically null.

5.5 Birth Weight and IQ

In OLS, each 100g increase in birth weight predicts +0.034 SD in WISC FSIQ. Under mother fixed effects ($n = 13{,}151$ siblings in 6,006 families), the coefficient drops by half but stays significant: +0.017 SD per 100g (SE = 0.002). The within-family effect points toward a genuine causal role for fetal growth.

Within-pair birth weight differences do not significantly predict IQ differences in either DZ ($n = 236$ pairs; $b = 0.11$ per 100g, $p = 0.65$) or MZ twins ($n = 132$ pairs; $b = 0.25$, $p = 0.20$), but neither sample is large enough to detect effects of this size.

A concern with the sibling FE estimate is that it may reflect genetic pleiotropy rather than a causal intrauterine effect: within a sibship, the child who inherits more "big baby" alleles may also inherit alleles that boost cognition (the GWAS-based genetic correlation between birth weight and educational attainment is $r_g \approx 0.10$; Warrington et al., 2019). If so, the within-pair coefficient should shrink as genetic relatedness increases—less segregating variation remains to confound the estimate—and should be approximately zero for MZ twins, who share 100% of segregating genes. We test this by estimating the within-pair birth weight–IQ slope across four kinship types of increasing relatedness. The coefficients (WISC FSIQ per 100g) are: half-siblings ($R = 0.25$, $n = 924$): $b = 0.31$ (SE = 0.08); full siblings ($R = 0.50$, $n = 7{,}313$): $b = 0.26$ (SE = 0.03); DZ twins ($R = 0.50$, $n = 236$): $b = 0.11$ (SE = 0.23); MZ twins ($R = 1.0$, $n = 132$): $b = 0.25$ (SE = 0.19). The point estimates are roughly constant across pair types—the MZ coefficient (0.25) is essentially identical to the full-sibling coefficient (0.26)—consistent with a genuine intrauterine effect rather than genetic pleiotropy. The joint interaction test does not reject slope equality across pair types ($F = 1.85$, $p = 0.086$), and no pairwise interaction is significant: half-sibling vs. full sibling ($\Delta b = -0.05$, $p = 0.54$), full sibling vs. MZ twin ($\Delta b = -0.01$, $p = 0.96$). The data are thus consistent with a constant within-pair effect, but the twin samples are too small to definitively rule out genetic pleiotropy.

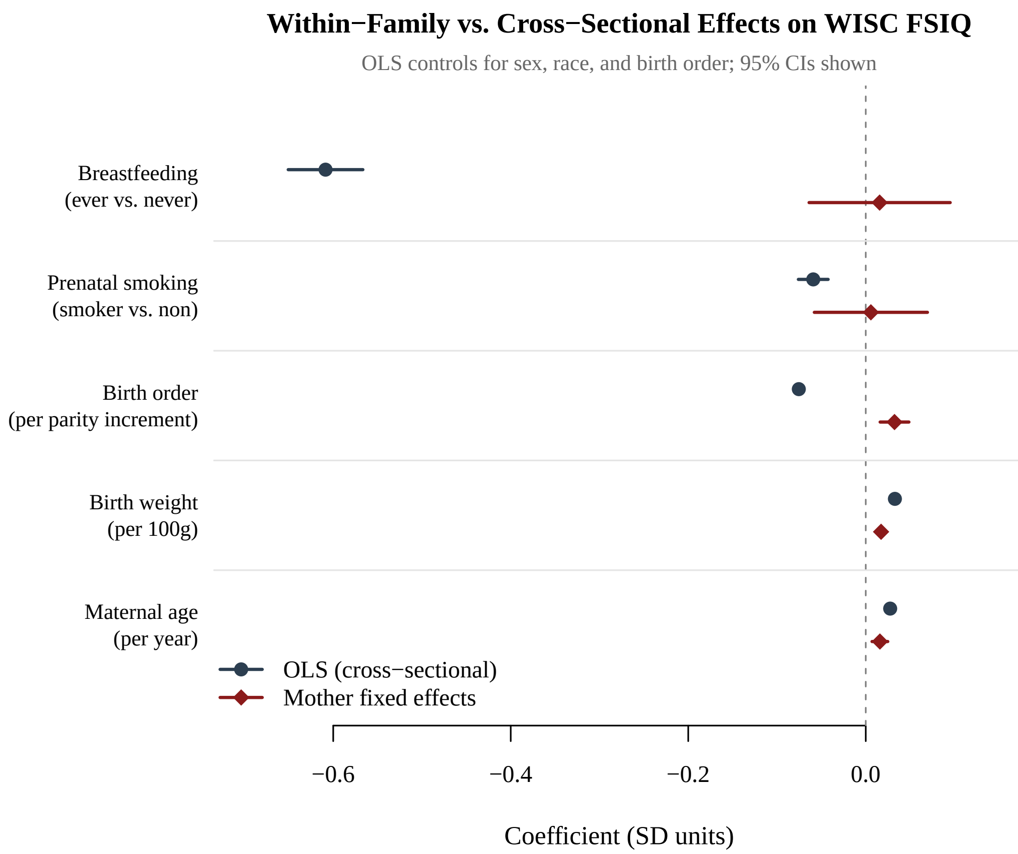
An exploratory four-subtest diagnostic correlates subtest g-loadings with adjusted low-birth-weight coefficients at $r = -0.51$; the analogous correlation for catch-up growth is $r = -0.09$. With only four outcomes, these are unstable descriptive pattern checks rather than inferential evidence for g-specificity, and we make no reliability-corrected or within-family subtest claim.

5.6 Maternal Age

Maternal age predicts child IQ positively between families and in a mother-FE model without rank ($b = +0.016$ SD/year, SE = 0.005). When birth order is included, the maternal-age coefficient is essentially zero ($b = +0.0006$, SE = 0.0092, $p = 0.95$), while birth order is +0.031 SD/rank (SE = 0.017). This is not a clean separation of effects: maternal age, birth year, and rank move together within families, so the conditional coefficients are collinearity-limited and require the strong assumption that no other time-varying family factor tracks the birth sequence.

Figure 6 summarizes the within-family analyses by comparing cross-sectional OLS and mother fixed-effects estimates for all five predictors of WISC FSIQ.

Figure 6: Forest plot of cross-sectional (OLS) vs. within-family (mother FE) effects on WISC FSIQ.



Breastfeeding and prenatal-smoking associations are near zero under mother FE; birth weight remains positive. The birth-order coefficient becomes positive, whereas maternal age becomes uninformative once rank is included because the two are nearly collinear within families.

5.7 Gestational Diabetes

The CPP recorded maternal diabetes mellitus (OB-60 disease codes) and glucose tolerance test results during pregnancy. We define gestational diabetes as any positive flag on either variable ($n = 1{,}124$; prevalence 2.1%). Prevalence varies by race (White: 2.9%, Black: 1.5%).

Cross-sectionally, gestational diabetes shows no significant association with offspring IQ: WISC FSIQ $b = -0.51$ (SE $= 0.43$) and Stanford–Binet IQ $b = +0.21$ (SE $= 0.56$). Mother-

FE estimates with mother-clustered standard errors remain imprecise: WISC $b = -1.03$ (SE = 1.49) and SB $b = +0.33$ (SE = 1.87). These estimates are compatible with both benefits and harms; the appropriate conclusion is low precision, not proof of no causal effect. Models control for child sex, maternal age, SEI, gestational age, and birth order.

For sex-adjusted head circumference, the mother-FE association at birth is -0.074 cm (SE = 0.132); at age 1 it is $+0.179$ cm (SE = 0.169). The wide within-family intervals again preclude a sharp conclusion. The IQ pattern is directionally consistent with Fraser et al. (2014), whose between-family deficit disappeared within sibships, but the CPP estimates should be read primarily as an imprecise replication.

5.8 Within-Family SES Variation

The CPP recorded family income, occupational status (SEI), and housing density at each pregnancy registration, so these indicators can change between siblings born to the same mother. Among 5,885 mothers with two or more children tested on the WISC (13,218 children total), 72% show within-family variation in income, 82% in SEI, and 85% in housing density. There is substantial within-family SES variation to leverage: the within-family standard deviation of income is 9.6 (vs. 18.6 between families), SEI is 0.72 (vs. 1.98), and housing density is 3.5 (vs. 6.1). The intraclass correlations are 0.81 for income, 0.90 for SEI, and 0.78 for housing density, meaning most SES variation is between families—but the within-family share (12–25%) is large enough to provide adequate statistical power.

Table 19 decomposes the SES–IQ gradient into between-family and within-family components. The between-family gradients are large: income $b = 0.291$, SEI $b = 3.422$, housing density $b = -0.687$, and education $b = 2.380$. Mother-FE coefficients with mother-clustered standard errors are all close to zero: income -0.004 (SE = 0.009), SEI -0.038 (SE = 0.123), housing density -0.021 (SE = 0.027), and education $+0.064$ (SE = 0.194). The between–within differences are significant, whereas no within-family coefficient is.

At one within-family SD, the between-family contrasts are approximately 2.79 IQ points for income, 2.47 for SEI, 2.37 for housing density, and 1.03 for education. The corresponding within-family contrasts are -0.04, -0.03, -0.07, and $+0.03$ points. These estimates control stable maternal characteristics, but causal interpretation additionally assumes no time-varying family shocks jointly affect SES and child cognition.

The own-birth SES estimates are reasonably precise relative to the between-family gradients. For example, the 95% interval for income is approximately $[-0.022, 0.014]$ IQ points per category, versus an OLS coefficient of $+0.230$. These are bounds on the conditional within-family association, not assumption-free bounds on causal SES effects: measurement error and time-varying confounding can attenuate or bias mother-FE estimates.

Because SES is measured at each pregnancy, we also construct a developmental exposure index from all sibling-birth updates falling within a child's birth-to-age-7 window. About 58% of children with an index have more than one update. Mother-FE estimates with clustered standard errors are income -0.023 (SE = 0.026), SEI -0.265 (SE = 0.318), and housing density -0.095 (SE = 0.068). The within-family SDs are 3.65, 0.30, and 1.47, yielding MDEs of approximately 0.26–0.28 IQ points per within-family SD. The estimates remain small, though housing density is less tightly centered on zero than the own-birth measures.

The estimates therefore place nearly all of the observed SES–IQ gradient between families rather than between siblings experiencing measured SES changes. They do not, by themselves, identify which stable familial factors generate that gradient.

Cross-sectional adjustment for income, SEI, and housing density reduces the Black–White mean difference substantially, whereas the within-family slopes are near zero. This contrast is evidence that the cross-sectional "explanation" is largely compositional. It is not a direct decomposition of the group gap: the mother-FE slopes are identified from changes within families, and transporting them to a between-group intervention requires invariance of the relevant causal response, adequate measurement, and absence of time-varying confounding.

These within-family results complement the measurement-invariance findings from Section 4, but the two address different questions. Measurement invariance supports score comparability across groups; it does not rule out environmental causes of latent mean differences or guarantee equal causal responses to SES. The race-stratified mother-FE coefficients are small and imprecise, providing no positive evidence of SES moderation in this sample.

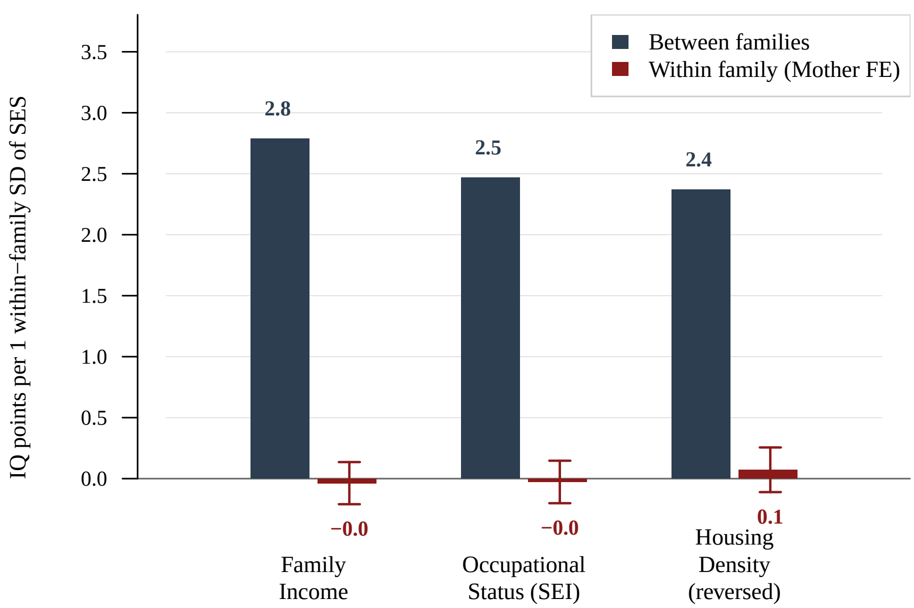
Combined with the sibling regressions, these results show that the measured SES gradient is overwhelmingly between families. Determining whether that pattern reflects genetic transmission, durable environmental differences, measurement, or their combination requires designs beyond mother fixed effects.

Table 19: SES–IQ Gradient: Between-Family vs. Within-Family (Mundlak Decomposition)

	Between		Within		OLS		
SES Indicator	b	SE	b	SE	b	SE	Atten.
Family income	0.291	0.007	−0.004	0.009	0.230	0.006	102%
SEI (decimal)	3.422	0.058	−0.038	0.123	3.014	0.053	101%
Housing density*	−0.687	0.027	−0.021	0.027	−0.498	0.022	96%
Education (years)	2.380	0.059	0.064	0.194	2.290	0.057	97%

Note. $N = 12{,}082$–$12{,}750$ children in 5,821–5,875 sibling families. Between-family coefficients are from a Mundlak correlated-random-effects specification. Within-family coefficients are mother fixed effects. All standard errors are clustered by mother, preventing repeated siblings from being treated as independent; models include sex and birth order. Attenuation $= 1 - b_{\mathrm{FE}}/b_{\mathrm{OLS}}$ and can exceed 100% when the FE sign reverses. Within-family SDs: income 9.57, SEI 0.72, housing density 3.45, education 0.43. *Housing density is partly endogenous to family size and relocation.

Figure 7: SES–IQ gradient scaled by within-family standard deviation.



Between-family coefficients (dark bars) reflect the IQ difference between families whose mean SES differs by 1 within-family SD. Within-family coefficients (red bars with 95% CIs) reflect the IQ difference between siblings in the same family who experienced that same SES change. All within-family estimates are indistinguishable from zero.

6 Developmental Trajectories

6.1 Family Structure

Half-sibling rates differ dramatically by race. Among classified sibling pairs, 19.6% of Black pairs are half-siblings, compared to 5.7% of White pairs ($\chi^2 = 470$, $p \approx 10^{-104}$). This difference is consistent with historical data on non-marital fertility and paternal instability in the 1960s. Blood type analysis reveals that all 68 impossible mother–child ABO combinations (0.14% of 50,201 tested) reflect serological typing errors rather than paternal misattribution.

6.2 Early Growth, Catch-Up, and IQ

6.2.1 Growth Predicts IQ

Early postnatal weight gain (birth to 4 months) predicts age-7 IQ: each SD increase is associated with +0.13 SD in FSIQ bivariately and +0.10 after adjustment. Under mother FE, the estimate is +0.041 SD (SE = 0.012). Weight gain from birth to 12 months is similar ($b = +0.045$, SE = 0.012). Head circumference at age 7 attenuates the mother-FE birth-weight coefficient by about 17%, but adding head-circumference gain to the 0–12 month weight-gain model changes the coefficient by -3% (a slight increase rather than mediation). Thus brain-size adjustment attenuates part of the birth-weight association but does not mediate the postnatal weight-gain association.

The within-family HC–IQ association is robust across HC measurement ages. With mother-clustered standard errors, the WISC FSIQ coefficient (sex-adjusted, per within-sex SD) is $b = 0.086$ (SE = 0.012) for birth HC, $b = 0.099$ (SE = 0.011) for 12-month HC, and $b = 0.091$ (SE = 0.011) for 7-year HC (N = 12,885–13,715). The pattern replicates for Stanford-Binet IQ at age 4: birth HC $b = 0.090$ (SE = 0.013), 12-month HC $b = 0.055$ (SE = 0.012), and 7-year HC $b = 0.038$ (SE = 0.012). These temporal patterns are consistent with HC indexing a stable trait, but mother FE does not rule out sibling-specific prenatal or postnatal confounding.

When all three HC ages are entered jointly in a mother-FE model predicting WISC FSIQ (N = 12,600, 6,531 families), all three remain positive with clustered inference: birth HC $b = 0.050$ (SE = 0.013), 12-month HC $b = 0.054$ (SE = 0.014), and 7-year HC $b = 0.052$ (SE = 0.013). The within-family (demeaned) correlations between HC ages are only moderate ($r = 0.28$ for birth–7yr, $r = 0.32$ for birth–12mo, $r = 0.53$ for 12mo–7yr), so the sibling born with the largest head is not necessarily the sibling with the largest head at 12 months or 7 years. Measurement reliability may be high in controlled studies, but these observational data alone cannot distinguish developmental reshuffling from residual measurement differences.

The biological basis for this reshuffling is well-established. HC heritability undergoes a dramatic developmental shift: at birth, shared (prenatal) environment accounts for ∼60–65% of variance and heritability is near zero, reflecting uterine constraint, placentation, and chorionicity effects; by 5–6 months, heritability rises to ∼90% as prenatal environmental influences wash out and genetic factors assert control over brain growth (Smit et al., 2010; Silventoinen et al., 2011). Twin studies confirm this pattern: monochorionic twins show 5.8% HC discordance at birth that resolves to 1.1% by 36 months (Cortez Ferreira et al., 2025). Birth HC and later HC therefore index a partly different mix of causes: birth HC captures genetic brain-growth potential overlaid with substantial prenatal environmental variance, while 12-month and 7-year HC are predominantly genetic.

The independent contributions of each HC age could reflect a stable trait measured with age-specific noise, distinct developmental processes, or both; the present regressions do not separate these mechanisms. For SB IQ, birth and 12-month HC remain positive in a joint mother-FE model (birth: $b = 0.071$, SE = 0.014; 12-month: $b = 0.035$, SE = 0.012). Sex adjustment (via within-sex standardization) makes negligible difference to the estimates.

6.2.2 Catch-Up Growth in Low Birth Weight Children

The CPP contains 5,562 low birth weight (LBW, <2,500g) children (10.4% of the sample), providing substantial power to study catch-up growth. LBW children show dramatic weight catch-up: the weight gap narrows from −1.85 SD at birth to −1.05 SD at 4 months and −0.33 SD at 7 years. Head circumference follows a similar trajectory: the gap narrows from −1.59 SD at birth to −0.52 SD at 12 months and −0.44 SD at 7 years.

The cognitive penalty also narrows with age on the IQ-point scale, although the standardized contrast is less decisive. Among the same 32,728 children observed on both tests, the adjusted LBW penalty is −3.90 IQ points (−0.237 SD) on the Stanford–Binet at age 4 and −3.11 points (−0.217 SD) on the WISC at age 7. The raw-point gap narrows by +0.79 points (SE = 0.24, $p < 0.001$), while the within-sample standardized narrowing is +0.021 SD (SE = 0.015, $p = 0.17$). In the same-sample mother-FE analysis ($N = 9{,}796$ children in 4,448 families), the penalty narrows from −2.58 points (−0.157 SD) at age 4 to −1.21 points (−0.084 SD) at age 7: a +1.37-point change (SE = 0.64, $p = 0.033$), or +0.073 SD (SE = 0.041, $p = 0.080$). The direction is consistent with partial cognitive catch-up, but LBW children retain a deficit at age 7 and the standardized change estimates remain imprecise.

Among LBW children, 0–12 month weight gain predicts WISC IQ between families. In the LBW sibling analysis, the mother-FE estimate is +0.086 SD (SE = 0.055). In the head-circumference-complete sibling sample, the weight-gain coefficient is +0.075 (SE = 0.058) without head-circumference gain and +0.083 (SE = 0.064) with it; the head-circumference-gain coefficient is −0.019 (SE = 0.045). The small increase in the weight coefficient is compatible with sampling variation or suppression, not mediation through head growth. These models concern 0–12 month growth and age-7 IQ level; because both growth measures precede the age-4 assessment, they do not test mediation of the age-4-to-age-7 narrowing in the LBW penalty.

7 External Validity

Follow-up rates vary dramatically across sites, from 28.5% (NY-Metropolitan) to 91.3% (Buffalo). Despite this site-level variation, SES-based attrition is minimal: children who completed the age-7 WISC differ from those lost to follow-up by only $d = 0.04$ in maternal education and $d = -0.02$ in occupational status. Maternal age shows a modest difference ($d = 0.12$). Black children have slightly higher follow-up rates (80.7%) than White children (75.7%), contrary to the typical pattern in longitudinal studies. Logistic regression confirms that race and maternal age are the strongest predictors of follow-up, while education and SES are not practically significant.

8 Replication of Published Findings

To validate our data extraction and demonstrate continuity with the prior CPP literature, we replicate eight landmark analyses that used the original CPP data. These replications serve a dual purpose: confirming that our modernized extraction produces results consistent with published benchmarks, and extending the original findings with methods unavailable to the original authors (e.g., mother fixed effects, multi-level behavior genetic decomposition).

8.1 Broman et al. (1975): Descriptive Benchmarks

Broman et al. (1975) reported foundational descriptive statistics for the CPP. Our extraction closely replicates their key benchmarks (Table 20). The White Stanford-Binet IQ mean of 104.5 is within 1.2 points of their published value (103.3), and the Black mean of 91.4 matches their 91.6 almost exactly. The WISC at age 7 shows similar racial gaps (White: 102.4, Black: 89.7). The education–IQ gradient rises monotonically from 89.8 (0–8 years of education) to 117.8 (16+ years) for SB IQ, and the birth weight–IQ gradient shows the expected monotonic increase with the steepest slope below 2,500g. These close replications validate our CPPVAR extraction pipeline.

Table 20: Replication of Broman et al. (1975): Mean IQ by Race

	Stanford-Binet (4yr)		WISC FSIQ (7yr)	
Race	Mean	n	Mean	n
White	104.5	17,124	102.4	18,681
Black	91.4	19,224	89.7	20,175
Puerto Rican	89.1	2,052	89.2	1,353
Other	108.4	76	112.0	84
Total	97.1	38,651	95.6	40,486
Published: White SB = 103.3, Black SB = 91.6				

8.2 Turkheimer et al. (2003): SES Moderation of IQ Heritability

Turkheimer et al. (2003) reported that IQ heritability was near zero in impoverished CPP families but substantial (∼0.72) in affluent families, with shared environment showing the opposite pattern. We replicate this using their original DeFries-Fulker (DF) method on twin pairs identified by Myrianthopoulos's blood-group zygosity classification—the same source Turkheimer et al. used—and extend it with sibling and multi-level analyses.

Close Replication

Using only twin pairs classified by Myrianthopoulos (zygosity determined from 9 blood group systems and placental examination), we obtain 327 pairs (118 MZ, 209 DZ) with complete WISC FSIQ and SEI data, closely matching the original 319 pairs (114 MZ, 205 DZ). In the base DF model, we estimate $c^2 = 0.40$ and $a^2 = 0.41$. Adding SES (standardized SEI) as a continuous moderator yields significant interactions ($p = 0.039$ for the three-way $\mathrm{IQ}_P \times R \times$ SES term): Table 21 shows ACE components evaluated at low (−1 SD), mean, and high (+1 SD) SES.

Table 21: Replication of Turkheimer et al. (2003): ACE by SES Level

	DF (Myrianthopoulos twins)			Classic Falconer		
SES Level	a^2	c^2	e^2	a^2	c^2	e^2
Low (-1 SD)	0.16	0.51	0.34	0.31	0.43	0.26
Mean	0.39	0.33	0.27	0.40	0.41	0.19
High (+1 SD)	0.63	0.16	0.21	0.59	0.24	0.17
Original: $a^2 \approx 0.00$ (low SES), $a^2 \approx 0.72$ (high SES)						
Original: $c^2 \approx 0.58$ (low SES), $c^2 \approx 0.00$ (high SES)						

We replicate the qualitative pattern: a^2 rises from 0.16 (-1 SD SES) to 0.63 (+1 SD) and c^2 falls from 0.51 to 0.16. The ICCs show why—MZ correlations are roughly flat across SES ($r = 0.72$ low, $r = 0.83$ high) while DZ correlations fall ($r = 0.61$ to $r = 0.50$). Our low-SES $a^2 = 0.16$ is larger than the original ($\sim$0) and the moderation gradient is shallower overall, which may reflect our different SES composite or differential attrition between the samples.

Extensions: Pairwise Contrasts and Multi-Level Analyses

Before introducing SES moderation with siblings, we validate the kinship structure by comparing unmoderated a^2 across all pairwise contrasts (Table 22). Under an additive model, $a^2 = (r_1 - r_2)/(R_1 - R_2)$ should come out roughly the same no matter which pair types go into the contrast.

Table 22: Heritability from Pairwise Kinship Contrasts (Unmoderated)

Contrast	r_1	r_2	R_1	R_2	$\hat{a}^2$	$\hat{c}^2$
MZ vs. DZ	.81	.61	1.00	0.50	0.41	0.40
MZ vs. FS	.81	.55	1.00	0.50	0.52	0.29
MZ vs. HS	.81	.35	1.00	0.25	0.62	0.19
DZ vs. HS	.61	.35	0.50	0.25	1.04	0.09
FS vs. HS	.55	.35	0.50	0.25	0.82	0.14

The a^2 estimates increase systematically as the more closely related group shifts from twins to siblings: 0.41 (MZ–DZ) $\rightarrow$ 0.52 (MZ–FS) $\rightarrow$ 0.82 (FS–HS). This gradient likely reflects assortative mating, which inflates the true genetic relatedness of non-MZ pairs above their nominal values, and the bias increases for more distant kinship types.

Joint assortative mating and twin-specific shared environment. With four observed correlations (MZ, DZ, FS, HS), we can exactly identify four parameters without assuming any value for the spousal IQ correlation μ (not to be confused with μ_g for group means in Section 4). The model (Figure 8) specifies:

$$r_{\mathrm{MZ}} = a^2 + c^2_{\mathrm{fam}} + c^2_{\mathrm{twin}} \tag{1}$$
$$r_{\mathrm{DZ}} = \tfrac{(1+m)}{2}\, a^2 + c^2_{\mathrm{fam}} + c^2_{\mathrm{twin}} \tag{2}$$
$$r_{\mathrm{FS}} = \tfrac{(1+m)}{2}\, a^2 + c^2_{\mathrm{fam}} \tag{3}$$
$$r_{\mathrm{HS}} = \tfrac{(1+m)}{4}\, a^2 + c^2_{\mathrm{fam}} \tag{4}$$

where $m = \mu a^2$ is the genetic spousal correlation induced by phenotypic assortative mating. Equations (2)–(3) immediately give $c^2_{\mathrm{twin}} = r_{\mathrm{DZ}} - r_{\mathrm{FS}}$, since DZ twins and full siblings share the same genetic relatedness and family environment, differing only in the prenatal twin environment. Letting $A = r_{\mathrm{MZ}} - r_{\mathrm{DZ}}$ and $B = r_{\mathrm{FS}} - r_{\mathrm{HS}}$, we obtain $m = (2B - A)/(A + 2B)$, $a^2 = 2A/(1-m)$, $\mu = m/a^2$, and $c^2_{\mathrm{fam}} = r_{\mathrm{HS}} - a^2(1+m)/4$.

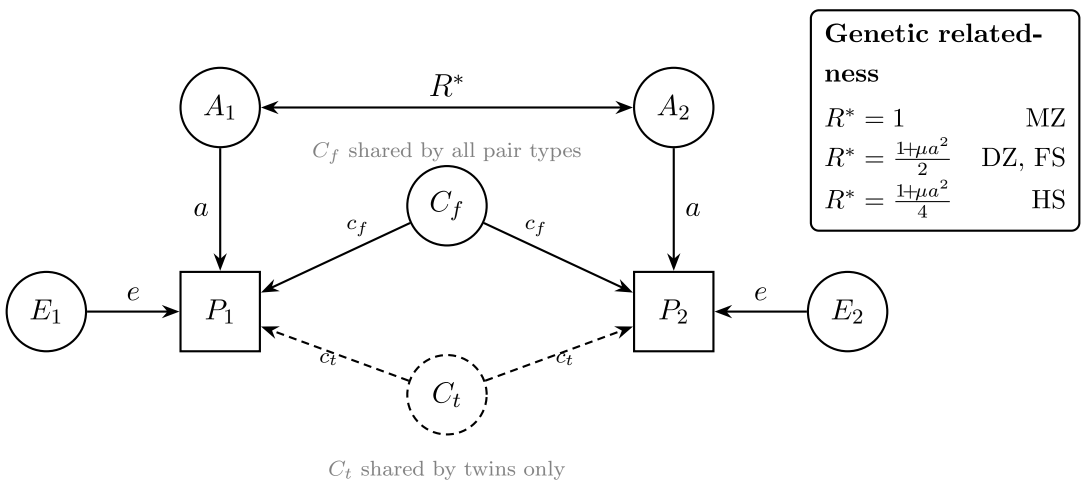


Figure 8: Path diagram for the extended kinship model. Latent additive genetic factors (A) are correlated at R^*, which depends on pair type and is inflated above nominal values by phenotypic assortative mating (μ). Family shared environment (C_f) affects all pair types; twin-specific shared environment (C_t, dashed) affects only co-twins. Each child has a unique environment (E). With four pair types, all parameters including μ are exactly identified.

The model specifies expected correlations as:

$$\begin{aligned} r_{\mathrm{MZ}} &= a^2 + c_f^2 + c_t^2, \\ r_{\mathrm{DZ}} &= R^* a^2 + c_f^2 + c_t^2, \\ r_{\mathrm{FS}} &= R^* a^2 + c_f^2, \\ r_{\mathrm{HS}} &= R^*_{\mathrm{HS}} a^2 + c_f^2, \end{aligned}$$

where $R^* = \frac{1}{2}(1 + \mu a^2)$ and $R^*_{\mathrm{HS}} = \frac{1}{4}(1 + \mu a^2)$ incorporate the first-generation assortative mating inflation, and $e^2 = 1 - a^2 - c_f^2 - c_t^2$. Four equations in four unknowns (a^2, c_f^2, c_t^2, μ) yield a unique solution. Table 23 presents the result. The implied spousal IQ correlation ($\mu = 0.55$) is consistent with the published literature: the meta-analytic spousal IQ correlation is $r = 0.44$ across 17 samples (Horwitz et al., 2023), while Bouchard & McGue (1981) reported $r = 0.33$ and Plomin & Deary (2015) summarized the consensus at $\sim$0.40. The observed CPP spousal education correlation is $r = 0.61$ ($N = 42{,}175$); since education correlates with IQ at $r \approx 0.55$–0.56 (Strenze, 2007), direct assortment on education alone would imply a spousal IQ correlation of $r^2_{\mathrm{edu}} \times \mu_{\mathrm{edu}} \approx 0.19$, while the full spousal education correlation sets an upper bound via phenotypic similarity. Our data-estimated $\mu = 0.55$ falls above the meta-analytic IQ value (0.44) but below the education correlation (0.61), suggesting that the CPP kinship structure is most consistent with spousal assortment operating through both cognitive ability and education. AM-corrected relatedness is $R^*_{\mathrm{DZ}} = R^*_{\mathrm{FS}} = 0.67$ (nominal 0.50) and $R^*_{\mathrm{HS}} = 0.33$ (nominal 0.25).

Table 23: Analytical Variance Decomposition: Observed Correlations and Exactly-Identified Solution

Observed correlations				
MZ ($N=129$)	DZ ($N=229$)	FS ($N=6{,}844$)	HS ($N=864$)	
.809	.607	.551	.347	
Variance components (exactly identified)				
a^2	c^2_{fam}	c^2_{twin}	e^2	μ
0.611	0.142	0.056	0.191	0.551
AM-corrected pairwise contrasts ($\hat{a}^2$)				
MZ–DZ	MZ–FS*	MZ–HS	DZ–HS	FS–HS
0.61	0.78*	0.70	0.78	0.61

Uncorrected range: 0.41–1.04. *MZ–FS inflated by c^2_{twin}.

The AM correction substantially improves convergence across contrasts: the uncorrected range of $\hat{a}^2 = 0.41$–1.04 narrows to 0.61–0.78. The MZ–FS contrast remains elevated because the MZ correlation includes c^2_{twin} that the FS correlation does not; subtracting $c^2_{\text{twin}} = 0.056$ from the MZ–FS numerator yields $a^2 = 0.61$, identical to the analytical solution. The twin-specific shared environment is modest ($c^2_{\text{twin}} = 0.056$), consistent with prenatal environment contributing 5–6% of IQ variance in twins but not siblings.

Dominance estimation. Dominance deviations are shared fully by MZ twins ($R_D = 1$), partially by DZ twins and full siblings ($R_D = \frac{1}{4}$), and not at all by half-siblings ($R_D = 0$). Since DZ and FS share the same R_D, c^2_{twin} remains identified from their difference. Fixing μ externally frees a degree of freedom for dominance; the resulting quadratic in a^2 has a closed-form solution: $a^2 = [-1+\sqrt{1 + 80\mu(3B - A)}]/(10\mu)$. At $\mu = 0.61$ (spousal education), the extended ADCE decomposition yields $a^2 = 0.59$, $d^2 = 0.02$, $c^2_{\text{fam}} = 0.15$, $c^2_{\text{twin}} = 0.06$, and $e^2 = 0.19$. At the meta-analytic spousal IQ correlation of $\mu = 0.44$ (Horwitz et al., 2023), the solution yields $a^2 = 0.67$, $d^2 = -0.04$; at the data-estimated $\mu = 0.55$, it gives $a^2 = 0.61$, $d^2 \approx 0.00$. Across this range, dominance is uniformly negligible ($|d^2| \leq 0.04$), and the crossover from slightly negative to slightly positive identifies $\mu \approx 0.55$ as the value at which the additive model fits exactly—consistent with the analytical solution that estimates $\mu = 0.55$ by constraining $d^2 = 0$.

SES matching. Half-siblings come disproportionately from low-SES, disrupted families: mean household SEI is 3.21 for HS pairs versus 4.96 for FS ($d = 1.0$). To ensure comparability, we compute FS-vs-HS heritability within SES quintiles. The FS–HS gap persists after matching—$a^2 = 0.74$ in the lowest quintile ($N_{\text{HS}} = 440$) and 0.49 in Q2 ($N_{\text{HS}} = 215$)—confirming that the correlation difference is genetic, not an artifact of SES-level differences.

We next examine SES moderation. Table 24 shows IQ correlations by pair type and SES.

Table 24: IQ Correlations by Kinship Type and SES

Type (R)	Overall	Low SES	High SES
MZ (1.0)	.81 (N=129)	.74	.83
DZ (0.5)	.61 (N=229)	.59	.54
FS (0.5)	.55 (N=6,844)	.48	.47
HS (0.25)	.35 (N=864)	.32	.38*

*Only 133 HS pairs at high SES (28 in top quartile).

MZ correlations increase with SES while DZ and FS correlations decrease, consistent with the Scarr–Rowe pattern. The MZ-vs-FS Falconer yields $a^2 = 0.51$ (low SES) to 0.73 (high

SES)—a stronger gradient than the MZ-vs-DZ estimate ($a^2 = 0.31$ to 0.59)—leveraging the much larger FS sample ($N = 6{,}844$) for the $R = 0.5$ reference.

However, half-sibling correlations *increase* with SES (0.32 to 0.38), driven by an extremely small high-SES HS sample. Including these noisy HS pairs in the multi-level DF produces unstable SES moderation estimates. Dropping HS pairs yields a cleaner three-level model (MZ + DZ + FS; $N = 7{,}202$ pairs) with a twin-specific shared environment term:

Table 25: Multi-Level ACE (MZ + DZ + FS, Dropping Half-Siblings)

SES Level	a^2	c^2_{fam}	c^2_{twin}	e^2
Low (-1 SD)	0.24	0.32	0.14	0.30
Mean	0.47	0.19	0.09	0.25
High (+1 SD)	0.70	0.06	0.04	0.20

The three-way interaction ($\text{IQ}_P \times R \times \text{SES}$) is significant ($p = 0.029$), confirming the Scarr–Rowe effect in the multi-level framework. Heritability increases from $a^2 = 0.24$ at low SES to 0.70 at high SES, while family shared environment drops from $c^2_{\text{fam}} = 0.32$ to 0.06. Twin-specific shared environment is modest ($c^2_{\text{twin}} = 0.09$ at mean SES, $p = 0.006$) and does not vary significantly with SES ($p = 0.11$ for the twin $\times$ SES interaction), consistent with the prenatal environment contributing approximately equally regardless of family socioeconomic status.

Robustness to assortative mating. Because AM correction rescales R linearly, the three-way interaction p-value is algebraically invariant to the choice of μ: it remains $p = 0.029$ across the full range from $\mu = 0$ (no AM) to $\mu = 0.61$ (spousal education upper bound). What changes is the magnitude of the estimated variance components: at $\mu = 0$, a^2 ranges from 0.24 (low SES) to 0.70 (high SES); at $\mu = 0.61$, it ranges from 0.37 to 1.09 (with out-of-bounds values at the extremes reflecting the narrower R range). The Scarr–Rowe interaction is thus fundamentally robust to AM.

Age specificity. The Scarr–Rowe effect is absent at age 4 (Stanford-Binet). In twin-only DF, the three-way interaction is non-significant for SB ($p = 0.81$; $\Delta a^2 = -0.06$, $N = 301$ pairs). In the multi-level model, the result is the same ($p = 0.57$; $\Delta a^2 = -0.12$, $N = 6{,}351$ pairs). The Scarr–Rowe interaction thus emerges between ages 4 and 7, consistent with a developmental model in which genetic and environmental influences on IQ differentiate as children enter formal schooling and the environmental demands on general cognitive ability intensify.

Within-race analysis. Because SES and race are confounded in the CPP (White mean SEI = 5.62, Black mean SEI = 3.79), the pooled Scarr–Rowe effect could in principle reflect a between-race artifact rather than genuine SES moderation. We therefore test the interaction separately within each racial group using multi-level DF (MZ + DZ + FS, concordant-race pairs only). Among White families ($N = 4{,}046$ pairs), the Scarr–Rowe effect is marginally significant ($p = 0.054$; $\Delta a^2 = 1.00$, from $a^2 = -0.22$ at -1 SD to 0.78 at $+1$ SD). Among Black families ($N = 3{,}013$ pairs), the effect is directionally consistent and also marginal ($p = 0.093$; $\Delta a^2 = 0.66$, from 0.32 to 0.98). Adding race as a covariate with main and interaction effects in the pooled model yields $p = 0.088$ for the three-way interaction. The within-race results are directionally supportive in both groups but individually underpowered.

Baseline heritability by race. Table 26 presents race-specific IQ correlations by pair type and the resulting Falconer ACE estimates. The multi-level WLS estimate of heritability is $a^2 = 0.76$ for White and $a^2 = 0.54$ for Black families, but the direction of the race difference is reversed in the classic twin estimate (MZ vs DZ: White $a^2 = 0.43$, Black $a^2 = 0.63$). The inconsistency across methods reflects the very small MZ twin samples within each race (49 White, 75 Black pairs) and differential confounding with SES. These results are consistent with the meta-analytic finding of Pesta et al. (2020) that race/ethnicity $\times$ heritability interactions do not exist: across 84,897 White, 37,160 Black, and 17,678 Hispanic individuals, heritabilities were consistently moderate to high and did not differ significantly across groups. The Scarr–Rowe interaction is thus better understood as SES moderation than as a race-mediated phenomenon.

Table 26: Race-Specific IQ Correlations and ACE Estimates

	White		Black	
	r	N	r	N
Correlations by pair type				
MZ	.797	49	.715	75
DZ	.580	104	.402	118
FS	.461	3,893	.436	2,820
HS	.249	204	.304	648
Method	a^2	c^2	a^2	c^2
MZ vs DZ	.43	.36	.63	.09
MZ vs FS	.67	.12	.56	.16
FS vs HS	.85	.04	.53	.17
Multi-level WLS	.76	.08	.54	.17

Generalization across cognitive measures. Table 27 presents the Scarr–Rowe interaction estimated via multi-level DF (MZ + DZ + FS) for seven cognitive measures. The effect is directionally positive (higher heritability at higher SES) for four of six age-7 measures, with Bender-Gestalt and Goodenough showing negligible negative Δa^2. However, only WISC FSIQ reaches statistical significance ($p = 0.029$). The WRAT Spelling ($\Delta a^2 = 0.24$, $p = 0.37$) and WRAT Reading ($\Delta a^2 = 0.11$, $p = 0.69$) show the largest effects after WISC, while the Stanford-Binet at age 4 again shows no effect ($\Delta a^2 = -0.12$, $p = 0.57$).

Table 27: Scarr–Rowe Interaction by Cognitive Measure (Multi-Level DF)

Measure	N_{pairs}	a^2_{low}	a^2_{high}	Δa^2	p
WISC FSIQ (age 7)	7,202	0.24	0.70	0.47	**.029**
WRAT Spelling (age 7)	4,593	0.41	0.64	0.24	.367
WRAT Reading (age 7)	4,591	0.42	0.53	0.11	.686
WRAT Arithmetic (age 7)	4,592	0.37	0.39	0.01	.966
Bender-Gestalt (age 7)	6,876	0.26	0.21	−0.06	.808
Goodenough (age 7)	4,602	0.38	0.25	−0.13	.669
Stanford-Binet (age 4)	6,351	0.29	0.16	−0.12	.572

g-Loading and SES Moderation

We further ask whether the SES moderation of heritability varies by the g-loading of the cognitive subtest. We compute g-loadings as Pearson correlations between each subtest and the PCA PC1 g factor from the full 12-test battery (N = 31,031; 43.7% variance explained), then run the moderated DF model separately for each subtest, estimating $\Delta h^2 = h^2_{\text{high SES}} - h^2_{\text{low SES}}$. Table 28 presents three specifications: twin-only (Myrianthopoulos pairs), and two multi-level designs pooling twins with full siblings. The multi-level models include a twin-specific shared-environment term and its SES interaction to prevent the extra prenatal environment shared by twins from biasing the SES moderation estimates; without this term, several per-subtest h^2 values exceed 1.0, indicating model misspecification.

Table 28: g-Loading and SES Moderation of Heritability by Cognitive Subtest

			Twin-only			Multi-level (MZ+DZ+FS)		
Subtest	g-ld	N_{tw}	h^2_{lo}	h^2_{hi}	Δh^2	h^2_{lo}	h^2_{hi}	Δh^2
WRAT Arithmetic	.80	250	0.51	0.79	0.27	0.37	0.39	0.02
WRAT Spelling	.80	251	0.33	0.87	0.54	0.40	0.64	0.24
WRAT Reading	.79	250	0.41	0.56	0.15	0.41	0.53	0.12
WISC Information	.73	321	0.07	0.69	0.61	0.18	0.65	0.47
WISC Vocabulary	.71	320	0.36	0.32	−0.03	0.39	0.38	−0.01
WISC Digit Span	.70	320	0.27	0.14	−0.12	0.15	0.12	−0.03
WISC Pic. Arrangement	.70	247	−0.03	0.10	0.13	−0.04	0.20	0.23
WISC Block Design	.61	247	0.07	0.27	0.20	0.17	0.58	0.41
Goodenough	.51	252	0.52	0.09	−0.42	0.38	0.25	−0.13
WISC Comprehension	.51	321	−0.35	0.00	0.35	−0.33	0.03	0.36
Bender Gestalt	.70	323	0.36	0.19	−0.17	0.27	0.21	−0.06
WISC Coding	.37	247	0.38	−0.36	−0.74	0.37	−0.35	−0.72
$r(g\text{-loading}, \Delta h^2)$			+0.70 ($p = 0.012$)			+0.53 ($p = 0.076$)		

The twin-only correlation between g-loading and Δh^2 is $r = +0.70$ ($p = 0.012$), indicating that higher-g subtests show greater SES moderation. However, with only ∼250–320 twin pairs per subtest, several per-subtest heritabilities are out of range (negative or > 1), and the correlation may be inflated by noise. Adding ∼4,500–7,100 full-sibling pairs per subtest with a twin-specific shared-environment term gives $r = +0.53$ ($p = 0.076$), attenuated but in the same direction. A complementary analysis using only full-sibling pairs—comparing sibling correlations above and below the median SEI—also yields $r = +0.50$ ($p = 0.099$) between g-loading and the SES-related increase in sibling resemblance. All three specifications agree that higher-g subtests tend to show more SES moderation, though the effect is only marginally significant in the better-powered analyses. Half-sibling pairs are excluded from the multi-level models because their severe SES restriction (mean SEI = 3.2 vs. 5.0 for full siblings) creates a confound between genetic relatedness and SES that biases the $R \times$ SES interaction; including them reverses the correlation to $r \approx 0$ (see Appendix A).

8.3 Willerman et al. (1970): Interracial Mating and IQ

Willerman et al. (1970) identified 129 children of Black–White interracial matings in the CPP and found that children with White mothers scored approximately 9 IQ points higher on the

Stanford-Binet at age 4 than children with Black mothers, regardless of the child's assigned race. We replicate this using father's race from the CPP SES detailed records.

We identify 142 children of interracial matings (106 with White mothers and Black fathers; 36 with Black mothers and White fathers), closely matching the original sample size. Table 29 presents the results.

Table 29: Replication of Willerman et al. (1970): IQ by Maternal Race in Interracial Matings

	SB IQ (4yr)		WISC FSIQ (7yr)	
Mating Type	Mean	n	Mean	n
White mother × Black father	103.0	72	100.1	74
Black mother × White father	96.4	22	97.0	22
White-mother advantage	6.6		3.2	
p (two-sample t)	0.044		0.176	
Original: ∼9 point White-mother advantage (SB at age 4)				

The White-mother advantage at age 4 (6.6 points, $p = 0.044$) is directionally consistent with the original but smaller in magnitude. At age 7 (WISC), the advantage narrows to 3.2 points and is not statistically significant. Adjusting for maternal education and SEI—which are similar across groups (education: 11.1 vs. 10.7 years; SEI: 4.7 vs. 4.7)—barely moves the estimates (adjusted SB: $b = 5.9$, $p = 0.067$; adjusted WISC: $b = 3.9$, $p = 0.12$). That the advantage is larger at age 4 than age 7 could mean the White-mother effect is prenatal or early-environmental and fades with time, though the small sample makes it hard to say much.

8.4 Myrianthopoulos (1976): Twin–Singleton Cognitive Gap

Myrianthopoulos & Keith (1976) found that CPP twins scored lower than singletons on cognitive tests, with the gap present in both races.

In our data, twins score 8.8 points below singletons on the Stanford-Binet ($p < 0.001$) and 5.6 points below on WISC FSIQ ($p < 0.001$). The gap appears in both racial groups (Table 30) and does not differ by race ($p_{\text{interaction}} = 0.36$). Controlling for birth weight and gestational age—twins are much smaller (mean birth weight: 2,351g vs. 3,169g)—cuts the gap roughly in half (-6.5 SB, -3.2 WISC), implicating lower fetal growth as a major mediator of the twin penalty. In the 41 families where the same mother had both twin and singleton births (91 children), mother fixed effects estimation yields a non-significant twin gap: $b = +0.104$

SD for WISC FSIQ ($p = 0.53$) and $b = -0.037$ SD for SB IQ ($p = 0.82$). Controlling for birth weight barely changes the estimate ($b = +0.125$, $p = 0.47$). The within-family null suggests that the cross-sectional twin–singleton IQ gap is largely driven by confounding rather than a direct twinning effect, though the small number of mixed twin-singleton families limits power.

Table 30: Twin–Singleton IQ Gap by Race (WISC FSIQ)

	Singletons		Twins		
Race	Mean	n	Mean	n	Gap
White	102.5	18,507	97.8	174	4.6
Black	89.8	19,958	84.3	217	5.5
Overall	95.7	40,083	90.1	403	5.6

8.5 Saha et al. (2009): Advanced Paternal Age and Neurocognitive Outcomes

Saha et al. (2009) reported that advanced paternal age was associated with impaired neurocognitive outcomes in CPP children, restricting to singleton term births (GA $\geq$ 37 weeks). We replicate their cross-sectional analysis and then add within-family estimation.

Restricting to singleton term births ($N = 34{,}407$) and using the published covariate structure, the unadjusted paternal-age coefficient is -0.056 IQ points per year. After full adjustment it is -0.036 (SE = 0.019, $p = 0.051$; $N = 25{,}803$), equivalent to about -0.36 points per decade.

In the full sample, the fully adjusted standardized WISC association is null ($b = -0.00048$ SD/year, SE = 0.00118, $p = 0.68$; $N = 29{,}754$). A mother-FE model without birth order is positive (+0.171 IQ points/year, SE = 0.074; $N = 30{,}597$), but adding birth order reduces it to +0.047 (SE = 0.114, $p = 0.68$; $N = 30{,}342$). For Stanford–Binet IQ, the corresponding FE coefficient changes from +0.621 (SE = 0.097) to -0.014 (SE = 0.142) after rank adjustment. The unadjusted FE reversal therefore tracks the birth sequence and should not be interpreted as a beneficial paternal-age effect.

Paternal age, maternal age, calendar time, and birth order are highly collinear within mothers. The conditional null is therefore low-precision evidence, not confirmation that paternal age has no effect; conversely, the positive unadjusted FE coefficient does not identify a paternal-age benefit.

8.6 Nelson & Ellenberg (1986): Antecedents of Cerebral Palsy

Nelson & Ellenberg (1986) identified risk factors for cerebral palsy in the CPP, finding that most CP cases had no identifiable perinatal cause and that low birth weight was a stronger predictor than birth asphyxia. We construct a CP indicator from five CPPVAR diagnostic codes for cerebral spastic paresis subtypes (right hemiparesis, left hemiparesis, tetraparesis, paraparesis, and other). Using definite diagnoses (code = 2), we identify 1,900 affected children (3.5%); an alternative definition using the 7-year neurological abnormality summary (v1156 = 2) yields a similar 1,725 cases (4.2%). Both rates substantially exceed the original 189 cases (0.35%), indicating that these variable definitions capture a broader spectrum of motor abnormalities than the clinician-confirmed CP diagnoses used by Nelson & Ellenberg (1986).

Despite this broader definition, the risk factor profile is qualitatively consistent: low birth weight is the strongest predictor (OR = 2.19 [1.91–2.52]), followed by low Apgar score (OR = 1.28 [1.14–1.44]) and preterm birth (OR = 1.31 [1.14–1.50]). In the full model with maternal and SES covariates, higher SEI is independently protective (OR = 0.92 per unit increase [0.89–0.94]). The finding that birth asphyxia (low Apgar) is a modest predictor relative to birth weight is consistent with the original conclusion that most cerebral palsy cannot be attributed to intrapartum events.

In cross-section, the CP indicator (definite spastic paresis) is associated with $b = -0.18$ SD in WISC FSIQ ($p < .001$). Under mother fixed effects—comparing CP-affected children to their unaffected siblings within 606 mixed families (1,354 children)—the coefficient shrinks to $b = -0.049$ (SE = 0.039, $p = 0.21$) for WISC FSIQ, though it remains significant for SB IQ at age 4 ($b = -0.098$, $p = 0.019$). Controlling for birth weight further attenuates the WISC estimate ($b = -0.035$, $p = 0.36$). By contrast, the broader 7-year neurological abnormality indicator (v1156) shows a much larger and persistent within-family deficit ($b = -0.535$, $p < .001$ for WISC; $b = -0.425$, $p < .001$ for SB), even with birth weight control ($b = -0.520$), indicating that severe neurological impairment has genuine causal effects on cognition beyond shared family confounds. The CP-specific indicator's attenuation under mother FE suggests that many of the milder spastic paresis diagnoses captured by the broad CPPVAR definition share familial confounds with IQ, consistent with the paper's observation that this definition captures a much broader motor spectrum than clinician-confirmed CP.

8.7 Naeye (1979): Maternal Weight Gain and Birth Outcomes

Naeye (1979) demonstrated that optimal maternal weight gain depended on pre-pregnancy body habitus: thinner women benefited more from weight gain than heavier women. We closely replicate this by computing pre-pregnancy weight as the difference between pre-delivery weight and weight gain (N = 47,729 singletons with complete data; mean pre-pregnancy weight = 127.7 lbs), then stratifying by Naeye's original weight categories (<110, 110–129, 130–149, 150+ lbs).

The weight gain gradient varies dramatically by pre-pregnancy weight: 20.9 g/lb for thin women (<110 lbs), 17.2 g/lb for 110–129 lbs, 12.4 g/lb for 130–149 lbs, and only 8.2 g/lb for heavy women (150+ lbs). This exactly replicates Naeye's core finding. LBW rates range from 31.3% among thin mothers with low weight gain (<16 lbs) to 2.6% among heavy mothers with high weight gain (≥36 lbs). In the full sample, overall weight gain is associated with 13.5g higher birth weight per pound ($p < 0.001$; N = 50,398) and reduced LBW risk (OR = 0.946 per lb; OR = 0.64 per 10 lbs).

Under mother fixed effects, maternal weight gain continues to predict birth weight within families (b = 8.76 g/lb, SE = 0.59, $p < .001$, N = 50,448), confirming a genuine causal pathway from weight gain to fetal growth. However, the within-family effect on child IQ is null ($b = -0.0006$ SD/lb, SE = 0.0011, $p = 0.57$, N = 38,399), indicating that maternal weight gain affects birth weight but not cognitive development independently of stable maternal characteristics.

8.8 Jensen & Johnson (1994): Head Circumference and IQ

Jensen & Johnson (1994) studied the relationship between head circumference (HC) and IQ in CPP children, reporting several key findings: (1) a modest within-subject HC–IQ correlation ($r \approx 0.17$ at age 7, adjusted for height and weight); (2) an asymmetry in racial matching—matching Black and White children on IQ eliminates the HC gap, but matching on HC leaves the IQ gap essentially intact; and (3) a within-family HC–IQ correlation among full siblings, indicating that the relationship is not merely an artifact of familial confounds. The matching asymmetry is consistent with what Pietschnig et al. (2015) describe as supervenience (many-to-one, not one-to-one): IQ reflects multiple neural and cognitive processes of which head size captures only a fraction, so matching on the more comprehensive downstream phenotype (IQ) necessarily equates the upstream components (including head size), whereas matching on any single upstream component leaves the others—and therefore IQ—unmatched. Jensen used only full siblings; we replicate and extend his analyses across all four kinship types available in the CPP.

Within-Subject Correlations

Table 31 reports correlations of age-7 anthropometrics with WISC FSIQ. Head circumference shows the strongest correlation with IQ ($r = 0.24$ White, 0.20 Black), exceeding both height ($r = 0.15$, 0.11) and weight ($r = 0.11$, 0.11). After adjusting HC for height and weight (i.e., correlating the HC residual with IQ), the HC–IQ correlation attenuates modestly to $r = 0.21$ (White) and 0.16 (Black), confirming that HC's relationship with IQ is not simply a proxy for overall body size. Jensen reported adjusted correlations of approximately 0.17 at age 7; our estimates bracket this value.

Table 31: Within-subject correlations of age-7 anthropometrics with WISC FSIQ ($N =$ 38,110). All values are sex-adjusted.

	Raw r		HC adjusted for Ht/Wt	
Measure	White	Black	White	Black
Head circumference	0.24	0.20	0.21	0.16
Height	0.15	0.11	—	—
Weight	0.11	0.11	—	—

Between-Family and Within-Family Correlations

A critical finding in Jensen & Johnson (1994) was that HC shows a within-family (WF) correlation with IQ—computed from sibling pair *differences*—in addition to the between-family (BF) correlation computed from sibling pair *means*. A non-zero WF correlation indicates a direct or pleiotropic relationship: the sibling who has the larger head also tends to score higher on IQ, even within the same family. By contrast, height and weight showed only BF correlations with IQ, suggesting that their cross-sectional associations with IQ are driven entirely by shared familial factors (e.g., SES, nutrition) rather than a direct biological relationship.

We replicate this decomposition across all four kinship types (Table 32). For full siblings—the comparison most directly comparable to Jensen—the HC BF/WF pattern replicates clearly: $r_{\mathrm{BF}} = 0.27$ (White) and 0.23 (Black), with $r_{\mathrm{WF}} = 0.15$ (White) and 0.06 (Black). The White FS WF/BF ratio of 0.55 closely matches Jensen's reported 0.57. The WF correlation for HC is consistently positive across kinship types ($b = 0.13$ for the pooled FS regression of HC differences on IQ differences, $p < .001$). Half-siblings, who share less genetic variance, show smaller but still positive WF correlations ($b = 0.08$, SE $= 0.05$).

Mother fixed effects estimation with the full sibling sample (N = 39,695) confirms a significant within-family HC–IQ association: $b = 0.082$ SD per SD of HC ($p < .001$). The effect is larger among White children ($b = 0.123$, $p < .001$, $N = 18{,}458$) than among Black children ($b = 0.046$, $p = 0.001$, $N = 19{,}652$). At the subtest level within families, HC predicts all WISC subtests except Coding ($b = -0.029$, $p = 0.07$) and Bender-Gestalt ($b = -0.001$, $p = 0.93$), with the largest effects on WRAT Arithmetic ($b = 0.155$), WRAT Spelling ($b = 0.130$), and WRAT Reading ($b = 0.121$).

Pooling across races and all kinship types ($N = 9{,}042$ pairs), the race-pooled WF correlation for HC–IQ is $r = 0.098$ ($t = 9.38$, $p = 8 \times 10^{-21}$). The BF correlation is $r = 0.272$ ($p < 10^{-150}$). By kinship type, the WF correlation is highly significant for FS pairs ($r = 0.107$, $p = 1.3 \times 10^{-19}$, $N = 7{,}182$) and ambiguous siblings ($r = 0.108$, $p = .008$, $N = 602$), marginally significant for HS ($r = 0.058$, $p = .083$, $N = 899$), and non-significant for twins (DZ: $r = 0.100$, $p = .13$; MZ: $r = -0.025$, $p = .78$), as expected given their small samples.

Height and weight show a contrasting pattern. Among Black siblings, WF correlations with IQ are near zero (FS: $r_{\mathrm{WF}} = 0.01$ for height, -0.003 for weight), confirming Jensen's finding that body size–IQ associations are primarily between-family phenomena. Among White siblings, WF correlations for height and weight are small but positive (FS: 0.12 and 0.09), suggesting some within-family relationship.

Table 32: Between-family (BF) and within-family (WF) correlations of anthropometrics with IQ by kinship type. BF = correlation of sibling pair means; WF = correlation of sibling pair differences.

		White			Black		
Measure	Kinship	N	BF	WF	N	BF	WF
Head circ.	DZ	109	0.29	0.07	116	0.27	0.12
	FS	4,035	0.27	0.15	2,993	0.23	0.06
	HS	207	0.10	0.06	679	0.11	0.08
Height	DZ	109	0.22	0.12	115	0.11	0.16
	FS	4,016	0.16	0.12	2,942	0.12	0.01
	HS	205	0.15	0.10	651	0.10	0.02
Weight	DZ	109	0.29	0.06	116	0.04	0.18
	FS	4,014	0.16	0.09	2,958	0.12	−0.003
	HS	206	0.04	0.10	664	0.12	−0.01

Matching Asymmetry

We replicate Jensen's matching asymmetry analysis ($N = 38{,}110$) using four complementary approaches. The raw racial gaps are $d_{\mathrm{IQ}} = 0.86$ SD and $d_{\mathrm{HC}} = 0.14$ SD (sex-adjusted).

Regression-based matching. Controlling for HC reduces the IQ gap by only 3.2% (from 0.860 to 0.833 SD), while controlling for IQ reduces the HC gap by 47.4% (from 0.137 to 0.072 SD) and reverses its sign: at matched IQ, Black children have slightly *larger* heads ($b = +0.072$ SD, $p < .001$). Adding height and weight controls makes the pattern even starker: the IQ gap barely changes (99.6% remains), while the HC gap reverses entirely ($b = -0.026$).

Jensen's regressed true-score method. To replicate Jensen's exact procedure, we computed regressed true scores $T = r_{XX} \cdot X + (1 - r_{XX}) \cdot \bar{X}_{\mathrm{group}}$ using WISC reliability $r_{XX} = 0.92$, then selected subjects whose regressed true IQ fell within ± 1 point of target values. Matching at the White mean IQ (102.4), the HC difference reverses: Black children's heads are 0.06 cm larger. Matching at the Black mean IQ (89.7), the reversal is larger (-0.24 cm). In the other direction, matching both groups on HC (using regressed true scores with $r_{XX} = 0.95$, per Jensen's estimate of HC reliability) leaves an IQ gap of 12.3 points ($d = 0.83$).

Binned matching. Matching on IQ in 0.5 SD bins ($n_W \geq 10$, $n_B \geq 10$), the weighted mean HC difference reverses to -0.074 SD (Black > White). Matching on HC, the IQ difference is 0.827 SD—96.2% of the raw gap.

Distribution matching via resampling. As a nonparametric alternative, we drew subsamples of $n = 1{,}000$ per race matched to a common IQ distribution (mean = 0, SD = 1 in z-scores) using the `RepsampleR` algorithm. In the IQ-matched subsamples, the HC gap reverses to -0.081 SD ($t = -1.77$, $p = .077$). Conversely, HC-matched subsamples retain 92.6% of the raw IQ gap ($d = 0.796$, $t = 20.4$, $p < .001$). Table 33 summarizes the asymmetry across all four approaches.

Table 33: IQ–head circumference matching asymmetry: controlling for one variable's effect on the other's racial gap.

	IQ gap (SD)		HC gap (SD)	
Specification	Estimate	% remaining	Estimate	% remaining
Raw (sex-adjusted)	0.860	100.0%	0.137	100.0%
Regression-based				
Controlling for HC	0.833	96.8%	—	—
Controlling for IQ	—	—	−0.072	reversed
Controlling for HC + Ht + Wt	0.857	99.6%	—	—
Controlling for IQ + Ht + Wt	—	—	−0.026	reversed
Binned (0.5 SD)				
IQ-matched	—	—	−0.074	reversed
HC-matched	0.827	96.2%	—	—
Resampling ($n = 1{,}000$ per race)				
IQ-matched	—	—	−0.081	reversed
HC-matched	0.796	92.6%	—	—

A test of common regression lines shows that Blacks and Whites fall on essentially the same HC-on-IQ regression ($\Delta b_{\text{slope}} = -0.008$, n.s.), but with a small intercept difference (+0.072 SD for Black; $F = 21.5$, $p < .001$). This replicates Jensen's finding that both groups follow a common regression of HC on IQ. Replacing WISC FSIQ with latent g extracted from the higher-order factor model (Section 4.1) gives nearly identical results: the race intercept at matched g is +0.064 SD ($p < .001$), and controlling for HC leaves 96.3% of the g gap intact.

One concern is Kelley's paradox: because the Black group has a lower mean IQ, matching on IQ could induce regression to the mean in HC, creating a spurious intercept shift. We test this by drawing 1,000 White subsamples matched to the Black IQ distribution and regressing HC on IQ plus a matched/unmatched indicator. The mean intercept shift within Whites is −0.004 SD (95% CI: $[-0.034, 0.026]$), far smaller than the +0.072 SD Black–White intercept and centred on zero. The matching asymmetry is therefore not a regression artifact.

Sibling Effect Sizes

Jensen reported that among full siblings differing by ≥ 1 SD in IQ, the higher-IQ sibling had a larger head by approximately ES = 0.18 SD. We replicate this analysis across kinship types. Among 1,746 full-sibling pairs differing ≥ 1 SD in IQ, the higher-IQ sibling has a head circumference advantage of 0.183 SD (SE = 0.026)—matching Jensen's estimate almost exactly. By contrast, the higher-IQ sibling's height advantage is only 0.107 SD and weight advantage is 0.071 SD, both smaller than the HC effect, consistent with HC having a more direct relationship with IQ than general body size.

The HC effect is smaller among half-siblings (0.097 SD, SE = 0.095), as expected given their lower genetic relatedness ($R = 0.25$ vs. 0.50), though the estimate is imprecise. The reverse analysis—pairs differing ≥ 1 SD in HC—shows that the larger-headed sibling scores 0.145 SD higher in IQ among full siblings (SE = 0.019), an effect that attenuates to 0.079 SD (SE = 0.051) among half-siblings.

Head Circumference and Subtest-Specific Variance Beyond g

Some studies have reported that brain volume retains associations with spatial/performance abilities after partialling out g, suggesting that head size captures something beyond general intelligence. We test this by extracting g via maximum-likelihood factor analysis (single common factor, regression scores) from the 12-test battery (N = 28,328; 37.7% of variance explained), residualizing each subtest of g, and correlating HC with the residuals. ML factor analysis targets common variance only, unlike PCA which includes specific and error variance; using ML g thus avoids over-partialling.

Cross-sectionally, HC correlates r = 0.05–0.20 with raw subtest scores, but all associations drop to near zero after partialling out g (range -0.035 to $+0.048$). There is no spatial/performance advantage: the mean net-of-g HC correlation is +0.006 for verbal subtests and +0.015 for performance subtests—both negligible. The largest residual associations are positive (Comprehension $r = +0.048$, Block Design $r = +0.045$), not spatially specific.

Within-family estimates (mother fixed effects) confirm this pattern. Predicting g-residualized subtest scores from HC, the mean coefficient is +0.002 for verbal subtests and −0.007 for performance subtests, with no individual subtest reaching significance. Regressing verbal and performance composites on HC while controlling for g produces within-family coefficients of $b = 0.011$ ($p = .135$) for verbal and $b = 0.005$ ($p = .556$) for performance—neither significant.

We confirm this within a structural equation model using a higher-order MIMIC specification: the same four-factor higher-order CFA from the measurement invariance analysis (Gc, Gv, Grw, Gs $\rightarrow g$), with HC added as an exogenous cause. Model 1 constrains HC to operate only through g ($\beta = 0.26$, $p < .001$; CFI = .932, RMSEA = .064). Model 2 frees direct HC $\rightarrow$ group-factor paths for all four factors; the ΔCFI is +0.002—far below the .01 threshold—and no direct path exceeds $\beta = 0.05$. Model 3 tests spatial specificity by freeing only the HC $\rightarrow$ Gv path: the direct effect is $\beta = 0.011$ ($p = .053$), with ΔCFI = 0.0001.

HC's entire association with the cognitive battery operates through g; there is no residual spatial or visual-processing specificity.

Sex Differences in Head Circumference at Matched IQ

Jensen & Johnson (1994) reported sex differences in head size alongside race differences. We extend their analysis by asking whether girls still have smaller heads than boys at the same IQ level, and whether girls score higher than boys at the same head size.

HC at matched IQ. The raw sex gap in HC at age 7 is 0.71 cm ($d = 0.45$; male $M = 51.66$, female $M = 50.95$). Controlling for WISC FSIQ barely changes this ($b = -0.71$ cm, $p < 10^{-16}$): boys and girls with identical IQs still differ by 0.71 cm in head circumference. Adding race, height, and weight reduces the gap only modestly to 0.65 cm ($p < 10^{-16}$). Within mixed-sex sibships under mother fixed effects—comparing brothers and sisters in the same family—the gap is essentially unchanged: -0.73 cm controlling for IQ ($p < 10^{-16}$; 3,211 families), or -0.66 cm with additional height and weight controls. The result is robust across cognitive measures: controlling for WISC VIQ (-0.56 cm), WISC PIQ (-0.60 cm), Stanford-Binet IQ at age 4 (-0.67 cm), or PCA g (-0.61 cm) all yield the same pattern, with within-family estimates (-0.64 to -0.73 cm) closely matching the cross-sectional values. The sex gap in head size at matched IQ is constant or slightly widening across the IQ range: 0.53 cm at IQ 65, 0.65 cm at IQ 95, 0.85 cm at IQ 115, and 0.92 cm at IQ 125. Girls achieve the same cognitive performance with substantially smaller brains.

IQ at matched HC. The reverse comparison asks whether girls score higher than boys at matched head circumference. The raw sex gap in IQ is trivial (-0.25 points, $p = 0.10$), but controlling for HC reverses it: girls score +1.3 WISC points higher than boys with the same head circumference ($p < 10^{-17}$), and adding race, height, and weight leaves this intact (+1.3, $p < 10^{-22}$). For Stanford-Binet IQ, the female advantage at matched HC is larger still (+3.9 points, $p < 10^{-100}$). Within mixed-sex sibships, the effect is +0.5 WISC points ($p = 0.05$).

However, HC-matched comparisons are subject to opposite-direction selection within each sex. At HC = 50.5–51.5 cm (near the overlap center), the matched boys are at the 47th percentile of the male HC distribution while the matched girls are at the 65th percentile of the female distribution. This asymmetric selection is visible in binned means: at HC below the male mean (∼51.7 cm), girls outscore boys by 1–3 IQ points; at HC above the male mean, boys outscore girls by 1–5 points. The crossover around the male mean indicates that the IQ advantage at matched HC reflects differential selection—boys with small heads are below-average for their sex, while girls with the same head size are above-average—rather than a uniform female neural efficiency advantage. The IQ-matched comparison (girls have smaller heads at every IQ level) is not subject to this confound and provides cleaner evidence that the HC–IQ relationship differs by sex.

HC–IQ slope invariance across sex. The raw HC–IQ slopes differ substantially by sex ($b_M = 2.50$, $b_F = 1.83$; interaction $p < 10^{-10}$). However, after controlling for body size (height, weight, and race), the slopes become identical: $b_M = 1.79$, $b_F = 1.78$ (interaction $p = 0.78$). The apparent slope difference is entirely driven by body-size confounding—boys are larger, and body size correlates with both HC and IQ. This parallels the race finding from Section 8.8: Jensen & Johnson (1994) showed that Blacks and Whites fall on the same HC-on-IQ regression with a small intercept difference, and here males and females likewise fall on the same body-size-adjusted regression with a +1.3 IQ-point intercept shift favoring females ($p < 10^{-22}$). The slope invariance holds within each race (White: $b_M = 2.33$, $b_F = 2.31$; Black: $b_M = 1.48$, $b_F = 1.60$) and within families (male same-sex sibs: $b = 1.23$; female: $b = 0.66$; pooled with sex control: $b = 0.82$). The HC–IQ relationship is thus a biological regularity that operates identically across sex, race, and family background, with group differences expressed as intercept shifts rather than slope differences.

Higher-order MI across sexes. The same 12-test higher-order CFA used for the racial MI analysis (Section 4.1) applied across sexes ($N = 31{,}031$; Male = 15,559, Female = 15,472) shows metric invariance holds (configural CFI = 0.946, ΔCFI = 0.002). Full scalar invariance fails (ΔCFI = 0.015), but freeing only the Goodenough intercept—the same parameter freed in the racial analysis—restores partial scalar invariance (ΔCFI = 0.005). From this partial scalar base, strict invariance holds (ΔCFI = 0.002) and latent variance homogeneity holds (ΔCFI = 0.001). Under partial strict invariance, the latent g mean gap is +0.19 male-SD favoring females, and the variance ratio is 0.96 (males slightly more variable). The Goodenough Draw-a-Person test thus shows sex-differential intercepts in both the racial and sex MI analyses, consistent with its known sensitivity to fine motor development, which differs across demographic groups independently of g.

9 Discussion

These twenty-five analyses demonstrate the CPP's unique research value. Several themes emerge.

Within-family designs change conclusions. The breastfeeding analysis is the clearest example: a −0.65 SD cross-sectional association becomes +0.016 (SE = 0.041) under mother fixed effects. The WISC smoking coefficient is similarly near zero. Observed-score birth-order coefficients become positive within mothers, although rank cannot be separated cleanly from maternal age and calendar time. Birth weight and early weight gain retain positive within-family associations. Head-circumference adjustment attenuates the birth-weight coefficient by about 17%, but does not attenuate the postnatal weight-gain coefficient. These designs remove stable maternal confounding; causal interpretation still requires no relevant time-varying sibling-specific confounding.

The Black–White gap is fully developed before schooling begins. The Stanford-Binet gap at age 4 ($d = 0.87$, 13.1 points) is at least as large as the WISC gap at age 7 ($d = 0.95$, 12.7 points). Among the 32,892 children tested at both ages, the gap narrows slightly from 13.1 to 12.3 points ($\Delta = -0.9$, $p < 0.001$): White children lose more IQ from age 4 to 7 (-2.1 points) than Black children (-1.3 points). By SES, the gap is completely stable from age 4 to 7 at low and middle SES ($\Delta < 0.1$ points) and narrows at high SES ($\Delta = -1.6$), consistent with regression to the mean being stronger for the more selected high-SES White group. The gap's full presence at age 4—before most children have entered formal schooling—is difficult to reconcile with theories that attribute it primarily to school quality or educational access. It is, however, consistent with the behavior genetics results (Section 3), which attribute ∼60–70% of IQ variance to genetic factors.

The multi-level relatedness result is robust to pedigree dependence. Classical twin-only ACE decomposition yields $a^2 = 0.40$ for FSIQ, while multi-level estimation with four kinship levels gives $a^2 = 0.71$ under its model assumptions. Connected-pedigree block bootstraps and one-pair-per-component samples preserve the raw kinship-correlation ordering. The 12-measure IP remains strongly preferred to the CP after each documented extended family contributes only one dyad, and the WISC-only result persists across 100 random one-dyad-per-component selections. The extended-kinship correlations still overlap the same pedigrees and are therefore a consistency check rather than a separate replication. Multivariate pathway estimates are independence-robust conditional on the documented genealogy but remain assumption-dependent; downstream Dolan, fourth-moment, and assortative-mating decompositions should be treated as sensitivity analyses rather than identified causal partitions.

Historical features create natural experiments. The reversed SES–breastfeeding gradient of the 1960s, the high smoking prevalence (46.7%), and the large half-sibling sample provide analytical opportunities unavailable in modern cohorts. The CPP's historical context is not a limitation but a scientific advantage.

Measurement invariance supports score comparability, not causal identification. The WISC models meet approximate invariance criteria across observed birth-order and family-size groups, and examiner ratings have a similar measurement structure across race. These findings argue against large distortions that operate by changing the fitted loadings or item intercepts. They do not rule out unmeasured causes of latent mean differences, approximately additive influences, selection, or different causal mixtures that leave the measurement model unchanged. Accordingly, the invariance results support comparing the modeled constructs across groups but do not establish why their means differ.

As Lubke et al. (2003b) emphasized, strict factorial invariance says nothing by itself about the sources of latent factor-mean differences. The Dolan exercise in Section 4.1 asks how well the observed mean pattern can be projected onto the fitted genetic and shared-environment loading profiles under a strong common-causation model. Its post-hoc $R^2 = 0.90$ is model fit, not validation that the latent profiles are causally responsible for the group mean difference. Because the projection inherits the ACE assumptions, loading collinearity, and absence of measured genetic data, we report it as an assumption-dependent decomposition and sensitivity analysis rather than an identified genetic/environmental attribution.

Attrition is not a major threat. Despite dramatic site-level variation in follow-up, the followed sample is remarkably representative on SES dimensions. This is likely because attrition was driven primarily by institutional factors (site closure, protocol changes) rather than participant self-selection on socioeconomic characteristics.

Replications validate the extraction and extend prior findings. The Broman descriptive benchmarks are replicated within 1–2 IQ points. The Scarr–Rowe SES moderation of heritability (Turkheimer et al., 2003) is qualitatively confirmed but less extreme than originally reported. For paternal age, the published-specification negative association is small; the positive mother-FE coefficient disappears after rank adjustment, leaving the age effect unidentified because paternal age and birth sequence are nearly collinear. The Willerman interracial mating analysis, the Myrianthopoulos twin–singleton gap, the Nelson & Ellenberg cerebral palsy risk factors, and the Naeye weight-gain gradients are directionally replicated. The Jensen & Johnson HC–IQ analysis extends especially well across kinship types. Together these checks support the extraction while also showing why source-code semantics and specification-level reproducibility matter.

Survey weighting does not overturn the main estimates. With combined inverse-probability and population-raking weights, the WISC birth-order coefficient changes from +0.045 to +0.038 SD, breastfeeding from +0.033 to +0.008, and birth weight from +0.0149 to +0.0156 SD per 100g. Across 39 checked coefficients, the median absolute percentage change is 12.1%. Three signs change, all for near-zero estimates. Weighting is therefore reassuring for these measured attrition dimensions, though it cannot correct selection on unmeasured variables. OpenMx structural models are reported unweighted.

The CPP remains a uniquely valuable resource for research on the developmental origins of cognitive ability. Its combination of large sample size, multi-ethnic composition, detailed obstetric and socioeconomic data, and rich family structure—including twins, half-siblings, and over 9,000 first-cousin pairs—is unmatched by any other single dataset. The extended kinship network opens several avenues for future work: cousin fixed effects (comparing children of siblings) can separate genetic transmission from shared-family environment more precisely than sibling comparisons alone; cousin-pair IQ correlations provide additional constraints on ACE models at the $R = 0.125$ level; and the racial stratification of the cousin network enables within-race cousin analyses that sidestep the population stratification issues that complicate cross-family comparisons. The modernized data release described in our companion paper makes this resource accessible for the first time to the broader research community.

Data Availability

The modernized CPP dataset and all analysis scripts used in this paper are available at https://osf.io/d3f9k/ (OSF project) and https://github.com/ljlasker/CollaborativePerinatalProject (documentation site). See the companion data paper (Lasker, 2026) for full documentation of the data rescue process, variable definitions, and tiered file descriptions. The analysis-ready file (`cpp_clean_v1.csv`), kinship links (`cpp_kinship_links.csv`), and extended kinship links (`cpp_extended_kinship_links.csv`) are sufficient to replicate all analyses presented here. All R scripts (numbered `01_data_prep.R` through `19_replication_naeye.R`) are included in the `analysis_paper/` directory of the release and can be executed sequentially via `00_run_all.R`.

References

Broman, S. H., Nichols, P. L., & Kennedy, W. A. (1975). *Preschool IQ: Prenatal and Early Developmental Correlates*. Hillsdale, NJ: Erlbaum.

Broman, S. H., Bien, E., & Shaughnessy, P. (1987). *Low Achieving Children: The First Seven Years*. Hillsdale, NJ: Erlbaum.

Chen, F. F., Sousa, K. H., & West, S. G. (2005). Testing measurement invariance of second-order factor models. *Structural Equation Modeling*, *12*(3), 471–492.

Chen, F. F. (2007). Sensitivity of goodness of fit indexes to lack of measurement invariance. *Structural Equation Modeling*, *14*(3), 464–504.

Cheung, G. W., & Rensvold, R. B. (2002). Evaluating goodness-of-fit indexes for testing measurement invariance. *Structural Equation Modeling*, *9*(2), 233–255.

Lasker, J. (2026). The Collaborative Perinatal Project: A modern data release. OSF Preprints. https://osf.io/4tna9.

Lubke, G. H., Dolan, C. V., Kelderman, H., & Mellenbergh, G. J. (2003a). On the relationship between sources of within- and between-group differences and measurement invariance in the common factor model. *Intelligence*, *31*(6), 543–566.

Lubke, G. H., Dolan, C. V., Kelderman, H., & Mellenbergh, G. J. (2003b). Weak measurement invariance with respect to unmeasured variables: An implication of strict factorial invariance. *British Journal of Mathematical and Statistical Psychology*, *56*(2), 231–248.

DeShon, R. P. (2004). Measures are not invariant across groups without error variance homogeneity. *Psychology Science*, *46*(1), 137–149.

Myrianthopoulos, N. C. (1975). An epidemiologic survey of twins in a large, prospectively studied population. *American Journal of Human Genetics*, *27*(2), 158–165.

Saha, S., Barnett, A. G., Foldi, C., Burne, T. H., Eyles, D. W., Buka, S. L., & McGrath, J. J. (2009). Advanced paternal age is associated with impaired neurocognitive outcomes during infancy and childhood. *PLoS Medicine*, *6*(3), e1000040.

Myrianthopoulos, N. C., & Keith, L. (1976). Distribution of intellectual potential in an infant population. In *Twin Research: Psychology and Methodology* (pp. 185–199). New York: Alan R. Liss.

Naeye, R. L. (1979). Weight gain and the outcome of pregnancy. *American Journal of Obstetrics and Gynecology*, *135*(1), 3–9.

Nelson, K. B., & Ellenberg, J. H. (1986). Antecedents of cerebral palsy: Multivariate analysis of risk. *New England Journal of Medicine*, *315*(2), 81–86.

Niswander, K. R., & Gordon, M. (1972). *The Women and Their Pregnancies*. Philadelphia: W.B. Saunders.

Bouchard, T. J., & McGue, M. (1981). Familial studies of intelligence: A review. *Science*, *212*(4498), 1055–1059.

Horwitz, T. B., Balbona, J. V., Paulich, K. N., & Keller, M. C. (2023). Evidence of correlations between human partners based on systematic reviews and meta-analyses of 22 traits and UK Biobank analysis of 133 traits. *Nature Human Behaviour*, *7*(9), 1568–1583.

Pesta, B. J., Kirkegaard, E. O. W., te Nijenhuis, J., Lasker, J., & Fuerst, J. G. R. (2020). Racial and ethnic group differences in the heritability of intelligence: A systematic review and meta-analysis. *Intelligence*, *78*, 101408.

Plomin, R., & Deary, I. J. (2015). Genetics and intelligence differences: Five special findings. *Molecular Psychiatry*, *20*, 98–108.

Strenze, T. (2007). Intelligence and socioeconomic success: A meta-analytic review of longitudinal research. *Intelligence*, *35*(5), 401–426.

Turkheimer, E., Haley, A., Waldron, M., D'Onofrio, B., & Gottesman, I. I. (2003). Socioeconomic status modifies heritability of IQ in young children. *Psychological Science*, *14*(6), 623–628.

Willerman, L., Naylor, A. F., & Myrianthopoulos, N. C. (1970). Intellectual development of children from interracial matings. *Science*, *168*(3938), 1329–1331.

Machts, N., Kaiser, J., Schmidt, F. T. C., & Möller, J. (2016). Accuracy of teachers' judgments of students' cognitive abilities: A meta-analysis. *Educational Research Review*, *19*, 85–103.

Hoge, R. D., & Coladarci, T. (1989). Teacher-based judgments of academic achievement: A review of literature. *Review of Educational Research*, *59*(3), 297–313.

Südkamp, A., Kaiser, J., & Möller, J. (2012). Accuracy of teachers' judgments of students' academic achievement: A meta-analysis. *Journal of Educational Psychology*, *104*(3), 743–762.

Mack, E., Jansen, M., & Lüdtke, O. (2025). How smart is my child? Parents' accuracy in judging their children's cognitive ability. *Intelligence*, *108*, 101876.

Freund, P. A., & Holling, H. (2024). Mirror, mirror on the wall, who is the smartest of them all? A multiverse meta-analysis of self-estimated intelligence. *Intelligence*, *106*, 101837.

Franić, S., Dolan, C. V., Borsboom, D., Hudziak, J. J., van Beijsterveldt, C. E. M., & Boomsma, D. I. (2013). Can genetics help psychometrics? Improving dimensionality assessment through genetic factor modeling. *Psychological Methods*, *18*(3), 406–433.

Molenaar, P. C. M., Boomsma, D. I., & Dolan, C. V. (1993). A third source of developmental differences. *Behavior Genetics*, *23*(6), 519–524.

Dolan, C. V., Molenaar, P. C. M., & Boomsma, D. I. (1989). LISREL analysis of twin data with structured means. *Behavior Genetics*, *19*(1), 51–62.

Dolan, C. V., Molenaar, P. C. M., & Boomsma, D. I. (1992). Decomposition of multivariate phenotypic means in multigroup genetic covariance structure analysis. *Behavior Genetics*, *22*(3), 319–335.

Dolan, C. V., Molenaar, P. C. M., & Boomsma, D. I. (1994a). Simultaneous genetic analysis of means and covariance structure: Pearson–Lawley selection rules. *Behavior Genetics*, *24*(1), 17–24.

Dolan, C. V., & Molenaar, P. C. M. (1994b). Testing specific hypotheses concerning latent group differences in multi-group covariance structure analysis with structured means. *Multivariate Behavioral Research*, *29*(3), 203–222.

Dolan, C. V., Huijskens, R. C. A., Minică, C. C., Neale, M. C., & Boomsma, D. I. (2020). Decomposition of mean sex differences in alcohol use within a genetic factor model. *Behavior Genetics*, *50*(6), 369–380.

Panizzon, M. S., Vuoksimaa, E., Spoon, K. M., Jacobson, K. C., Lyons, M. J., Franz, C. E., Xian, H., Vasilopoulos, T., & Kremen, W. S. (2014). Genetic and environmental influences on general cognitive ability: Is g a valid latent construct? *Intelligence*, *43*, 65–76.

Rowe, D. C., Vazsonyi, A. T., & Flannery, D. J. (1994). No more than skin deep: Ethnic and racial similarity in developmental process. *Psychological Review*, *101*(3), 396–413.

Rowe, D. C., & Cleveland, H. H. (1996). Academic achievement in Blacks and Whites: Are the developmental processes similar? *Intelligence*, *23*(3), 205–228.

Rowe, D. C. (2005). Under the skin: On the impartial treatment of genetic and environmental hypotheses of racial differences. *American Psychologist*, *60*(1), 60–70.

Meredith, W. (1993). Measurement invariance, factor analysis and factorial invariance. *Psychometrika*, *58*(4), 525–543.

Rijsdijk, F. V., Vernon, P. A., & Boomsma, D. I. (2002). Application of hierarchical genetic models to Raven and WAIS subtests: A Dutch twin study. *Behavior Genetics*, *32*(3), 199–210.

Lipovetsky, S., & Conklin, M. (2001). Analysis of regression in game theory approach. *Applied Stochastic Models in Business and Industry*, *17*(4), 319–330.

Jensen, A. R., & Johnson, F. W. (1994). Race and sex differences in head size and IQ. *Intelligence*, *18*(3), 309–333.

Pietschnig, J., Penke, L., Wicherts, J. M., Zeiler, M., & Voracek, M. (2015). Meta-analysis of associations between human brain volume and intelligence differences: How strong are they and what do they mean? *Neuroscience & Biobehavioral Reviews*, *57*, 411–432.

Nye, C. D., & Drasgow, F. (2011). Effect size indices for analyses of measurement equivalence: Understanding the practical importance of differences between groups. *Journal of Applied Psychology*, *96*(5), 966–980.

Nye, C. D., Bradburn, J., Olenick, J., Bialko, C., & Drasgow, F. (2019). How big are my effects? Examining the magnitude of effect sizes in studies of measurement equivalence. *Organizational Research Methods*, *22*(3), 678–709.

Fraser, A., Almqvist, C., Larsson, H., Långström, N., & Lawlor, D. A. (2014). Maternal diabetes in pregnancy and offspring cognitive ability: sibling study with 723,775 men from 579,857 families. *Diabetologia*, *57*(1), 102–109.

DeFries, J. C., & Fulker, D. W. (1985). Multiple regression analysis of twin data. *Behavior Genetics*, *15*(5), 467–473.

Scarr-Salapatek, S. (1971). Race, social class, and IQ. *Science*, *174*(4016), 1285–1295.

Wechsler, D. (1949). *Wechsler Intelligence Scale for Children*. New York: The Psychological Corporation.

Jastak, J. F., & Jastak, S. R. (1965). *The Wide Range Achievement Test*. Wilmington, DE: Guidance Associates.

Jensen, A. R. (1973). *Educability and Group Differences*. New York: Harper & Row.

Jensen, A. R. (1998). *The g Factor: The Science of Mental Ability*. Westport, CT: Praeger.

Lewontin, R. C. (1970). Race and intelligence. *Bulletin of the Atomic Scientists*, *26*(3), 2–8.

Mundlak, Y. (1978). On the pooling of time series and cross section data. *Econometrica*, *46*(1), 69–85.

Blake, J. (1981). Family size and the quality of children. *Demography*, *18*(4), 421–442.

Downey, D. B. (1995). When bigger is not better: Family size, parental resources, and children's educational performance. *American Sociological Review*, *60*(5), 746–761.

Zajonc, R. B. (1975). Birth order and intellectual development. *Psychological Review*, *82*(1), 74–88.

Ulijaszek, S. J. & Kerr, D. A. (1999). Anthropometric measurement error and the assessment of nutritional status. *British Journal of Nutrition*, *82*(3), 165–177.

Fok, T. F., et al. (2003). Updated gestational age specific birth weight, crown-heel length, and head circumference of Chinese newborns. *Archives of Disease in Childhood: Fetal and Neonatal Edition*, *88*(3), F229–F236.

Smit, D. J. A., Luciano, M., Bartels, M., van Beijsterveldt, C. E. M., Wright, M. J., Boomsma, D. I., & Posthuma, D. (2010). Heritability of head size in Dutch and Australian twin families at ages 0–50 years. *Twin Research and Human Genetics*, *13*(4), 370–380.

Silventoinen, K., Karvonen, M., Sugimoto, M., Kaprio, J., Dunkel, L., & Yokoyama, Y. (2011). Genetics of head circumference in infancy: A longitudinal study of Japanese twins. *American Journal of Human Biology*, *23*(5), 630–634.

Cortez Ferreira, R., et al. (2025). Growth trajectories in monochorionic and dichorionic twins with weight discordance: A cohort study. *European Journal of Pediatrics*, *184*, 319.

Stiles, J. & Jernigan, T. L. (2010). The basics of brain development. *Neuropsychology Review*, *20*(4), 327–348.

Sattler, J. M. (1974). *Assessment of Children's Intelligence*. Philadelphia: W. B. Saunders.

Koppitz, E. M. (1963). *The Bender Gestalt Test for Young Children*. New York: Grune & Stratton.

Harris, D. B. (1963). *Children's Drawings as Measures of Intellectual Maturity*. New York: Harcourt, Brace & World.

Warrington, N. M., Beaumont, R. N., Horikoshi, M., et al. (2019). Maternal and fetal genetic effects on birth weight and their relevance to cardio-metabolic risk factors. *Nature Genetics*, *51*(5), 804–814.

A Supplementary: Proportionality of g-Loadings and Variance Components

Under a strict Common Pathway model, phenotypic g-loadings should predict both per-test heritability (h^2) and shared environmentality (c^2) equally, because both variance sources flow through the common factor proportionally. Under the Independent Pathway, g-loadings may correlate more strongly with one component than the other, reflecting the structural non-proportionality of the genetic and environmental loading vectors. We test this by correlating PCA g-loadings with the Cholesky per-test ACE estimates and with the IP common-factor loading vectors.

g-Loadings vs. Cholesky Per-Test ACE

Table 34 reports PCA g-loadings alongside Cholesky per-test h^2, c^2, and e^2 for all twelve measures.

Table 34: PCA g-loadings and Cholesky per-test ACE estimates.

Subtest	g-loading	h^2	c^2	e^2
Information	0.688	.39	.20	.41
Comprehension	0.577	.11	.19	.71
Vocabulary	0.714	.26	.43	.31
Digit Span	0.570	.36	.18	.46
Picture Arr.	0.594	.32	.26	.42
Block Design	0.562	.24	.24	.52
Coding	0.370	.27	.09	.64
WRAT Reading	0.791	.71	.08	.21
Bender-Gestalt	0.695	.36	.17	.47
WRAT Spelling	0.803	.63	.06	.31
WRAT Arithmetic	0.799	.64	.05	.31
Goodenough	0.520	.36	.12	.52

Pearson correlations between g-loadings and the Cholesky variance components are $r(g, h^2) = +0.557$ ($p = 0.060$), $r(g, c^2) = +0.064$ ($p = 0.844$), $r(g, e^2) = -0.773$ ($p = 0.003$), and $r(g, h^2/(h^2 + c^2)) = +0.549$ ($p = 0.065$). Spearman rank correlations tell the same story: $\rho(g, h^2) = +0.573$, $\rho(g, c^2) = -0.014$, $\rho(g, e^2) = -0.678$. Higher-g subtests tend to be more heritable and have less unique-environmental variance, but g-loadings are essentially unrelated to c^2. A strict CP would predict that g-loadings track both h^2 and c^2 proportionally, since both flow through a single common factor. That g predicts h^2 but not c^2 fits the IP picture instead: genetic influences are organized along g, but shared-environmental influences follow a different subtest profile.

g-Loadings vs. IP Common-Factor Loadings

The IP genetic common-factor loading vector ($\mathbf{a}_c$) lines up almost perfectly with phenotypic g: $r(g, \mathbf{a}_c) = 0.853$, Tucker $\phi(g, \mathbf{a}_c) = 0.989$. The environmental vector ($\mathbf{c}_c$) does not: $r(g, \mathbf{c}_c) = 0.240$, $\phi(g, \mathbf{c}_c) = 0.725$. Put plainly, the genetic common factor *is* phenotypic g—the factor that falls out of the phenotypic covariance matrix is the same one that organizes genetic variation across these twelve tests. The environmental common factor, by contrast, loads on subtests in a markedly different pattern, which is exactly why the CP (which forces a single loading vector for both) fits poorly.

B Supplementary: Survey-Weighted Sibling Death Decomposition

Table 35 repeats Table 18 with survey weights calibrated to the age-7 follow-up population. Weighted mother-FE estimates are surviving +0.52 (SE = 0.20, $p = 0.011$) and dead +0.82 (SE = 0.41, $p = 0.048$). Weighting does not generate the negative surviving-sibling association predicted by simple dilution, but it does not make the channel contrast causally identified.

Table 35: Survey-Weighted Mean WISC FSIQ by Biological Birth Order and Prior Sibling Deaths

	Prior siblings who died			
Bio BO	0	1	2	3+
1	101.7			
2	100.8	101.7		
3	99.2	100.0	99.3	
4	96.9	98.7	99.8	106.2
5	95.6	97.4	99.6	98.1

Note: Weighted using combined IPW × population-raking weights. Compare with unweighted Table 18. Mother-FE estimates are surviving +0.52 (SE = 0.20) and dead +0.82 (SE = 0.41). Sparse high-death cells remain unstable.